An Annotated Bibliography and Information Summary on The Fisheries Resources of the Yukon River Basin in Canada

G.L. Ennis, A. Cinader, S. McIndoe and T. Munsen
(Yukon River Basin Study Project No. 31)

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AN ANNOTATED BIBLIOGRAPHY AND INFORMATION SUMMARY ON THE FISHERIES RESOURCES OF THE YUKON RIVER BASIN IN CANADA

by

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TERMS OF REFERENCE

This study was undertaken to provide background information for the Fisheries Work Group of the Yukon River Basin Study. The Basin Study is a joint study by Canada, B.C., and Yukon of the water and related resources of the Yukon River Basin.

The fisheries information study forms project No. 31 of the Yukon River Basin Study. The Terms of Reference (Project Description) for the fisheries information study follows:

PROJECT DESCRIPTION: This study will provide a description of the fisheries resources of the Yukon Basin by:

(1) Compiling existing fisheries information from periodicals, university theses, government and industrial technical reports, mapping projects, and government records (open files, field notes, unpublished or incomplete studies, etc.).

(2) Reviewing and evaluating the quality of the information.

(3) Producing a document which:

(a) provides an annotated bibliography of the information (by sub-basin) including comments on the quality and usefulness of the existing information;

(b) discusses and synthesizes the reports focusing on fish processes existing in the Basin, the applicability of the existing information, data gaps, and a summary of relevant recommendations and conclusions from the reports that were reviewed;

(c) provides documentation of other data basis for reference, the location of referenced material and scope of the material referenced;

(d) maps of the watershed displaying fish species distribution and abundances.

The annotated bibliography and information summary on the following pages fulfills objectives 1, 2 and 3, a, b, c of the Terms of Reference. As a point of clarification on objective 3b we found it impractical to summarize data gaps and recommendations from individual reports and have instead made general statements on data gaps and recommended studies for each sub-basin after reviewing the relevant information. The summaries and recommended studies are those of the authors and not necessarily those of the Yukon River Basin Committee, Parties to the Basin Agreement or the Department of Fisheries and Oceans.

Fisheries resource maps were separately produced to fulfill objective 3d of the Terms of Reference. These maps may be ordered from the Department of Fisheries and Oceans (contact G.L. Ennis).

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ABSTRACT

Ennis, G.L., A. Cinader, S. McIndoe, and T. Munsen. 1982. An annotated bibliography and information summary on the fisheries resources of the Yukon River Basin in Canada. Can. MS Rept. Fish. Aquat. Sci. 1657: V + 278 p.

This annotated bibliography provides a listing of 223 fisheries reports pertaining to the Canadian portion of the Yukon River and its tributaries (exclusive of the Porcupine River). Sources for the bibliography include periodicals, government and industrial technical reports and government records. Identification of source materials was facilitated by a computer search of several data bases.

The bibliographic information is listed alphabetically by author and cross-referenced under the following sub-basin headings: Yukon River mainstem, White River, Stewart River, Pelly River and Teslin river. The Aishihik River Basin was also referenced as there are existing proposals to divert water bodies of the Yukon system into the Aishihik system. Information collected for each sub-basin has been summarized and evaluated according to its applicibality to the Yukon River Basin Study objectives. Data gaps are identified and some additional studies suggested for each sub-basin.

References which were not received in time to be annotated are listed alphabetically.

Alternative sources of fisheries information for the Yukon Basin such as other bibliographies and mapping projects are listed in a separate section.

RÉSUMÉ

Ennis, G.L., A. Cinader, S. McIndoe, and T. Munsen. 1982. An annotated bibliography and information summary on the fisheries resources of the Yukon River Basin in Canada. Can. MS Rept. Fish. Aquat. Sci. 1657: V + 278 p.

Cette bibliographie annotée donne une liste de 223 exposés en pêcherie ayant rapport à la rivière Yukon et ses tributaires (à l'exclusion de la rivière Porcupine). Les références bibliographiques proviennent de périodiques, de rapports techniques governementals et industriels ainsi que des archives de governement. L'identification du matériel de référence fut facilité par une recherche sur ordinateur de plusieurs lignes de repère.

L'information bibliographique est organisée en ordre alphabétique par auteur et les renvois sont catalogués sous les titres des sous-bassins suivants: la parti principale de la rivière Yukon, la rivière White, la rivière Stewart, la rivière Pelly et la rivière Teslin. Le bassin de la rivière Aishihik fut aussi classifié car il existe des soumissiones pour dévier des masses d'eau du réseau Yukon dans le réseau Aishihik. L'information recueilli de chaque sous-bassin fut résumé et évalué selon l'applicabilité aux objectifs de l'Étude du Bassin de la Rivière Yukon. L'écart des donnés fut identifié et des études additionnelles furent suggèrées pour chaque sous-bassin.

Les références reçues trop tard pour être annotées sont cataloguées en ordre alphabétique.

Des soures alternatives d'informaiton relié au sujet de pèche pour le bassin du Yukon -- telles que certaines bibliographies additionnelles et des projets cartographiques -- sont cataloguées dans une section supplémentaire.

ACKNOWLEDGEMENTS

A number of people have helped with this report. We would like to thank the librarians at Fisheries and Oceans, Paulette Westlake and Marcia Waldron, for their kindness and assistance in locating references. We also thank the many librarians from other Fisheries' libraries, other agencies and private industry that supplied us with information. Staff of the Habitat Management Division, Department of Fisheries and Oceans, provided reference material not otherwise available. Sandy Johnston of Fisheries in Whitehorse made valuable suggestions for summarizing the sub-basins, for which we are grateful. Bill Parker, of Woodward Library, University of British Columbia, conducted the computer searches. We thank Donna Lee, Thelma Collins and Patricia Benoit for typing the bibliography.

Funding for the project was provided by the Yukon River Basin Study and we thank the Study Director, W.G. Whitley, for his help and encouragement.

INTRODUCTION

This annotated bibliography provides a listing of fisheries and fish-related information pertaining to the Canadian portion of the Yukon River and its tributaries (exclusive of the Porcupine River). Sources for the bibliography include periodicals, government and industrial technical reports and government records.

Identification of source materials was facilitated by a computer search of the following data bases: QL, CANOLE, and DIALOG. Three QL files were searched: Canadian environment (CENV), Arctic Sciences and Technology (AST) and Boreal Northern Titles (BNT). Four CANOLE files were searched: Biological Abstracts (BA), NTIS, Environment Canada Departmental Library (ELIAS), and Directory of Federally Supported Research in Universities from the Information Exchange Center (IEC). One DIALOG file was searched: Pollution Abstracts.

The bibliographic information is listed alphabetically by author and cross-referenced under the following sub-basin headings: Yukon River mainstem, White River, Stewart River, Pelly River and Teslin River. The Aishihik River Basin was also referenced as there are existing proposals to divert water bodies of the Yukon system into the Aishihik system. Information collected for each sub-basin has been summarized and evaluated according to its applicabality to the Yukon River Basin Study objectives. Data gaps are identified and some additional studies suggested for each sub-basin.

References which were not received in time to be annotated are listed alphabetically after the first list of references.

Alternative sources of fisheries information for the Yukon Basin such as other bibliographies and mapping projects are listed in a separate section.

Most of the abstracts in this bibliography were prepared by the authors of the articles. However, studies for which no abstracts were available were summarized by the authors of this report. In some cases, abstracts did exist but did not outline the important information from a fisheries point of view. In these cases, the studies were summarized by the authors of the bibliography.

The authors apologize for any errors or ommissions that may have occurred and would appreciate receiving comments and other relevant Yukon River Basin information and reports.

ANNOTATED YUKON BASIN TITLES

#1
Acres Consulting Services in association with IEC International Environmental Consultants Ltd. 1982. McInytre Creek Power Plant No. 3 - feasability study. Final report prepared for the Yukon Electrical Company Ltd. January, 1982. 77 pages.

Abstract

Yukon Hydro is proposing a hydro-electric development which would include a forebay formed by an earthfilled dam located between the Whitehorse pumping station and the Alaska Highway on McIntyre Creek. Water would be conveyed via a penstock to the powerhouse located below the junction of Range and Porter Creek roads. The powerhouse would be located on the valley slope of the Yukon River floodplain. A tailrace channel would convey water from the powerhouse to the main channel of the Yukon River.

This report summarizes the engineering and environmental feasibility of the potential development, and includes the results of a habitat assessment and fisheries survey conducted between September 14th and 19th on McIntyre Creek.

Juvenile chinook salmon were found in the area below Range Road and above the estuary. The most abundant species in McIntyre Creek, other than sculpin, were rainbow trout. Trout were found in all reaches, except in reach 1 where juvenile salmon were found, and pool and riffle areas were thought to provide good trout spawning and nursery habitats. Grayling didn't appear to be abundant in the study area, and no juveniles were captured. Barriers in McIntyre Creek prevent upstream migration of fish.

The report concludes that fish habitat, including some salmon rearing habitat in lower McIntyre Creek, would be lost with the development. It is suggested that fish be stocked in and above the proposed headpond as this area will have a greater sports angling potential.

An appendix lists species, weight, age, length and location of fish captures.

#2

Acres Consulting Services Ltd. in association with IEC International Environmental Consultants Ltd. 1981. McIntyre Creek Power Plant No. 3 Prefeasibility Study Report prepared for the Yukon Electrical Company Limited. 42 pages.

Abstract

The Yukon Electrical Company has made a proposal to place a dam at one of three alternative dam sites on McIntyre Creek. The first and second alternatives would see a dam located adjacent to the city pumping station (Alternative 1) or slightly upstream (Alternative 2) with a penstock leading to a power plant located at the foot of the escarpment in the vicinity of the new Porter Creek access road cut along the escarpment. The third alternative (Alternative 3) is downstream from the city pumping station.

This study is a preliminary evaluation of Power Plant No. 3 and includes a short section on fisheries concerns.

Existing data indicates that McIntyre Creek supports a grayling population and provides some rearing habitat for Chinook salmon fry.

McIntyre Creek was surveyed on foot from the Alaska Highway to the Yukon River and suitable fish habitat was evaluated in terms of the rate of water flow and slope gradient of various sections of the creek. It was concluded that the dam will result in the loss of some suitable grayling and rainbow trout habitat but the head pond may provide some replacement habitat.

The Department of Fisheries and Oceans plans to construct a salmon hatchery in the vicinity of the pumphouse. While this will not neccessarily be incompatible with the ECL hydro proposal a number of key issues will require resolution. These issues are summarized in the report.

#3
Alaska Highway Pipeline Panel 1979. Initial impact assessment: Dempster Corridor. 300 pages.

Abstract

This report deals with the impacts on the biological, physical and human environments which would result from the construction and operation of the proposed Dempster Highway gas pipeline. The proposed pipeline will be at least 30 inches in diameter and will require initially 4 compressor stations. The construction work force will approximate 3000 people. The permanent work force will consist of 100 people. The environmental evaluation has been approached by examining three major components of the physical environment; land, water, and air; four major components of the biological environment; mammals, birds, fish and vegetation; and a heritage and land-use component of the human environment. Each component has been reviewed in terms of the changes it will undergo as a result of pipeline construction and highway use. In addition resource management problems and prospects are reviewed. The report stresses the need to view the pipeline development as just one part of an ongoing development process for the Mackenzie Delta. The effect of the pipeline must also be considered in respect to the already constructed Dempster highway which has on its own, a potential impact on the porcupine caribou herd. The single most important action required is to establish immediately a regional development planning and impact management process for the entire area that would permit anticipatory response or direction to events before the event occurs. Appendices to the report discuss the following topics: 1. the porcupine caribou herd and the impact of the Dempster corridor, 2. public involvement in managing the impact of the pipeline, 3. the impact of the Dempster corridor on the Mackenzie Delta, and 4. socio-economic data of the Delta and Dempster segments.

#4
Archibald, P. L. and B. E. Burns. 1981. A baseline survey of the water quality and biological conditions in the streams of the Howard's Pass area, Yukon Territory. Canada, Department of the Environment, E.P.S. Manuscript Report 81-13. 63 pages.

Abstract

A pre-development survey of water quality, sediment composition and biological conditions in the streams of the Howard's Pass area was conducted in July, 1980. The purpose of the survey was to obtain baseline information on the environmental quality of streams in the area around a large lead/zinc ore body which is expected to be developed as a mine. The study area incorporated three contiguous blocks of mineral claims belonging to Placer Developments Limited, and the streams draining them. Fish were sampled at three of the seven sampling stations using the angling method. Station 1 was located on an unnamed stream draining into the Pelly River, station 4 was situated on an unnamed stream just above the confluence with Don Creek and station 7 was located on Don Creek. The objective of the fish sampling was to obtain fish samples for metals analysis, rather than to assess the abundance and diversity of fish. However, the fact that grayling were collected so easily indicates that the area, particularly near station 4, may be a very productive grayling area.

The benthic macroinvertebrates and fish sampling results were representative of generally clear, unpolluted mountain streams. The high zinc concentrations in the stream sediments sampled reflected the presence of zinc ore deposits in the area.

#5

Arthur, J. R., L. Margolis and H. P. Arai, 1976. Parasites of fishes of Aishihik and Stevens lakes, Yukon Territory, and potential consequences of their interlake transfer through a proposed water diversion for hydroelectrical purposes. J. Fish, Res. Board Can. 33:2489-2499.

Abstract

Forty-six species of parasites were recovered from 383 specimens of 8 species of fishes collected from Aishihik and Stevens lakes during the summer of 1973. Substantial differences in the parasite fauna of Aishihik and Stevens lakes are noted; the possible consequences of the introduction of species of potential economic and pathogenic importance into new areas as a result of the proposed diversion of Stevens Lake (Yukon River system) into Aishihik Lake (Alsek River system) are discussed. Factors that must be given attention include preventing the transfer of infected fishes, infected intermediate hosts, free-swimming larval stages, and contaminated host waste products.

#6
Baker, S. 1980. Environmental concerns and recommendations: Alaska Highway reconstruction and maintenance, Kilometre 1008-1635. Environmental Impact and Assessment Report. EPS 8-PR-80-1. 94 pages.

Abstract

Environmental concerns associated with Alaska Highway reconstruction and maintenance practices in the Yukon and practices which might be useful in resolving concerns are identified in this report.

Although this information is applicable to Yukon highways in general, the focus of this report is on:

(i) Alaska Highway widening and paving between Watson Lake and Haines Junction,

(ii) Relocation of certain sections of Alaska Highway between Watson Lake and Haines Junction,

(iii) Alaska Highway maintenance pactices between Watson Lake and Haines Junction.

Recommendations for specific sites on this stretch of highway were made during the study and may be obtained from the Environmental Protection Service, Whitehorse, along with brief site descriptions which complement the more general information contained in this report. No new fisheries data is included.

#7
Baker, S. A. 1980. In situ bioassay study at a gold placer mining area in Yukon using Arctic grayling. In: Weagle, K. 1980, Report on the technical workshop on suspended solids and the aquatic environment. Department of Indian Affairs and Northern Development. 10 + pages.

Abstract

The study was conducted primarily as an experiment with the technique of in-situ bioassays. Eight stations were selected on placer mined creeks in the Dawson area, including Hunker Creek, Bonanza Creek, and Klondike River. Bioassay cages were set up for the time period of September 23, 1979 to September 19, 1979 and monitored, as were various water quality parameters including dissolved oxygen, suspended sediments and metals.

The results, although by no means conclusive, indicated an inverse correlation between elevated suspended sediment concentration, fish survival and dissolved oxygen concentration.

#8

Baker, S. A. 1979. An environmental survey of selected Yukon river headwater lakes impacted by communities. Canada Department of Environment. EPS Regional Program Report No 79-26. 75 pages.

Abstract

Water chemistry data and bottom fauna samples were collected from six Yukon lakes in the summer of 1977. The lakes sampled were; Tagish Lake on August 11 and September 26, at five sampling stations, Bennett Lake and Nares Lake on July 4 and September 9, at eight stations, Teslin Lake on July 2 and September 7, at eight stations, Marsh Lake on August 8 and September 13 at seven stations, Lake Laberge on July 3 and August 29 at eight stations, Kluane Lake at Burwash Landing on July 7 at eight stations, and Kluane Lake at Destruction Bay on July 9, at eight stations.

Water samples were analyzed for bacteria, nutrients, metals, mercury and arsenic content. Invertebrate data was collected and diversity and evenness values were calculated. High dissolved oxygen contents and the absence of any significant concentrations of deleterious substances characterized the waters of these oligotrophic lakes. All exhibited low concentrations of metals with only a few exceptions, probably attributable to natural mineralization. The more populated areas such as the Carcross area (Nares Lake), Marsh Lake cottage areas, Lake Laberge (Deep Creek), and Kluane Lake near Burwash Landing all had countable coliform results. Most invertebrate communities shared the characteristics of low diversity (relative to southern waters) and moderate evenness. The benthic community composition described in this survey was found to be similar to results found by Withler (1956) on Atlin and Tagish Lakes, Clemens et al (1968) on Teslin Lake and Kussat (1972) on Aishihik Lake. The biological data may be considered to be typical of an unpolluted northern freshwater system. Invertebrate species presence and water chemistry data are summarized in tables. This report contains no fisheries data.

#9
Baker, S. A. 1979. A survey of environmental mercury concentrations in Yukon Territory. Canada Department of Environment, E.P.S. Regional Program Report No. 79-21. 74 pages.

Abstract

The Yukon mercury survey was set up to establish a data base of mercury concentrations throughout the Territory. Samples of fish muscle tissue, sediment and water were obtained from thirty eight water bodies around the Yukon. The fish species found included arctic grayling, lake trout, northern pike, lake whitefish round whitefish, burbot, longnose sucker, coho salmon, sockeye salmon, chum salmon, chinook salmon and lake cisco.

Several moose tissue and waterfowl tissue samples were also analyzed for mercury concentration. The general finding of the survey was that the Yukon is problem free in terms of mercury contamination, although there are a few areas with naturally elevated concentrations. Areas of possible concern were labelled as those where sediment concentrations were greater than or equal to 500 ppb Hg and/or fish tissue mean concentrations were greater than or equal to .30 ppb. The areas identified by these criteria were Fox Lake, Francis Lake, Lake Laberge, Little Atlin Lake, Mayo Lake, Porcupine River, Quiet Lake, Simpson Lake, Teslin Lake and, possibly Atlin Lake. Each of these lakes supports either sport fishing, commercial fishing or native fishing. A discussion of the interpretation of the impact of mercury on fish, and the interpretation of the results of mercury on fish as documented in previous studies, is included. There is no evidence of anthropogenic mercury pollution in Yukon.

#10

Baker, S. A. 1977. Environmental quality of Rose Creek as affected by Cyprus Anvil Mining Corporation Ltd., (Survey data from 1974, 1975 and 1976). Canada, Department of Environment, E.P.S. Regional Program Report No. 79-25. 138 pages.

Abstract

Water quality, benthic invertebrates and fish surveys were conducted within the Rose Creek watershed during 1974 through 1976. The ten sampling sites were located upstream and downstream of the mine on Rose Creek and on Anvil and Pelly Creeks. Fish were collected using an electrofisher, with a barrier net downstream. The fish obtained were counted and identified and the dorsal tissue and liver tissue were sampled for heavy metal concetrations. Very few fish were collected in Rose Creek, probably due to the effect of the continuous tailings pond discharge into the creek. The low number of fish in 1974 was probably due to a high silt load from Faro Creek prior to the completion of the diversion channels. Stations on Anvil Creek and Pelly River located at some distance from the mine produced the greatest number of fish and the most species. The fish species collected at all sampling sites from 1974 to 1976 included 30 arctic grayling, 48 slimy sculpin, 19 round whitefish, 2 humpback whitefish, 7 longnose suckers, 4 burbot and 11 chinook salmon.

Results of the study indicated that the water quality of Rose Creek has been adversely affected on many occasions by the activity of the mine. In some cases the effluent concentrations were within acceptable limits.

#11
Beak Consultants Ltd. 1980. An initial environmental assessment of the Jason property, MacMillan Pass area, Yukon Territory. Prep. for Pan Ocean Oil Limited. 66 pages.

Abstract

The Jason property is situated in the vicinity of two major drainage systems, the South MacMillan and Hess Rivers. A preliminary environmental assessment is presented, based primarily on the literature and on a one-day visit on July 8, 1980.

Water quality data collected in 1976, 1977 and 1978 from the South MacMillan River and its immediate tributaries indicates that, in general, water chemistry does not appear to be limiting to fish life. Examination of substrates at several MacMillan sites as well as on Sekie Creek indicated the presence of stoneflies, mayflies and Dipteran larvae, all of which may serve as food for resident fishes. Fisheries and Marine Service information indicates that Arctic grayling, round whitefish, slimy sculpin and lake trout are present in the South MacMillan system. Beak observed the South MacMillan from below to above the property perimeter by helicopter but no fish were observed. There is potential for northern pike and chum salmon to be found in the South MacMillan as they have been collected in the MacMillan River proper. Sekie Creek, near the base camp exhibited good fish habitat up to the Canol Road, above which gradients increased markedly and substrates consisted primarily of large boulders. A tributary to the Hess River draining northwest from Jason Property consisted of a narrow, relatively steep gradient channel, flowing over primarily large cobble/boulder substrates. Given the elevation and configuration of this tributary, it is doubtful that fish would be present in this river near the property.

#12
Beak Consultants Ltd. 1980. Summary of fisheries investigations of new crossing locations, Alaska Highway Gas Pipeline, Yukon Territory, 1979. Prepared for Foothills Pipe Lines (Yukon) Ltd. 92 + pages.

Abstract

Recent changes in the prime route alignment of the Foothills Pipe Line have resulted in new waterbody crossings at both the eastern and western extremities of the pipeline route in the Yukon Territory. Beak Consultants Limited were retained to conduct fisheries investigations, during the 1979 open-water season, of the new watercrossing locations. The quality of the habitat available for utilization by fish was determined at each watercrossing investigated. The fishery sampling effort expended on each stream was determined by the number and proximity of previous crossings of the same stream, and the quality of habitat and fish fauna present. the species considered important in this area of the Yukon Territory are chinook salmon, chum salmon, Arctic grayling, lake trout, lake whitefish, and Dolly Varden char. Northern pike and burbot were also included in discussions of the fish fauna of the streams. In general, streams not previously crossed exhibited limited utilization by fish.

#13
Beak Consultants Ltd. 1979. Fisheries investigations along the Dawson Trail segment of the Dempster Lateral gas pipeline route, Yukon Territory, 1979. Report prepared for Foothills Pipe Lines (Yukon) Ltd., Calgary, Alberta. 32 + pages.

Abstract

Between KP 1100 and KP1200 of the proposed Dempster Lateral gas pipeline route, the alignment would proceed through valleys drained by the Klusha and Thirty-seven Mile Creek systems. Klusha Creek would be crossed six times and Thirty-seven Mile Creek crossed once by the proposed route. Both watercourses would be paralleled for several kilometres and some of the lakes associated with these drainages would be bypassed within several hundred metres. Surveys of proposed crossing sites, selected areas of Klusha and Thirty-seven Mile Creeks and representative lakes were conducted in May and September, 1979, in order to assess potential impacts of the proposed development on the regional fisheries resource.

Klusha Creek provides excellent rearing and summer habitat for Arctic grayling and burbot. Data suggest that Arctic grayling spawn near the proposed crossing site at KP1124.2, but suitable spawning substrate was not observed at any site farther upstream on this system. Lakes provide year-round habitat for several species, most noticeably lake trout, lake whitefish, Arctic grayling, northern pike and burbot. Pothole lakes not connected to the creek appear to be void of fish. The proposed crossing site on Thirty-seven Mile Creek does not provide suitable habitat for use by fish species. The lower reaches of this watercourse provide rearing and summer habitat for Arctic grayling. Lakes within this system are utilized by lake trout, lake whitefish, arctic grayling and burbot.

#14
Beak Consultants Limited 1979. Fishery resource investigations of waterbodies within the influence of the Alaska Highway gas pipeline. Alternative alignments, 1978. 2 Volumes. Report prepared for Foothills Pipe Lines (South Yukon) Ltd. Calgary, Alberta. 267 + pages.

Abstract

This report presents the results of fisheries surveys conducted in waterbodies within the influence of alternative pipeline routes.

Alternative alignments in the Kluane Lake, Whitehorse and Mount Michie Squanga areas were examined in the spring (May 15 - June 7), summer (August 3 - 25) and autumn (September 22 - October 7, and October 24, - November 6). Additional alternative alignments in the Kluane Lake, Whitehorse regions and Pickhandle Lakes area were examined only in the autumn.

The purpose of the study was to collect data on species distribution and habitat utilization.

Results are presented and maps of the alternative alignments are included.

Volume 2 contains data summary sheets for waterbodies examined during the studies.

#15
Beak Consultants Ltd. 1979. Surveillance of selected watercourse crossings along the Alaska Highway gas pipeline route in Yukon Territory, Year 1 (1978). Prepared for Foothills Pipe Lines (South Yukon) Ltd., Calgary. 57 + pages.

Abstract

A field survey of six watercourses along the Alaska Highway gas pipeline was carried out to collect data on water quality, benthic macroinvertebrates and fish species. The six watercourses studied were Beaver Creek (KM 23.8), Marshall Creek (KM 309.8), M'Clintock River (KM 472.8), Koidern River (KM 101.4), Morley River (KM 630.1), Rancheria River (KM 730.6).

Fisheries information on species presence, abundance and habitat utilization was collected during weekly visits to the watercourses between May 1 - June 30, 1978 and bi-weekly investigations between July 1 - September 30, 1978.

#16
Beak Consultants Ltd. 1977. A preliminary inventory of fish resouces in southern Yukon Territory, 1976. Prepared for Foothills Pipe Lines (South Yukon) Ltd. 13 pages.

Abstract

Foothills have proposed the development of a natural gas pipeline that would traverse the southern section of the Yukon Territory. The construction, maintenance and operation of this pipeline has the potential for serious impairment of aquatic environments. Two field surveys were conducted by BEAK from 22 September to 1 October, and 22 - 27 October, 1976. The purpose of these surveys was to provide additional information on the fish resources of selected aquatic systems in southern Yukon Territory, with emphasis on the identification and enumeration of fall spawning habitat and its utilization. Although all fall spawning fish species were considered in this study, of particular concern was the determination of whether chinook and chum salmon spawning areas could be impacted by the proposed pipeline. Rivers surveyed were restricted to those that were known or suspected to support fall spawning activity. Methods employed to locate fall spawning habitat included the use of seine nets, electrofisher, angling, aerial reconnaissance and the location of redds at the proposed pipeline river crossing and downstream locations. Numbers of fish collected in this study were low, but included a total of eight species from the 24 aquatic systems sampled. Arctic grayling were the dominant species collected, found typically in most streams characterized by gravel bed material. No fall spawning habitat was located in the vicinity of the proposed pipeline river crossings except for chum salmon. Chum salmon were found in the Kluane River, a river not crossed by the proposed pipeline, and one chum carcass was located in the Duke River, a short distance down-stream of the proposed pipeline crossing. No chinook or chum salmon redds were found. In 1976, chinook and chum salmon returned in abnormally low numbers. This would result in an underutilization of available spawning habitat. Although the possibility exists that little fall spawning habitat is located within the zone of influence of Foothills' proposed pipeline, it would be premature at this early stage in the study program to arrive at this conclusion confidently. This study represents the initial phase in a comprehensive study program of the aquatic resources. Results of studies scheduled in spring 1977 to collect emerging fry of fall spawners will provide additional data on fall spawning use of aquatic systems in southern Yukon Territory.

#17
Beak Consultants Ltd., 1977. A survey of fall spawning fish species in waterbodies within the influence of the proposed Alaska Highway Pipeline in Yukon Territory, 1977. Prepared for Foothills Pipe Lines (Yukon) Ltd. 40 pages.

Abstract

Foothills Pipe Lines (Yukon) Ltd. have proposed the development of a natural gas pipeline that would traverse the southern section of Yukon Territory. The construction, maintenance and operation of this pipeline has the potential for impairment of aquatic environments. An inventory of the fish fauna in selected watercourses within the influence of this proposed pipeline was conducted from August 20 to November 10, 1977. The purpose of this investigation was to document fishery utilization of the aquatic environment during the fall season. This program was designed to identify waterbodies which were used by fall spawning fish populations, and to identify migration times, routes and spawning areas as they relate to possible impact from the proposed pipeline. Potential fish overwintering areas in watercourses directly affected by the proposed pipeline alignment were also identified. These data will be used to supplement existing information on utilization and sensitive periods of fish fauna and to assess fully the impact of the proposed pipeline on the aquatic resource. Chinook and chum salmon were collected or observed in many waterbodies during this investigation. A summary of salmon utilization in regions within the influence of the proposed pipeline is presented. In the southern section of Yukon Territory, chinook salmon began arriving in spawning areas near the middle of August. Spawning activity peaked near the end of August in 1977. Chum salmon were first detected near spawning sites in southern Yukon Territory on September 27, 1977. Numbers of fish in these areas increased on subsequent days and spawning activity appeared to peak about October 22, 1977. In Bear Creek, Dolly Varden spawning activity began in early September when water temperatures were approximately 5.0 C. Areas which possessed suitable spawning substrate were found throughout the creek. Most spawning areas were small and no more than three Dolly Varden were observed in any of these regions. Many fish appeared to have migrated out of the region of this watercourse above the Alaska Highway. These fish may have moved to other spawning sites. In Kluane Lake, lake trout spawning appeared to have occurred from mid-September to late October. No spawning areas were identified during this investigation.

#18
Beak Consultants Ltd., 1977. Winter fish investigations of selected watercourses in Yukon Territory, 1977. Prepared for Foothills Pipe Lines (Yukon) Ltd. 13 + pages.

Abstract

Watercourses along the proposed pipeline route which will be crossed during winter construction were investigated from March 16 to March 20, 1977. A total of seventeen watercourses were investigated. Of these seventeen, six watercourses were judged to exhibit overwintering potential for fish fauna. These rivers will require further investigation during the winter of 1977 - 1978 to determine the fish fauna present during the winter season, the physical location of overwintering areas, if any, and the relative abundance of the fauna present. The majority of watercourses investigated in this survey did not exhibit overwintering potential for fish fauna. Of the seventeen rivers and streams surveyed, eleven were of this category with winter pipeline construction preferred. Five rivers (White, Koidern, Donjek, Duke and Kluane Rivers) and one creek (Snag Creek) were judged capable of supporting overwintering populations of fish, and additionally, appear suitable for overwinter egg incubation of fall spawning fish species. Further study is needed of these six watercourses to determine if actual fish overwintering occurs, and to assess the implications of winter pipeline construction. The data collected from this survey reduces the number of streams of potential concern to winter pipeline construction.

#19

Bodaly, R. A. 1979. Morphological and ecological divergence within the lake whitefish (Coregonus clupeaformis) species complex in Yukon Territory. J. Fish. Res. Board Can. 36:1214-1222.

Abstract

Eighty-nine lakes in the Yukon Territory and adjacent areas in northern B.C., in the drainage basins of the Yukon, Alsek, Upper Liard and Peel rivers were fished by gillnet between 1968 and 1975. Two forms of lake whitefish were found in each of 5 lakes in Yukon Territory: Squanga, Little Teslin, Teenah, Dezadeash and Hanson lakes. They are characterized by differences in gill raker counts. The naturally occurring fish fauna of Hanson Lake was exterminated in 1963 for the purpose of planting rainbow trout. Only northern pike are found in the lake at present.

#20
Bodaly, R.A. and C.C. Lindsey. 1977. Pleistocene watershed exchanges and the fish fauna of the Peel river basin, Yukon Territory. J. Fish. Res. Board Can. 34 : 388-395.

Abstract

The Peel River basin is a unique Canadian glacial refugium containing many relict fish populations. Peel River is presently tributary to the Mackenzie River system, but at least twice during Pleistocene glaciations it was diverted into headwaters of the Yukon River system, offering the possibility of two-way transfer of aquatic organisms between the Mackenzie and Yukon. Present fish distributions in the Peel basin are summarized. Biochemical and morphological evidence suggests that races of at least six species now inhabiting the area (Coregonus clupeaformis, Prospium coulteri, Thymallus arcticus, Salvelinus namaycush, Esox lucius, Cottus cognatus) originated from types which either came from the Yukon River system or developed in situ in unglaciated parts of the Peel.

#21
Boland, J. P. 1973. The Yukon fishery resource. Its existing role and its future potential. Economics Unit, Northern Operations Branch, Fish. Mar. Ser., Pacific Region, D.O.E. NOB/Econ. 1-73, 43 pages.

Abstract

This study presents an overview of the fishery resource activities of people living in the Yukon. It includes a section on each of the three main fishing activities: commercial fishing, sport fishing, and subsistence food fishing. It specifically focuses on the influence these activities have on the residents of the Yukon.

The sport fishery makes a considerable contribution to the Yukon economy with a significant number of residents receiving a portion of their income as a result of this fishery.

At present the Yukon's commercial fishery is very small but opportunities for future growth do exist. The main method used for catching salmon are fish wheels.

A significant number of the Yukon's native Indian people make use of their right to subsistence fish. The report suggests that Indians living in more remote locations are more dependent on fishing than Indians living in the more populated areas.

#22
Boughton, R.V. 1966. The annual production of the whitefishes *Coregonus clupeaformis* and *Coregonus nasus* in Teslin Lake, B.C. Northwest Sci. 40: 147-154.

Abstract

Teslin Lake supports eleven species of fish. Of these *Coregonus clupeaformis*, the lake whitefish, *C. nasus*, the broad whitefish, *Prosopium cylindraceum*, the round whitefish, and *Catostomus catostomus*, the longnose sucker, are bottom feeders. The calculation of the annual yield, however, has been restricted to that for *Coregonus clupeaformis* and *C. nasus* for two reasons: first, because they are of commercial importance owing to their sizes and numbers, and second, because they feed on bottom organisms for which quantitative data have been secured. Several assumptions have beem made in the calculations. Reserving 10,000 to 15,000 for the local population, it would appear that 40,000 to 50,000 pounds might be taken from the lake annually in a commercial fishery. If Teslin Lake is ever opened up for commercial fishery, the authors suggest that it be kept under direct scientific control.

#23
Brock, D. N. 1976. Distribution and abundance of chinook and chum salmon in the upper Yukon River system in 1974, as determined by a tagging program. Fish. Mar. Ser. Tech. Rep. PAC/T-76-3. 56 pages.

Abstract

During the 1974 Yukon River tagging program a total of 727 chinook salmon (Oncorhynchus tshawytscha) and 1,276 chum salmon (Oncorhynchus keta) were tagged with Peterson disc and spaghetti tags 48.3 km. below Dawson City. The tag recovery program showed a greater incidence of Peterson disc-tagged salmon than spaghetti-tagged salmon. The program indicated that a large portion of the chinook and chum salmon populations are mainstem spawners. Mainstem chinook salmon were found spawning at the Ingersol Islands, Yukon Crossing, and the outlet of Lake Laberge while chum salmon were found spawning in a clear water side channel of the Yukon River adjacent to kilometer 212 of the Klondike Highway. Population estimates for chinook range from 11,000 to 36,700 and for chum salmon from 8,960 to 31,352. Information on length, sex, age composition, gear selectivity and migration rates for both salmon species is included in this report.

#24
Brown, R. F., M. S. Elson, and L. W. Steingenberger, 1976. Catalogue of aquatic resources of the upper Yukon River drainage (Whitehorse area). D.O.E. Fish. Mar. Ser. Tech. Rep. PAC/T-76-4. 149 pages.

Abstract

Eighteen species of fish have been recorded from the upper Yukon River drainage, of which 3 species have been introduced and one species is probably no longer present. Spawning periods for these species in the upper Yukon River drainage are given. Physical, chemical and biological information is given for each of 7 major tributary systems: the Atlin Lake, Tagish Lake, Bennett Lake, Marsh Lake, Takhini River, Lake Laberge and Yukon River drainages. Exploitation of the fish stocks is discussed. A description of dams present in the system and their effect on fishery stocks is given. A summary of fish stocking programs in the drainage is presented.

#25
Buckles, D. 1982. (In preparation). Allocating fisheries resources for fly-in sports fishing camps. Information and Policy Branch. Department of Renewable Resources. Y.T.G.

Abstract

This report summarizes the history of allocations decisions with regards to the fishery resource. The author argues that before fisheries resources can be allocated determinations must be made of (1) what the resource can yield and (2) which user groups have priority.

NOTE: This information was gathered in a telephone conversation with the editor. The report has not been seen.

#26

Burns, B. 1981. Methods, environmental impacts, treatment and control, reclamation and revegetation and management of placer mining in Yukon Territory. Draft. Prepared for Westwater Institute, University of British Columbia.

Abstract

This brief report summarizes information on placer mining in the north. Five areas were explored (1) methods of placer mining, (2) environmental impacts, (3) treatment and control, (4) reclamation and vegetation, (5) management.

A section on aquatic life outlines the direct and indirect effects of placer mining on the fisheries resource. No new information is presented.

#27
Canada, Department of the Environment. 1979. Water chemistry and biological conditions in the watershed near Mount Nansen mines Ltd. Yukon Territory. E.P.S. Regional Program Report: 79-12, 16 pages.

Abstract

Mount Nansen mines are located in the Dawson range about 30 miles west of Carmacks. The area surrounding the mine is drained by the headwaters of the Nisling River. Effluent from the tailings pond flows into an unnamed creek which travels about 5 kilometers before draining into Victoria Creek.

The water chemistry and biological conditions (benthic invertebrates, fish) in the Victoria Creek watershed near Mount Nansen Mines were examined in 1976 and 1977. Fish were collected from two sites on the Victoria Creek using an electrofisher, to determine which species were present and to analyze fish tissue for heavy metals. Twenty-four grayling were captured at station 4 and thirty-five grayling were captured at station 5. The results of fish sampling for heavy metals did not indicate any metal levels of concern. There was a marked improvement in the quality of tailings decant from 1976 to 1977 as well as in te toxicity of the decant. The creek receiving the decant had a much reduced bottom fauna population but the bottom fauna and fish population in Victoria Creek showed little impact from the decant.

#28

Canada, Department of the Environment. 1978, Assessment of the water quality and biological conditions in watersheds surrounding the United Keno Hill Mine, Elsa, Yukon, during the summers of 1974 and 1975. Canada, Department of the Environment. E.P.S. Regional Program Report: 78-14. 26 pages.

Abstract

Surveys were conducted during the early summer of 1974 and 1975 at United Keno Hill Mines in the Yukon Territory. The parameters examined included water chemistry, trace metals, benthic species diversity, fish concentrations and toxicity testing. Sampling sites were located in two watersheds. The main watershed under consideration was associated with the South McQuesten River and received the decant water from the tailings pond below the mine's milling operation. The other watershed examined briefly was the Lightning Creek/Thunder Gulch watershed where a small placer gold mining operation was located. Electrofishing with barrier nets downstream was used to capture fish.

The greatest number of fish were collected along the South McQuesten River, while only one pike was caught in Flat Creek. One grayling was caught on Lightning Creek, downstream from the placer mining operation.

The results indicated that zinc concentrations as high as 2.0 ppm were present in the waters of Flat Creek below the tailings pond decant. Copper, cadmium, and lead concentrations were found to to be above the limits of the water use licence, intermittently. Benthic species diversity and fish density were reduced in Flat Creek below the tailings pond decant. Bioassay samples collected from the tailings pond decant were found to be toxic on two occasions out of twelve sampling times.

#29
Canada, Department of the Environment, 1976. Environmental Design for Northern Road Developments Report. EPS-8-EC76-3.

Abstract

This report recommends environmental protection measures to minimize adverse effects on the environment from northern road developments. The measures presented are based on experience gained in the environmental review of the MacKenzie Highway and proposed MacKenzie Valley Pipeline projects and on the advice of environmental and engineering experts.

The road development process is presented in six sequential phases: preliminary route selection, the environment survey, the alignment selection and route field survey, design, construction and finally, maintenance. In the initial phases of development the participation of those responsible for environmental protection is emphasized. In the latter phases specific measures for environmental protection are given for individual highway activities.

Includes guidelines for necessary fisheries studies, watercourse selection, stream crossing design, construction of drainage and erosion, stream crossings, drainage and erosional control maintenance, stream crossings maintenance.

Appendix B includes erosion and drainage control practices. Appendix C includes guidelines for fish passage through culverts.

#30
Canada, Department of Environment, 1973. Catalogue of fish and stream resources of the Teslin Watershed. Tech. Rep., Fish. and Mar. Ser. PAC/T-73-13, 47 pages.

Abstract

Fourteen species are represented in the fish stocks of the Teslin Watershed. These are: humpback whitefish, broad whitefish, lake herring, round whitefish, inconnu, grayling, lake trout, chinook salmon, chum salmon, longnose sucker, northern pike, burbot, slimy sculpin and Arctic lamprey.

Physical, chemical and biological characteristics are presented for each of the 8 major tributary systems: Gladys River, Hayes River, Jennings River, Morley River, Nisutlin River, Swift River, Teslin River and Upper Teslin River. Detail on life history and abundance is known to a limited degree for chinook salmon only. The total run of chinook in the Teslin system is estimated to be 3,700 - 8,600 fish annually. Relatively good numbers spawn in the Teslin River (2,000 - 5,000) and others pass through Teslin Lake to spawn in the drainages of the Nisutlin River (1,000 - 2,000), Morley River (200 - 500), Swift River (100 - 300), Upper Teslin river (100 - 200), Gladys River (100 - 200), Hayes River (100 - 200) and Jennings River (100 - 200). Unconfirmed sightings of chinook salmon have been reported in a number of smaller tributary streams. Chum salmon allegedly spawn in a slough upstream of Roaring Bull Rapids but their numbers are unknown.

#31
Canada, Department of the Environment, 1973. Statement of Fisheries Service concerns and responsibilities presented to the Yukon Territorial Water Board Inquiry into the placer mining industry. Canada, Department of the Environment, Fish. Mar. Ser., November, 1973. 19 pages.

Abstract

This brief outlines the main implications that placer mining activities have on the indigenous fish stocks of the Yukon gold fields. Placer mining activities as they are presently conducted would destroy fish production in the assiciated streams as illustrated by the examples of Bonanza and Hunker Creeks. Spawning and fish food producing substrates can be blanketed or buried with silt so that the streams are no longer capable of producing either fish or fish food. Available technology to overcome the detrimental effects on fish associated with placer mining is presented. The water use situations which the Fisheries and Marine Service may control under the Fisheries Act are given. Recommendations concerning the control of placer mining activities in the Yukon as they relate to fish and fish habitat protection are included.

#32
Canada, Department of Environment, 1972. Fish power studies northern B.C. and Yukon. Preliminary Report, April 1972. Energy Study Group, Environmental Quality Unit, Fish. Mar. Service. 31 pages.

Abstract

The report deals with 6 hydro power projects in B.C. and Yukon which are under investigation by the Fisheries Service: Alcan, Skeena, Stikine-Iskut, Aishihik, Whitehorse and Mayo. Fisheries problems posed by the developments are described and planned studies in each area are outlined. No new fisheries data is presented.

#33
Canada, Department of Environment and Department of Fisheries and Oceans, 1979. Submissions to the Yukon Public Hearings Panel. 150 pages.

Abstract

This brief to the Yukon Public Hearings Panel reviews the deficiencies to the Foothills Pipe Lines Ltd. Environmental Impact Statement on the question of fisheries. The DOE/DFO Task Force provides advice on methods of mitigating potential fisheries problems, particularly through construction scheduling. The need for fair and adequate compensation where fishery stocks and resource users are damaged is discussed.

Knowledge of the fisheries resources along the pipeline route is very limited. Insufficient information is provided on stream crossing scheduling, alternative route alignments, required aggregate volumes and borrow sites, parasite transfer during hydrostatic testing, and methanol disposal. Several contradictions on construction schedules and critical life stages of fish stocks are also evident throughout the E.I.S.

Appendices include Comments on Fisheries Related Information Included in the Environmental Impact Statement, Summary of Fish Species and Life Stages Present in Streams on the Proposed Pipeline, and Comments on Proposed Scheduling of Stream Crossings. Future requirements are outlined and there is a list of studies and reports related to the Alaska Highway Gas Pipeline.

#34

Canada, Department of Fisheries and the Environment. 1978. Recommended environmental practices for the proposed Alaska Highway Gas Pipeline. Interim. Environmental Protection Service. EPS-1-PR-77-1. 70 pages.

Abstract

This document represents consolidated advice from the Department of Fisheries and the Environment on measures necessary to maintain environmental quality during the construction, operation and abandonment of the proposed Alaska highway gas pipeline. It outlines general measures which would protect the environment from unnecessary disturbance, as well as specific constraints which would protect components of the environment during particularly sensitive periods and in particularly sensitive locations.

Contained herein are recommended environmental design practices covering selection of pipeline alignment and locations of ancillary facilities, pipeline construction and construction of ancillary facilities, field services, operation and maintenance, inspection and reporting requirements and clean-up and restoration of disturbed areas.

#35
Canada, Department of Fisheries and the Environment, 1977. Alaska Highway Pipeline - Interim report of the Environment Assessment Panel. 55 pages.

Abstract

In this report the Environmental Assessment Panel presents its conclusions for the Alaska Highway Pipeline with respect to physical and engineering concerns, biological concerns, unique and sensitive areas, and aesthetics and recreation. Associated developments and other possible pipeline routes are discussed. Numerous persons appearing before the Panel expressed their belief that insufficient information had been gathered on which to base the design and scheduling of water crossings. Similarly, the Panel was advised that there were a lack of baseline data on the locations, use and extent of spawning, rearing and overwintering areas at, and downstream, from proposed water crossings. Salmon, lake trout, whitefish and grayling were of particular concern. The Panel concluded for fisheries that with proper scheduling and construction techniques, further site specific data on fish, their habitat and their food chains, will be required. Proper management of fish populations could be achieved through the enforcement of appropriate regulatory controls.

#36
Canada, Department of Fisheries and Oceans, 1982. Commercial catch reports indigenous species 1980-82. File 5160-5, D.F.O. Whitehorse.

Abstract

Cards note lake, month and fish caught under commercial license during 1980 and 1981 (untabulated data).

#37
Canada, Department of Fisheries and Oceans, 1982. Domestic catch reports indigenous species 1980-82. File 5161-2, D.F.O. Whitehorse.

Abstract

Domestic catch cards record fish caught under domestic licence in Yukon waterbodies during 1980 and 1981 (untabulated data).

#38
Canada, Department of Fisheries and Oceans. 1982. Lake and Stream Files, D.F.O. Whitehorse, Yukon.

Abstract

The files contain information for many waterbodies within the Yukon. Fish species presence and sampling location are noted. Maps are sometimes included. Substrate information and data on general waterbody characteristics are sometimes noted.

#39
Canada, Department of Fisheries and Oceans. 1982. Reports - General and annual narrative, district 10. File 5871-BC1-10. D.F.O. Whitehorse.

Abstract

Annual narratives are included for the Northern B.C. - Yukon South sub-district. These reports contain information on commercial salmon catch statistics, number of commercial licences issued, fresh water catch data, Indian food fishery catches ad data on sports fishing activity. In addition the reports summarize environmental conditions in the area and note significant developments in the area (ie. logging, mining, highway construction) as well as listing enforcement actions. A final section deals with administration in the sub-district.

Another file containing annual statistics includes similar information for most years between 1958 and 1967.

#40
Canada, Department of Fisheries and Oceans, 1981. Domestic and commercial questionnaires. District 10, Files. D.F.O. Whitehorse.

Abstract

This file includes responses to questionnaires distributed to commercial and domestic fisheries in the Yukon Territory (untabulated data).

The domestic fishing questionnaires includes information on location, season and amount of time spent fishing, species caught, number of lakes fished, amount of fish needed to meet the fisherman's needs in the course of a year, observations on the status of fisheries in the waterbodies fished, fishermen's opinions on fish management etc.

The commercial fishing questionnaires include similar types of data and in addition provide information on expenditures and income related to fishing, species sold, and fish marketing.

#41
Canada, Department of Fisheries and Oceans, 1981. Public meeting to discuss commercial and domestic fishing in the Yukon. March 26, 1981. In file 5161-1. Domestic fishing general. D.F.O. Whitehorse.

Abstract

The history of the freshwater commercial and domestic fishery in the Yukon Territory was summarized. Significant harvests were taken by Klondike miners at the turn of the century. From 1925 to 1935 large quantities of fish were taken from Bennett, Atlin and Tagish Lakes. The history of fisheries administration and regulations, and the present management and operations are described Charts list whitefish and lake trout commercial and domestic catches from 1972 to 1980. Schedules of lakes open to commercial fishing in 1961 and those currently open to commercial fishing are listed.

#42
Canada, Department of Fisheries and Oceans. 1981. Salmon stream and spawning ground reports. File 5871-BC3-10 Northern British Columbia/Yukon Subdistrict, Yukon/Arctic Subdistrict, Alsek-Taku Subdistrict, D.F.O. Whitehorse.

Abstract

These files contain the results of salmon spawning surveys conducted on selected watercourses throughout the Yukon. Species and peak of spawning, physical condition of spawning grounds, and obstructions to fish passage where present, are identified.

#43
Canada, Department of Fisheries and Oceans. 1980. Survey of sportsfishing in the Yukon territory. Recreational Fisheries Branch, Ottawa.

Abstract

A computer print-out presents information derived from a 1978 sports fishing survey. Included are catch statistics for various districts in the Yukon, classification of fishermen, frequency of fishing activity, expenditure on sportsfishing by residents and non residents.

#44
Canada, Department of Fisheries and Oceans. 1978. Survey of sportfishing in the Yukon Territory in 1975. Recreational Fisheries Branch, Ottawa.

Abstract

This report presents the results of a survey conducted in 1975. Information was provided by the 1,056 anglers who returned questionnaires.

The results of the survey indicated that over 12,000 licensed anglers fished in the Yukon in 1975. They spent over 134,000 days fishing and caught an estimated 185,000 fish. Although over half of the anglers were non residents of the Yukon, they accounted for only 35 percent of days fished and 23 percent of fish caught. Angling was found to be an important economic activity in the Yukon. Anglers invested substantial amounts of money on items associated with sportsfishing (accommodation, travel, licences, fishing gear, boats).

The number and characteristics of anglers, place of residence, number, month and locations of fish caught, and annual investments are all detailed in the report.

#45
Canada, Department of Indian Affairs and Northern Development and the Government of the Yukon, 1979. Volume I - Review Framework and Resource Data - Carcross Valley - Mash/Tagish Lakes - Atlin Road Management Planning Project, Volume II - Resource Data Maps.

Abstract

The purpose of the report is to recommend land management principles and a land management plan that will meet the anticipated demands on Federal and Commissioners lands south of Whitehorse, while respecting the inherent limitations of the landscape. Land management proposals are made for the Carcross Valley, Marsh/Tagish Lakes and Atlin Road areas and the present resources and land uses are summarized including agriculture, forestry, recreations, heritage, fish and wildlife, and minerals.

Volume II presents a series of land use and potential land use maps for the area, including 2 maps that outline the fisheries and recreation resources for Carcross, Marsh/Tagish Lakes and Atlin, indicating camp sites and areas of commercial, sport and subsistence fishing.

The report includes a 2 page summary of fish species presence in the area noting sports, commercial, domestic and Indian fisheries. No new information is presented.

#46
Canada, Department of Public Works and U.S. Department of Transportation Federal Highway Administration 1978. Environmental Impact Statement Shakwak Highway Improvement Volume 1 and 2 Errata. 74 pages.

Abstract

This report includes corrections in the text of the Shakwak Project Environmental Impact Statement issued in December 1977. Corrections included in this errata are those that could affect the interpretation or ease of interpretation of that document's findings. Includes a section of fisheries information.

#47
Canada, Department of Public Works and U.S. Department of Transportation, 1978. Environmental Impact Statement Shakwak Highway Improvement, British Columbia and Yukon Canada; Statement of social, economic and environmental impacts incorporating issues presented at public hearings. 156 pages.

Abstract

A comparison document to the original EIS in which the project planners address the issues raised during the public review and summarize the pertinent information contained in the first two technical volumes of the EIS. Fisheries sections summarize fisheries resources, resource utilization, impacts on fishery resources and investigative measures. Fish resources may be temporarily affected by the siltation that accompanies instream construction. Increased sport fishing in the corridor may deplete the fisheries resource. Existing culverts that do not meet Northern Canadian guidelines for fish passage, will be replaced or improved.

#48

Canada, Department of Public Works and U.S. Department of Transportation, 1977. Shakwak Highway improvement British Columbia and Yukon Territory, Environmental Impact Statement. 2 volumes.

Abstract

The Shakwak Project consists of the paving and upgrading of the Haines Road from the Alaska/British Columbia border to Haines Junction, and the Alaska Highway from Haines Junction to the Yukon/Alaska border, a distance of approximately 520 kilometres.

Volume I of the Environmental Impact Statement includes chapters on: project description, summary of project purpose and need, and description of the social, economic and environmental context. Environmental context is divided into subsections on climate, terrain, hydrology, vegetation, social and economic considerations, land status, historic sites, recreation and tourism, aesthetics and fisheries resources. Data on fish species composition and habitat utilization is summarized from previous studies. In addition, a list of streams crossed with unsuitable culverts is presented in tabular form, and includes a description of the culvert problem and an indication of game fish species presence. Maps indicate fish species presence in waterbodies along the route of the proposed Shakwak Highway alignment.

Volume II includes chapters on (1) the relationship of the proposed project to land use plans, policies and controls, (2) environmental impacts, (3) alternatives to the project, (4) unavoidable adverse effects, (5) relationship between local short-term uses of man's environment and the maintenance and enhancement of long-term productivity, (6) irreversible and irretrievable commitments of resources, (7) other interests and considerations of national policy which are thought to offset the environmental effects of the proposed action.

A sub-section on fish resources outlines the general and site specific impacts and includes the preferred timing of instream construction, on the basis of periods critical to fish. A variety of methods for improving presently unacceptable culverts are suggested.

#49
Carey, L.A. 1960. Whitehorse Rapids power development. Eng. J. vol. 43(10) Oct. 1960. 4 pages

Abstract

This short technical report describes the structure and location of the Whitehorse Rapids power development. It includes descriptions of the hydrology of the system, the river diversion, the dam structure and the fish ladder.

#50
Carr, D.W. 1968. Hunting, trapping and fishing in the Yukon Territory. In: The Yukon economy - its potential for growth and continuity. Volume IV: reference studies on social services and resources industries. Prepared by Wm. Carr and Associates Ltd., Ottawa, Queens Printer. p. 175-202.

Abstract

This is a survey of the economic role of hunting, trapping and fishing in the Yukon. Includes an overview of the status of commercial, sports and domestic fisheries, including tables showing commercial licences issued, market value of produce, angling licences issued. The report concludes that commercial fisheries may be expected to contract while sport fishing, particularly by non-resident anglers, is expected to increase. Recommends (1) that priority be given to sports fishing and the related tourist industry, (2) that special provisions be made by the Department of Fisheries to meet the requirements of native fishermen, (3) that the Fisheries Research Board be requested to carry out a survey of the present and potential fish resources of the Yukon.

#51
Clemens, W. A., R. V. Boughton, and J. A. Rattenbury, 1968. A limnological study of Teslin Lake, Canada. B.C. Fish. Wildl. Br., Managem. Publ. (12).

Abstract

In July and August, 1944, a limnological study was made of Teslin Lake which lies across the British Columbia - Yukon boundary line. The lake is 78 miles long and 2 miles wide on the average. The greatest depth obtained was 702 feet. It is evidently a lake with limited productiveness because of a short growing season and a large volume of cold water in the hypolimnion; probably a great run off from an extensive watershed may be a contributing factor. The summer heat budget in 1944 was 20,000 cal. $cm.^{-2}$. Plankton and bottom organisms are limited in species and quantity. Thirteen species of fish occur in the lake of which the lake whitefish, Coregonus clupeaformis and the broad whitefish C. nasus might be used commercially in a small fishery of 40,000 to 50,000 pounds annually. The lake trout, Salvelinus namaycush is fairly abundant but its use should be restricted for sport fishing.

#52
Cleugh, T. 1978. Hydro-electric proposals of selected sites in Yukon Territory. D.F.O. Fisheries and Marine Service, Habitat Protection, Water Use Division. 75 pages.

Abstract

This report is intended to acquaint the Fisheries Service personnel with some of the proposed hydroelectric schemes in the Yukon Territory. The hydro projects discussed include (1) Yukon-Taku, (2) Bates River, (3) Primrose River, (4) Alsek River watershed, (5) Kusawa Lake, (6) Teslin River, (7) Pelly River, (8) Five Fingers, (9) Stewart River project, (10) Peel River watershed and Aberdeen Falls.

The summarized data presented for each system consists of the site location, project description and alternative plans and area hydrology. No fish or fish related data is included.

#53
Cleugh, T. R. and L. R. Russell, 1980. Radio tracking chinook salmon to determine migration delay at Whitehorse Rapids Dam. Fish. and Mar. Ser. Man. Rept. #1459: 43 pages.

Abstract

Migration pattern of radio-tagged adult chinook salmon (Oncorhynchus tshawytscha) were monitored during August, 1979 in the upper Yukon river, Yukon Territory. Escapement, migration pattern, travel time, spawning areas and delays caused by the hydroelectric facilities and physical obstructions were assessed.

Eleven hundred and eighty-four chinook salmon migrated through the Whitehorse Rapids Fishway in August, 1979 and twenty-one fish were radio tagged. Migrating salmon used both sides of the river and the hydro facilities and physical obstructions resulted in some delay in upstream migration. The minimum and maximum observed delay which occurred at the hydro facilities was 10 hours and 9 days respectively, with a mean delay of 3 days; average delay through the fishway was 7 hours. Fifteen tagged fish were located on the spawning grounds.

#54
Cleugh, T. R., W. J. Schouwenburg and L. R. Russell, 1978. Michie Creek watershed, 1976 - 1977 biological investigations and biophysical stream inventory. Canada, Department of Fisheries and Environment, Fish and Mar. Ser., Habitat Protection Division, Internal Report. 39 pages.

Abstract

In order to determine the presence of juvenile salmon, 2 gillnetting series were undertaken in March and July of 1976. The nets were set overnight at various depths. In addition to gillnets, wire minnow traps were fished at 3 shoreline locations in March and a beach seine was operated in June. Inclined plane live traps and minnow traps were used to estimate the abundance of migrating juvenile chinook salmon in Michie Creek in June and July. Two divers swam the creek between Michie Lake and M'Clintock River on August 23 - 24 and September 16 to make visual estimates of young salmon abundance. Adult salmon were estimated by aerial observation and the stream swimming. A biophysical stream inventory of the creek from Michie Lake downstream to the confluence with Byng Creek was prepared in order to provide an accurate estimate of the amount of spawning gravel and rearing habitat available to salmon in the Michie watershed. In 1977, the number and distribution of juvenile chinook salmon were estimated by stream swimming.

Out of a chinook escapement of 120 at Whitehorse Rapids fishway, only 21 spawners were observed in Michie Creek in 1976. The majority of chinook in Michie Creek spawned between August 15 and September 15 in 1976. The majority of immature chinook migrated out of the Michie system after August 23 in 1976. An estimated 77,228 sq. m. of spawning gravel in the main system could accommodate 7,700 adult chinook spawners based on 20-1 sq. m. of gravel required per spawning pair. Incubation and rearing potential for juvenile salmon is also excellent.

#55

Craig, P. and G. Clark, 1978. Winter and summer fisheries surveys for the Shakwak Highway Improvement Project (Public Works Canada) British Columbia and the Yukon Territory. Report submitted to Thurber Consultants Ltd. 64 pages.

Abstract

In response to concerns raised by the Environmental Impact Statement, fisheries inventories were conducted in twenty-two streams crossed by Highway Construction Segment one (Rainy Hollow area) and Segments seven to ten (near Haines Junction and Kluane Lake) of Haines Road/Alaska Highway. The streams were surveyed during late winter (12-18 April, 1978) and early summer (12-19 June, 1978) to assess their use by overwintering fish and spring-spawning species. A variety of sampling methods, including electrofishing, siene nets, gill nets and minnow traps were used to capture fish above and below highway crossings. Existing highway culverts were also evaluated using fisheries criteria.

The timing of the late winter survey was several weeks prior to spring breakup, although some streams, particularly those in the Haines Junction area, were in the initial stages of breakup. During this survey overwintering fish were caught in seven streams: Granite, Five Mile, Seltat, Schultz, Bear, Summit and Christmas Creeks. In almost all cases, the species present at this time of year was Dolly Varden.

The timing of the early summer survey corresponded with spring freshet conditions in the Rainy Hollow area and the end of the freshet period in the Haines Junction area. Under these conditions sampling for fish was often restricted to stream edges. Fish were captured in thirteen streams; Granite, Three Mile, Five Mile, Seltat, Schultz, Ames, Tina, Pine, Bear, Summit, Christmas and Topham Creeks and the Dezadeash River overflow channel #2. Of the three fish species caught at this time (Dolly Varden, grayling, slimy sculpin) only the grayling is a spring spawner. Grayling appear to use only Pine and Christmas creeks for spawning in the study area.

Finally, updated information is presented for culverts which do not meet Canadian fisheries guidelines and preferred seasons of construction for in-stream activities during the upgrading and paving of the highway are listed.

#56

Craig, P. and G. Glova, 1977. Catalogue of streams crossed by the Haines Road/Alaska Highway in British Columbia and the Yukon Territory. Prepared for Public Works Canada and U.S. Federal Highway Admin. LGL Ltd. Environmental Research Associates, Nanaimo, B.C. 80 pages.

Abstract

This catalogue was prepared in response to the proposed Shakwak Highway Improvement Project by Public Works Canada and the U.S. Federal Highway Administration. Previous fisheries surveys in the vicinity of the existing highway are summarized and new data are presented for 47 streams. Small streams, including many unnamed creeks, were emphasized for two reasons: (1) many had never been sampled, particularly along the Haines Road and (2) existing culverts on many streams do not meet environmental guidelines for fish passage. The fisheries data were gathered from August 26 - 31, 1977. Fish were sampled by means of electrofishing, seining or gillnetting.

#57
Davies, D. J. and C. D. Shepard, 1981. Preliminary fisheries report for North Canol Road. Prepared for Northern Roads and Airstrip Division D.I.A.N.D. 46 pages.

Abstract

This preliminary fisheries survey of the river systems along the north Canol Road was undertaken in the summer of 1981. The study area extended from the town of Ross River to the Yukon-Northwest Territories border. Objectives of the survey were to determine fish species composition and distribution; to examine stream utilization by various life stages of fish for spawning, rearing or migration; to evaluate existing road crossing structures in terms of their effect on fish movement and stream habitat; and to identify areas where additional studies may be required to complete a timetable for instream construction. Streams were sampled for fish by electroshocking, angling and swim surveys where applicable. The potential fisheries habitat was evaluated and given a relative rating of nil, low, moderate or high significance primarily based on the measurements or descriptions of % pools, overhanging vegetation, debris, flow character, substrate and invertebrates. Five species of fish were encountered in the study area. These were chinook salmon, Arctic grayling, northern pike, burbot, and slimy sculpin. A more comprehensive survey is required to establish the distribution of spawning areas in the systems examined.

#58
DeGraaf, D. A. 1981. First Canadian record of Bering Cisco (Coregonus laurettae) from the Yukon River at Dawson, Yukon Territory. Canadian Field Naturalist 95 (3): 365.

Abstract

On September 21, 1977, a specimen of Bering Cisco was captured by gillnet during fish inventory studies in the Yukon River at Dawson. This constitutes the first record of the species in Canadian waters and a range extension of 500 km. upstream in the Yukon River from previously recorded catches at Fort Yukon. This range extension resulted from impact-related studies funded by Foothills Pipe Lines (Yukon) Ltd.

#59
Department of Indian and Northern Affairs, Northern Environmental Protection Branch, 1979. Technical review comments dealing with the environmental impact statement for the Alaska Highway gas pipeline project. 10 pages.

Abstract

This report was presented to the Yukon Public Hearings on the environmental assessment of the proposed Alaska Highway Pipeline, March 19, 1979. The brief is a critique of the Environmental Impact Statement (EIS) prepared by Foothills Pipelines Ltd., and assesses the problems and gaps in the data base of the EIS. The report analyses the EIS in terms of its general coverage, construction project descriptions and alternatives, geotechnical assessments, hydrology, flora and revegetation, fish and wildlife, water crossings, and resource use assessments. In the general section the report points out that the EIS is non specific in discussing principles and measures for the project design, construction, and operation; that there are no contingency plans applicable to worst case scenarios, and that the description of monitoring programs are temporary roads, alternative pipeline alignments have not been thoroughly discussed, and there is no discussion of impacts related to alternate construction modes. Geotechnical aspects of the EIS are discussed in reference to problems in the discussion of frost heave, thaw settlement, permafrost construction, borrow material, active layer reduction, construction, borrow material, active layer reduction, permafrost distribution, and erosion control. The section on hydrology reviews the EIS coverage of clearing and revegetation, test sites, unique and sensitive areas, and taxa. Fish and wildlife coverage in the EIS is analysed in respect to the data base used, hunting and fishing, raptors, and other species such as wolves, wolverines, and foxes. No mitigative measures are outlined which will protect fish habitat during gravel removal from stream floodplains. The section on resource use covers omissions in the EIS concerning land use, and errors in the interpretation of land management and planning policies.

#60
D'Estimauville, M., 1982, Unpublished draft. Report on sportfish harvest of selected Yukon lakes in 1981. Department of Fisheries and Oceans, Whitehorse.

Abstract

A program was initiated in the summer of 1981 to obtain information on freshwater sports fishing in the Yukon. A variety of methods were used to collect creel information and the methods are discussed. All retained fish catch were sampled for weight, fork length, sex and maturity. Growth curves and frequency distributions have been plotted for the lake trout populations from Marsh Lake, Fox Lake, Little Salmon Lake and Watson Lake. A list of recommendations dealing with creel study methodology are made. Fish species identified in this report include northern pike, lake trout, arctic grayling, whitefish.

#61
Doyle, P. M. S., 1977. Winter Survey, Alcan Pipeline. Department of Fisheries and Environment, Fish. & Mar. Ser., Vancouver, B.C. (Unpublished memorandum directed by C. E. Walker.)

Abstract

From January 16 to March 19, 1977, surveys were conducted on streams along the right-of-way of the proposed Alcan pipeline in the Yukon Territory. This memorandum report presents information obtained from these field activities but does not include a review of information on file with Fisheries Service. Firstly, the surveys were made to determine presence or absence of water, and, where present, to collect samples for quality analysis. Secondly, the field work involved establishing cross-sections, measuring flow and other physical characteristics, and determining fish presence or absence (where possible) at the pipeline crossing site. Fishing was only successful at 2 sites - Morley river and Hazel Creek. The Yukon drainage is subdivided into Lower Yukon and Upper Yukon. The Lower Yukon system represents those streams draining into the White River which then drains into the Yukon while the Upper Yukon system represents streams draining into the Yukon and Teslin Rivers.

#62
Eby, P. and Associates. 1980. Economic feasibility of a freshwater fish commercial fishery and processing plant in the Yukon. Prepared for Council of Yukon Indians.

Abstract

The report examines the feasibility of expanding the commercial freshwater fishery and establishing a processing plant for freshwater species in the Yukon.

There are several constraints to increased commercial harvesting by native fishermen. These are: 1. low productivity of lakes 2. inaccessibility of many lakes 3. competition for available stocks from non-native subsistence fishermen 4. parasite infestation of some whitefish stocks. The low wholesale prices for Freshwater Fish Marketing Corporation fish represent a serious impediment to the establishment of processing facilities in the Yukon.

The author of the report concludes that while it doesn't seem feasible to establish a processing plant based solely on freshwater fish, assistance to individual native fishermen in the purchase of gear, equipment, and packaging materials, and instruction in handling, processing and storage, would improve the potential for native fishermen to increase sales within the Yukon. This report contains no new fisheries data.

#63
Eby, P. & Associates Ltd., 1977. Potential impacts of gas pipeline construction and operation on use and value of Yukon fisheries. Report submitted to the Department of Fisheries and Environment, July 1977. 55 pages.

Abstract

This report provides a preliminary overview of the socio-economic effects which may result from construction and operation of Foothills' proposed gas pipeline upon utilization of the Yukon's fisheries. Major emphasis has been placed on the Alaska Highway route but a number of possible alternate routes, including the Tintina Trench, Robert Campbell Highway, Klondike Highway and Dempster Highway Lateral, are also discussed.

The report is mainly compiled from existing biological data and studies and available studies and statistics relating to tourism and fishing activity in the Yukon. Published data was supplemented by interviews with Federal and Territorial officials in Whitehorse an fishermen utilizing the fish resource for sport, commercial or subsistence purposes.

The study indicates that the tourist and resident sport fisheries would be most heavily impacted by the Alaska Highway Route. Tourist fishing is concentrated in streams along the Alaska Highway, while residents fish heavily in lakes bordered by the highway and the proposed pipeline. Habitat degradation, loss of access and overfishing by workers at construction camps are identified concerns. Throughout the study, lack of site-specific data is identified as a major problem in coming to any definite conclusions.

#64
Elson, M. S., 1974. Catalogue of fish and stream resources of east central Yukon Territory. Fish. Mar. Ser. Tech. Rep. No. PAC/T-74-4.

Abstract

This catalogue is a historical record and information source concerning fish species and their habitats in the Selwyn Mountains of East Central Yukon. Physical, chemical and biological data are presented for 16 streams within the study area: Peel River, Bonnet Plume River, Snake River, Wind River, Hart River, Stewart River, Beaver River, Hess River, Lansing River, S. Macmillan River, Rogue River, South McQuesten River, Pelley River, Ross River, S. Macmillan River and N. Macmillan River. Capture methods during the 1973 survey were restricted to beach seining, gill-netting and angling. Information on specific biology, relative and absolute abundance and spawning times and locations is limited. Sixteen non-anadromous and 4 anadromous fish species are represented in the area.

#65
Elson, M. S. and L. W. Steigenberger, 1977. Measurements of chinook salmon collected in the Yukon River commercial fishery near Dawson City in 1975 and 1976. Prepared for Northern British Columbia and Yukon Fisheries Branch. 27 pages.

Abstract

This report presents data collected from chinook salmon taken in the 1976 commercial fishery operating approximately twenty-five miles downstream of Dawson City at Fresno Creek. Some catch data is included for chinooks taken in 1975 in the same location and mention is made of other species taken incidentally in the chinook salmon fishery.

Gillnets were set continually at the mouth of Fresno Creek from July 20 to August 6 and for each salmon captured the following information was recorded: date, species, sampling time, fork length, sex, maturity, mesh size, scale reference number. The data collected on chinook salmon in 1975 was restricted to numbers of fish taken per day in the last half of the run. A sample of chinook salmon collected near Fort Selkirk from the domestic fishery conducted there is used in statistical comparisons in this report. Results are presented. Possible explanations are suggested for the difference in rates of chinook migration at Fresno Creek and Fort Selkirk. The average number of days difference between the levels of 10, 50 and 90% harvest at Fresno Creek and Fort Selkirk was 4, 6 and 11 days, respectively. These differences in the rates of migration may be explained by a reduction in total numbers of migrating salmon because of establishment of spawning grounds between these two points. Another possible explanation is a decrease in the rate of migration with an increase in distance travelled from the Bering Sea. There is a discussion of possible spawning areas downstream of Dawson City. The spawning areas identified are similar to documented chinook salmon spawning areas near the Ingersoll Islands (Walker, 1976).

Four deformed male chinook salmon were captured near the mouth of Fresno Creek and it is suggested that the deformities may be either genetically caused or they may be the result of passage through the turbines at Whitehorse Rapids dam. Further investigation of this phenomenon, of migration rates and of spawning ground locations throughout the Yukon River are recommended.

#66
Elson, M. S., and L. W. Steigenberger, 1976. A biological investigation related to placer mining operations in the Moosehorn Range in Yukon Territory. Prepared by Northern Natural Res. Serv. Ltd. for Claymore Resources.

Abstract

The purpose of this study was to assess the environmental impact of placer mining in Swamp, Discovery and Claymore Creeks. Fish sampling was by beach seine. No fish were captured in Discovery or Swamp Creeks. Adult Arctic grayling were observed in Claymore Creek but none were captured. At the mouth of Claymore Creek in the Ladue River 60 juvenile Arctic grayling were captured. Based on inspection of longitudinal profiles, low stream discharges, extreme gradients and the presence of obstacles to fish migration it is not possible for the headwaters of Swamp, Discovery or Claymore Creeks to support fish life in any season. The lower reaches of Claymore Creek exhibit a more gentle meandering form but the gravel composition is not ideal as a spawning substrate for Arctic grayling. It is probable that lower Claymore Creek serves as a feeding area for limited populations of adult Arctic grayling in the summer months, that spawning occurs in the Ladue River (as evidenced by juvenile fish captures and presence of suitable gravel), and that adult fish in Claymore Creek migrate back into the Ladue River prior to freeze-up.

#67

Envirocon Ltd. 1982. (in prep.) Overwintering habitats of juvenile chinook salmon in tributary streams along the North Canol Road right-of-way Yukon Territory (exact site unknown). Department of Indian and Northern Affairs, Whitehorse.

Abstract

The objectives of the study are to identify waterbodies where the construction of culverts and bridges might affect areas critical to juvenile chinook salmon overwintering and egg incubation. Where possible (depending on ice conditions and open water areas) an inventory of fish species composition, benthic invertebrates and potential overwintering habitats for salmonids will be collected. Sampling techniques will include electrofishing and minnow trapping. Tentative study sites will include Tenas Creek, Blue Creek, Pup Creek, Twin Creek No. 1, Riddell Creek, Boulder Creek plus an additional four or more streams. Field studies are being conducted from March 1 - March 15, 1982.

NOTE: This information was taken from a letter. The study has not been completed, and therefore the results have not been seen.

#68
Envirocon Ltd., 1981. Environmental study: An overview of fish resource use in Atlin Lake and Surprise Lake, 1980. In: Klohn Leonoff Consultants Ltd., Stage II Adanac Mine environmental and socio-economic assessment. Volume I Environmental and engineering assessment. Appendix II. Report prepared for Placer Development Ltd.

Abstract

This investigation was carried out to assess the influence of an expanded population in the Atlin area on the harvest of fish resources from Atlin and Surprise Lakes. A recreational creel census was carried out on these lakes in August and September, 1980. This report also contains a brief review of the history of the fisheries and previous aquatic ecological studies conducted on Atlin and Surprise Lakes.

The data gathered indicate that the current annual recreational harvest of sport fish from Atlin Lake is in the order of 600 - 900 lake trout and 100 - 150 Arctic grayling per year. The total annual harvest of lake trout from Atlin Lake - the sum of the commercial, domestic and recreational fishery landings - is estimated at approximately 1,964 kg. Until additional biological data are gathered and analyzed, the answer to the question of whether or not the lake trout and Arctic grayling population in Atlin Lake can compensate for the potential increase in fishing pressure will remain uncertain. Since no quantitative information on the recreational fishery has been gathered previously, a creel census was conducted in 1980 to determine the annual harvest of Arctic grayling. Results indicated that the Arctic grayling population in Surprise Lake provides good recreational opportunities for both Atlin residents and tourists. Since fishing effort will inevitably increase following mine construction, the B.C. Fish and Wildlife Branch should be encouraged to develop and maintain a recreational fishing management program.

#69
Envirocon Ltd., 1981. Environmental Study: Arctic grayling spawning and rearing in Surprise Lake: a preliminary assessment of the effects of hydroelectric power development. In: Klohn Leonoff Consultants Ltd. Stage II Adanac Mine environmental and socio-ecoomic assessment. Volume I Environmental and engineering assessment. Appendix III. Report prepared for Placer Development Ltd.

Abstract

Investigations were conducted during the periods May 16 to 19 and June 23 to 27, 1980 to collect data on Arctic grayling spawning and rearing near inflowing streams not surveyed in 1979. Fishing was conducted in the littoral areas at the mouths of 3 creeks and at several locations in Pine/Cup Creek. In June, fishing was conducted to locate alevins or fry in the vicinity of known or suspected spawning and rearing areas.

The data suggest that Pine/Cup Creek is the principal spawning area utilized by the Arctic grayling population in Surprise Lake and that the shoreline habitat at the mouths of other tributaries such as Cracker, Union and Otter Creeks is of secondary importance. The report also includes potential impacts and mitigation for Surprise Lake and a plot contour of future reservior level in May. The shoreline of Surprise Lake at the outlet of inflowing creeks along the new shoreline contour was mapped to identify suitable spawning gravels and new littoral areas.

#70
Envirocon Ltd., 1980. Matson Creek Environmental Assessment. Prepared for Goldmark Minerals Limited, Calgary. 57 + pages.

Abstract

Goldmark Minerals Limited proposes to obtain water from Matson Creek and its tributaries for use in its placer gold mining operation located 90 km. southwest of Dawson City. The purpose of this report is to provide environmental information required by the Yukon Territorial Water Board for issuance of a water-use licence.

Data on aquatic habitat and fish resources of the Matson Creek drainage were obtained during field investigations carried out over three periods (May 31 to June 6, July 16 to 19, and August 23 to 24). Fish were captured by a variety of methods (beach seine, gill nets, electrofishers). For each sample collected the species, fork length, weight, sex and stomach contents were recorded.

Arctic grayling, round whitefish, longnose sucker and slimy sculpin were captured in Matson Creek. In the upper reaches of the creek, where placer mining is underway, only Arctic grayling and slimy sculpins were captured. Due to the small sampling size and brevity of the study it is not known whether the grayling population is naturally small or whether it has been affected by previous placer mining activities. At times, the suspended sediment concentrations in the stream were between 500 and 1,000 mg/1 in the upper reaches of Matson Creek. However the high suspended sediment concentration did not prevent some arctic grayling from inhabiting this area throughout the summer.

Arctic grayling do not spawn in the upper reaches of Matson Creek but may spawn near the inflow to the Sixty Mile River. These spawning salmon are unlikely to be affected by the placer mining activities of the upper reaches of Matson Creek.

The number of Arctic grayling and other species utilizing Matson Creek in summer may be reduced since feeding and rearing habitat will be destroyed or degraded by the mining activities.

#71
Envirocon Ltd., 1977. An environmental comparison of alternative pipeline corridors in the Yukon Territory. Report prepared for Environmental Assessment Panel, Alaska Highway Pipe Line Project. 2 Volumes.

Abstract

Envirocon Ltd. was commissioned by the Environmental Assessment Panel to conduct an independent comparison of the environmental implications of major alternative gas pipeline corridors through the Yukon. The four major corridors are the Alaska Highway, the Klondike Highway, the Tintina-Lower Robert Campbell, and the Tintina-Liard. A program was designed which weighted the various environmental components according to the estimated importance of each component. The corridors were analyzed using aerial reconnaissance and existing data, the impact values were computed and the corridors compared. No new fisheries data were produced for this report. According to land based resource impacts, the preferred corridor is Alaska Highway, with Klondike Highway the second choice, Tintina-Lower Robert Campbell a close third, and the Tintina-Liard a definite last. Water-based resource impacts indicated a different preference ranking, with Tintina-Liard the preferred alternative, the Tintina-Lower Robert Campbell a close second, the Klondike Highway a definite third, and the Alaska Highway a close last. The conclusion is that other factors, notably socio-economic, engineering, and national interest should be considered to arrive at a definitive decision regarding pipeline corridors in the Yukon Territory.

#72
Environmental Assessment Panel, 1981. Alaska Highway Gas Pipeline - Routing Alternatives Whitehorse/Ibex Region. July, 1981. 40 pages.

Abstract

In accordance with the Federal Environmental Assessment and Review Process, the Environmental Assessment Panel on the Alaska Highway Gas Pipeline Project completed a review of the pipeline routing alternatives in the Whitehorse/Ibex Region. This review led to the conclusion that the Ibex Pass route, which is preferred by Foothills Pipe Lines (South Yukon) Ltd., should be rejected in favour of the First Whitehorse Route with the West Whitehorse Cut-off. Despite the projected extra cost associated with it, this route offers significant environmental advantages, preserves acceptable options for the eventual link with a Dempster Lateral, and adheres to the existing Alaska Highway corridor. It was recommended that Government agencies take early and positive action to preserve the present wildlife and environmental values in the Ibex Valley area and to preserve future options for this unique area.

#73
Environmental Assessment Panel, 1979. Alaska Highway Gas Pipeline - Yukon Hearings (March - April, 1979). August 1979. 61 pages.

Abstract

This report of the Environmental Assessment Panel came in response to the Environmental Impact Statement prepared by Foothills Pipe Lines (Yukon) Ltd. The information needs described in the 1977 Panel Report still remain, as do many of the deficiencies identified at the recent Yukon Public Hearings. Since so many issues still remain unsolved, this Panel Report is similar to the 1977 Interim Report. This report outlines the requirements for the completion of the assessment of the project.

Foothills' current construction plan has potential for significant impact on Yukon fish populations in the vicinity of the pipeline. The absence of a detailed construction schedule makes it impossible to determine the extent of impact on fish habitat in the vicinity of the pipeline. The Panel concluded that it is not possible to complete an environmental impact assessment review at this time and that the information requirements should be provided in a revised Environmental Impact Statement to be prepared by Foothills Pipe Lines Ltd.

#74
Environmental Assessment Panel, 1978. Shakwak Highway Project. June, 1978. 60 pages.

Abstract

This report is the result of the review by the Environmental Assessment Panel. The report uses the Environmental Impact Statement, in addition to information presented by public interest groups, individuals and government agencies. It is divided into chapters on project management issues, physical and engineering issues, ecological issues, social and economic issues, and a summary of conclusions and recommendations.

The panel calls for the upgrading of inadequate culverts on the Shakwak route, following additional fisheries studies to determine the timing and location of fish migration, spawning, fry emergence and overwintering at watercrossings that may be impacted by the proposed Shakwak project. These studies will be used to determine stream crossing design and construction timing.

#75
Environmental Management Associates. 1981. Fisheries resources of selected watercourses within the influence of the Alaska Highway gas pipeline in southern Yukon Territory - Update to January 1, 1981. Prepared for Foothills Pipe Lines (South Yukon) Ltd., Calgary. 19 pages.

Abstract

Five separate fisheries resource studies were conducted along the Alaska Highway gas pipeline route in 1980. These include studies of the Nisutlin Bay area, the Beaver Creek area, and the Kluane Lake area, salmon streams along the pipeline route, and selected areas with new stream crossing sites along the pipeline route. The purpose of this report is to present a brief summary of the results of the 1980 fisheries investigations and, where applicable, to compare these results with those aspects of fisheries habitat utilization documented during earlier programs.

#76
Environmental Management Associates. 1981. Spring fisheries investigations of selected watercourses along the Alaska Highway gas pipeline in southern Yukon Territory. Prepared for Foothills Pipe Lines (South Yukon) Ltd., Calgary. 42 pages.

Abstract

Spring fisheries investigations were conducted in 1981 in order to obtain information for the purpose of environmental planning during the design phase of the pipeline. Eight creeks were examined within 1 km downstream of proposed pipeline crossings for the presence of Arctic grayling spawning activities. Sampling was conducted from May 16 - 21 and May 22 - 26. Streams were sampled using electrofishers or angling. Sampling locations were at: Jarvis River (KP 263.9), Judas Creek (KP 495.9) unnamed creek (KP 381.3), Bear Creek (KP 273.1), Arkell Creek (KP 390.8), Grayling Creek (KP 477.2), Seaforth Creek (KP 532.3), and Big Creek (KP 787.9). These eight watercourses were evaluated for the feasibility of using mitigative methods such as fluming and check dams. In the event that such techniques are required during construction Bear Creek, Grayling Creek and the unnamed creek were classified as being suitable for fluming activities, however, a minor creek diversion might be required at Grayling Creek. At Bear Creek, relatively extensive bank modifications would be required for flume installation.

A spawning area used by one or two pairs of Arctic grayling, existed downstream of the crossing on the Jarvis River. A minor spawning area was identified downstream of the crossing on Judas Creek. A small unnamed creek (KP 381.3) was recorded as being heavily utilized for spawning by arctic grayling. No such utilization was observed in the vicinity of the proposed crossing locations on Bear Creek, Arkell Creek, Grayling Creek, Seaforth Creek or Big Creek.

Bear Creek was sampled for Dolly Varden char fry and evaluated for the presence of suitable Dolly Varden spawning areas. No fry were captured but suitable spawning substrats were common commencing 300m downstream of the proposed pipeline crossing.

Chinook salmon fry were sampled at a chinook spawning area located 8.25 km downstream of the pipeline crossing, on the Teslin River. Emerged fry were present on both sampling dates: however no alevins were found by digging in spawning gravels. Because actual redds had not been marked during the previous spawning period, no conclusions could be drawn regarding the time at which fry emergence is complete in Teslin River.

#77
Environmental Management Associates. 1981. Winter studies of selected watercourses along the Alaska Highway gas pipeline in the southern Yukon Territory, 1981. Prepared for Foothills Pipe Lines (South Yukon) Ltd. Calgary. 28 pages.

Abstract

Selected watercourses which are included in spreads presently identified for winter construction were investigated during February and March, 1981. Three watercourses were sampled to determine the presence or absence of fish and the location of overwintering areas in the vicinity of proposed pipeline crossings. The proposed crossing locations studied were the second crossing of the Koidern River (KP 98.7), the Donjek River (KP 129.0) and the Swift River (KP 687.8). No concentrations of overwintering fish were identified in the crossing regions on the Koidern River (KP 98.7) or the Donjek River. An overwintering area was identified upstream of the proposed crossing on the Swift River.

The physical characteristics of the proposed crossing locations on Logjam Creek (KP 671.3), McNaughton Creek (KP 689.2) and Plate Creek (KP 708.0) were recorded to provide quantitative information regarding water quality, discharge rates, ice conditions and bank and substrate conditions. These data are presented in the text.

#78
Environmental Management Associates. 1980. Enumeration of spawning salmon in aquatic systems along the Alaska Highway gas pipeline in southern Yukon Territory, 1980. Prepared for Foothills Pipe Lines (South Yukon) Ltd., Calgary, Alberta. 48 pages.

Abstract

Watercourses that are likely to be affected by the Alaska gas pipeline and that are utilized for spawning by chinook or chum salmon were surveyed to gather information on the distribution and numbers of proposed pipeline crossing locations. Chinook salmon were enumerated in the Takhini, Ibex, Yukon, Teslin, Morley, Swift and Smart rivers and Michie and McNaughton creeks. No chinook salmon were observed in M'Clintock River between Marsh Lake and the confluence of Michie Creek. Only one chinook was observed in the Yukon river, in the region between the Lewis Dam and the confluence with Takhini River.

No chinook spawning activity was observed in the regions immediately downstream of the proposed pipeline crossings of the Yukon, M'Clintock, Smart or Swift rivers, or McNaughton Creek. Spawning areas do exist in downstream regions of tributaries to the Swift River which are crossed by the pipeline route. A group of four chinook salmon were observed 2.8 km. downstream of the proposed pipeline crossing on the Takhini River; however, these fish appeared to be migrating upstream rather than spawning in this location.

Previously undocumented spawning areas in the Smart and Swift and McNaughton Creek were observed. Chum salmon spawning was documented in Kluane and Koidern rivers during 1980. No chum salmon were observed in the White River upstream of the proposed pipeline crossing on that watercourse, or in Christmas Creek during the aerial survey. No chum salmon were observed upstream of the mouth of Burwash Creek in Kluane River during 1980, nor were any observed in the upper reaches of the Koidern River. New chum spawning areas were documented at the mouth of Koidern River.

Appendix I of the report includes an information pamphlet on the chinook salmon of the upper Yukon River. Appendix II reports on a one day survey conducted by boat on a section of the Teslin River to locate and enumerate salmon.

#79
Environmental Management Associates. 1980. Winter studies of aquatic systems along the Alaska Highway gas pipeline in Southern Yukon Territory - Beaver Creek Area (KP 0 to KP 219). Prepared for Foothills Pipe Lines (Yukon) Ltd., Calgary, Alberta. June 1980. 39 + pages.

Abstract

Selected waterbodies between KP 0 and KP 219 of the Alaska Highway Gas Pipeline in southern Yukon Territory were examined between March 4 and 21, 1980. The unnamed creeks at KP 60.5 and 107.8, the unnamed lake at KP 114.2, the unnamed lake adjacent to KP 114.3 and Swede Johnson Creek at KP 149.7 were found to exhibit no overwintering potential for fish. The potential of the White River (KP 72.1), the lower reaches of the Koidern River (KP 79.2) and Grafe Creek (KP 108.5) to support fish during winter months was judged to be limited by the discharge characteristics of these watercourses.

The most suitable overwintering conditions were observed in Mirror Creek (KP 7.2), and at the second crossing of the Koidern River (KP 99.8). Overwintering is less likely to occur at the crossing sites in the upper reaches of the Koidern River (KP 107.9) and on Halfbreed Creek (KP 189.6), but the presence of developing eggs in downstream regions of these watercourses has been documented.

Pertinent physical, chemical and fishery resource data for these watercourses are summarized in tabular form.

#80
Environmental Management Associates. 1980. Fall fisheries investigations of Kluane Lake and north-shore creeks. Prepared for Foothills Pipe Lines (South Yukon) Ltd. Calgary. 29 + pages.

Abstract

A gillnetting program was conducted in Kluane Lake from September 23 to October 9, and October 21 to November 5, 1980, to identify lake trout and lake whitefish spawning areas in the region of the lake between Destruction Bay and Congdon Creek. Information on the location of spawning areas is to be utilized in route selection of a lake crossing for the Alaska Highway gas pipeline.

Three lake trout spawning areas were conclusively identified, while data indicated the likely existence of three other such areas in the study region. The majority of these spawning areas are located on the north side of the lake. Identification of lake whitefish spawning areas was difficult due to the low numbers of sexually mature whitefish taken during the spawning period. Six lake whitefish spawning areas were identified based on numbers of ripe and spent fish taken in each net set. Four of these areas are on the south shore, while two are located on the north shore of Kluane Lake. The data collected regarding lake whitefish spawning locations are not conclusive, however. It was determined that not all adult whitefish spawn each year in Kluane Lake.

Streams entering the north side of Kluane Lake between Long Point and Cultus Bay were examined in order to provide baseline data for these watercourses if they should be crossed by a new pipeline alignment. Only two watercourses, Clear and Cultus creeks, were found to provide habitat suitable for use by fishes. Clear Creek exhibited habitat with the potential for use as spawning, nursery and rearing areas and summer feeding areas for Artic grayling, but not overwintering. Cultus Creek was suitable for use by all life history stages of Artic grayling. The presence of a resident population of Artic grayling in Cultus Creek was indicated during the study. No other important fishes are anticipated to use these watercourses.

#81
Environmental Management Associates. 1980. Winter studies of aquatic systems along the Alaska Highway gas pipeline in southern Yukon Territory - Nisutlin Bay area (KP 586 to KP 649). Prepared for Foothills Pipe Lines (Yukon) Ltd., Calgary. 37 + pages.

Abstract

Nine watercourses between KP 586 and KP 649 of the Alaska Highway Gas Pipeline in southern Yukon Territory were examined between February 12 and 25, 1980. The unnamed creeks at KP 623.8 and 624.2, and the upper crossing of Hazel Creek (#2) at KP 633.7 were found to exhibit no fisheries overwintering potential. The potential of Andy Creek (KP 625.4) to support fish was judged to be very low.

Hays Creek (KP 603.5), Strawberry Creek (KP 608.7), Morley River (KP 620.2), the lower crossing of Hazel Creek (#1) (KP 631.7) and Smart River (KP 647.0), exhibited suitable overwintering conditions. With the exception of Strawberry Creek, important fish species were collected in each of these watercourses.

Pertinent physical chemical and fishery resource data are summarized in tabular form.

#82
E.V.S. Consultants Ltd., 1981. Assessment of the effects of the Clinton Creek Mine waste dump and tailings, Yukon Territory. Prepared for Cassiar Resources Ltd., Vancouver. 97 pages.

Abstract

During September 1980, a biological assessment of the Clinton Creek watershed was undertaken. The objective of this survey was to inventory the indigenous fish stocks and critically evaluate the influence of mining activities on the instream habitat and fishery resource. Clinton Creek from Hudgeon Lake to Forth Mile River was subdivided into six sections and sampled over a six day period. Sampling techniques included minnow trapping, electroshocking, gillnetting, beach seining, and the installation of a two-way weir to determine the movement of fish utilizing Clinton Creek during the sampling period. Arctic grayling, juvenile chinook salmon and sculpins were sampled in Clinton Creek. Fish distribution was governed by habitat characteristics and food availability. Large numbers of grayling inhabited upper Clinton Creek, below the outlet of Hudgeon Lake. This section supported a large biomass of invertebrate drift organisms and represented optimal stream habitat. Detrimental impacts on fishery resources were observed in a 0.5 kilometer section immediately downstream of the mined area. In this section, outwash of waste rock destabilized the streambed and reduced fish habitat. Also included is a section on the biological effects of asbestos fibers on aquatic organisms and Clinton Creek water quality - asbestos fibre content.

#83
Foothills Pipe Lines (South Yukon) Ltd. 1981. Addendum to the Environmental Impact Statement for the Yukon section of the Alaska Highway Gas Pipeline. Submission 5-3. Design and use of culverts, 10 pages.

Abstract

The addenda submission reviews project plans for permanent access roads, temporary access roads and culverts as these relate to fish.

#84

Foothills Pipe Lines (South Yukon) Ltd. 1981. Addendum to the Environmental Impact Statement for the Yukon section of the Alaska Highway Gas Pipeline. Submission 5-2. Development of construction schedules in relation to fisheries and wildlife issues. 36 pages.

Abstract

This submission presents a detailed construction schedule indicating pipelining steps, anticipated site specific impacts and planned investigations for two typical construction sections. The typical summer section analyzed is Construction Section 8, KP 452-537 which covers the area from the Yukon River, east around the north and east shores of Marsh Lake and south of Squanga Lake to a point west of Teslin River. The winter construction section analyzed is section 3, KP 110-164.5 which includes the section of the pipeline west of Donjek River to a point between Quill and Burwash Creeks. Details of the fisheries and wildlife issues encountered, the use of scheduling as a mitigative device and the potential impacts remaining after application of scheduling changes are included.

Charts show fish species presence, critical periods, and scheduled instream construction periods.

#85
Foothills Pipe Lines (South Yukon) Ltd. 1981. The Alaska Highway pipeline project. Addendum to the Environment Impact Statement for the Yukon section of the Alaska highway gas pipeline. Submission 3 - 5. Examination of routing alternatives for the Alaska highway gas pipeline in the Kluane Lake region. 71 + pages.

Abstract

An examination of four pipeline routing alternatives in the Kluane Lake region. Includes a discussion of the conditions along the four routes, an evaluation of the alternatives, and an environmental impact assessment of the preferred route #3. This route enters Kluane Lake 2 km. south of the mouth of Congdon Creek, crosses the lake and exits at the eastern shore, 3 km. south of Cultus Bay. It then proceeds southeasterly to the Alaska Highway near Sulphur Creek.

Fisheries habitat utilization in waterbodies along each of the proposed routes is presented in tabular form. Fisheries data is summarized from earlier reports done for the Department of Fisheries and Foothills Pipe Lines. The major fisheries impact from the proposed route #3 will be the disruption of one trout and one lake whitefish spawning area.

#86
Foothills Pipe Lines (South Yukon) Ltd., 1981. Addendum to the Environmental Impact Statement for the Yukon section of the Alaska Highway Gas Pipeline. Submission 3-3 Examination of routing alternatives for the Alaska Highway Gas Pipeline in the Marsh Lake/Squanga Lake region. 51 + pages.

Abstract

In selecting the route followed by the Alaska Highway Gas Pipeline in Yukon Territory, a multi-disciplinary approach was used, involving construction, engineering, environmental, socio-economic and operations evaluation. As a result, certain segments of the pipeline route were located in areas considered to be sensitive for environmental reasons, and criticisms of these routes have been voiced by individuals and groups with environmental interest. One such area is in the viscinity of Marsh and Squanga Lakes. This report gives details of the selection processes involved in identifying a route from three alternatives passing through this area, and describes potential impacts, mitigative measures and residual impacts along the preferred route. No new fisheries information is presented.

#87

Foothills Pipe Lines (South Yukon) Ltd., 1981. Addendum to the Environmental Impact Statement for the Yukon section of the Alaska Highway Gas Pipeline. Submission 3-4. Examination of routing alternatives for the Alaska Highway Gas Pipeline in the Swift River/Rancheria Valley Region. 48 + pages.

Abstract

This report gives details of the process involved in selecting a route from two alternatives passing through the Swift River and Rancheria Valley area. This area is considered to be sensitive for environmental reasons. Potential impacts, mitigative measures and residual impacts along the preferred route are described. No new fisheries information is presented.

#88
Foothills Pipe Lines (South Yukon) Ltd., 1981. Addendum to the environmental impact statement for the Yukon section of the Alaska Highway gas pipeline with respect to alternative routes. Submission 3-1. Examination of routing alternatives for the Alaska Highway Gas Pipeline for the Whitehorse/Ibex region. 61 + pages.

Abstract

This report was drawn together to examine the routing alternatives for the Alaska Highway gas pipeline in the Whitehorse/Ibex region. Foothills reached the conclusion that the Ibex Pass route was superior to all other alternatives. The basis for this decision was that concern for environmental factors present in the Ibex was outweighed by socio-economic, land use construction and safety factors associated with alternative routings. This report once more reviews the advantages and disadvantages of the alternative routings in the region. There is an appendix which tabulates fishery resources along the 11 alternative routes. This data was compiled from previous Foothills Pipe Lines reports.

#89
Foothills Pipe Lines (South Yukon) Ltd. 1981. Addendum to the Environmental Impact Statement for the Yukon section of the Alaska Highway Gas Pipeline. Submission 5-4. Exploitation of fish and wildlife. 15 pages.

Abstract

The potential for over exploitation of fish and wildlife stocks as a consequence of the influx of people to Yukon Territory during construction of the Alaska Highway Gas Pipeline was identified as a possible problem.

This report presents recent fish and wildlife data from the Yukon, and a comparable set of data for Alaska. The impact of the Trans-Alaska Oil Pipeline on these non-renewable resources and the management strategies followed are analyzed. This analysis is then applied to the situation in the Yukon Territory. It is concluded that over exploitation of fish and wildlife associated with the construction of the pipeline in the Yukon Territory can be adequately handled through existing government agencies. A number of regulatory measures are suggested.

#90
Foothills Pipe; Lines (Yukon) Ltd. 1981. Catalogue of fisheries resource information for waterbodies crossed by the Alaska Highway gas pipeline route in southern Yukon Territory. 14 + pages.

Abstract

Information concerning the fisheries resources of the waterbodies crossed by the Alaska Highway Gas Pipeline route was initially synthesized into catalogue format in 1978 by Slaney and Company Ltd. on behalf of Foothills Pipe Lines (South Yukon) Ltd. The catalogue was revised and updated in May 1980 by Foothills Pipe Lines (Yukon) Ltd. and in November 1980 by Environmental Management Associates. This 1981 catalogue is similar in format to the 1979 catalogue and the content is identical except where clarifications or revisions of data were necessary. In addition, this catalogue includes references to all field surveys conducted during 1980 and the winter and spring of 1981.

A data sheet is used to present information for each waterbody and its format has been designed so as to highlight the important fish species present and their habitat utilization. Any other fish species observed and the potential of the waterbody to support overwintering fish are also indicated.

Of the 285 waterbodies crossed by the route, 51 have been found to support an important fish species utilizing habitat that is sensitive to pipeline-related disturbance. These waterbodies are listed in a table.

#91
Foothills Pipe Lines (Yukon) Ltd., 1981. Draft. Fisheries Protection Plan December, 1981.

Abstract

This document has been prepared in compliance with the requirement of sections 179 and 155 "Submission and Implementation of Environmental Plans" of the Northern Pipeline Agency's Environmental Terms, Conditions and Related Guidelines for Yukon Territory and Swift River Portion of the Pipeline in the Province of British Columbia. During the design, construction and maintenance phases of the project, this document will be utilized as a source and summary of: fisheries information, regulatory requirements pertaining to fishery resources, and standards and guidelines directed at minimizing adverse impacts on fish and fish habitat. Protection plans will be revised from time to time to reflect new information. No new fisheries information is presented in this report.

#92
Foothills Pipe Lines (Yukon) Ltd. 1979 The Dempster Lateral Gas Pipeline Project, Environmental Impact Statement, Volume 4. 632 pages.

Abstracts

A report on the proposed Dempster Lateral gas pipeline which identifies environmental concerns and suggests mitigation measures to offset potential problems. It is divided into the following chapters; project setting, project description, environmental setting, environmental concerns and possible mitigations and residual impacts. Included is a summary of existing data on fishery resources and fishery utilization of waterbodies along the pipeline route. Types and duration of possible impacts at streamcrossings are discussed and corrective measures are suggested.

#93
Foothills Pipe Lines. 1979. Dempster Lateral Gas Pipeline Environmental Impact Atlas, Volume 5.

Abstract

A series of maps showing the pipeline route and associated facilities. Fish, birds, mammals, historic resources, and land and resource use are all indicated. Fishery habitat utilization is indicated, but there is no species inventory. The maps graphically illustrate areas of conflict between environmental components and the pipeline. Predicted envionmental impacts are identified at the base of each map and defined according to environmental component affected, dominant concerns and predicted duration, location and level of impact.

Map Set I - Fish and Wildlife
Map Set II - Historic Resources and Land and Resource Use

#94
Foothills Pipe Lines (South Yukon) Ltd., 1979. Environmental impact statement for the Alaska Highway gas pipeline project. 450 pages.

Abstract

This is the Environmental Impact Statement for the Yukon portion of the Alaska Highway Gas Pipeline, as proposed by Foothills Pipelines Ltd. This document is part of the environmental assessment and review procedure required prior to project approval. Most of the proposed route in the Yukon Territory crosses federally administered territorial lands. The environment impact statement includes a description of the proposal, and construction, operation and maintenance activities associated with the proposal. The environmental setting is described, in terms of climate, terrain, hydrology, vegetation, fish resources, birds and mammals. A description of land and resource use within the area, and of socio-economic conditions in affected communities, is included. Statements of environmental concerns related to the pipeline include concern over the effect of a warm pipe in permafrost soils, concerns over the effects of a pipe containing chilled gas in unfrozen soils; concerns over the effects of stream crossings on fish populations; problems relating to siltation during construction or the introduction of organic matter into streams; effects on Dall's sheep and mountain caribou populations; disturbance of nesting falcons and other raptors; and the potential for overexploitation of wildlife resources due to a large influx of people during construction. Areas traversed by the pipeline which are particularly sensitive to disturbance include: an area surrounding Sheep Mountain in Kluane National Park, Ibex Pass; the wetland area near Pickhandle Lake; and an area near Mount Mitchie and Squanga Lake. This document contains responses to all environmental concerns, and proposals for measures mitigating or eliminating disturbance.

#95
Foothills Pipe Lines (South Yukon) Ltd., 1979. Overview summary of the environmental impact statement for the Alaska Highway gas pipeline project. 57 pages.

Abstract

This report summarizes the Environmental Impact Statement prepared by Foothills Pipe Lines (South Yukon) Ltd. for the Environmental Assessment and Review Process Panel. The projected Alaska Highway gas pipeline is described and there is a section on the evaluation of alternatives. Environmental impacts are stated and assessed. Sections are included on biological concerns and fish resources.

#96
Foothills Pipe Lines (Yukon) Ltd. 1978. Fisheries investigations along the Klondike Highway section of the prospective Dempster Lateral Pipeline Route, Yukon Territory - Summer and Fall, 1977. Prepared by Beak Consultants Ltd., Calgary. 34 + pages.

Abstract

An inventory of fish fauna in waterbodies within the influence of the Klondike Highway was conducted from July 29 to November 1, 1977. Electrofishing, gillnetting, beach seining, minnow trapping and angling were used to capture fish. In addition, helicopter surveys were flown August 18 and 19 to identify spawning salmon areas and August 28 and September 6 to locate chinook carcasses. An average of six visits was made to each prospective pipeline crossing which exhibited fishery potential.

Information on fish habitat utilization (spawning, rearing, migration) was collected with special emphasis given to the Pacific salmon. Chinook salmon were found in the study area between July 15 and September 6. They migrated through or spawned in the vicinity of the prospective pipeline crossings in eight watercourses; Yukon River, Klondike River, McQuesten River, Stewart River, Tatchun River and the Nordenskiold River. In addition, most of the streams along the prospective route were utilized as nursery or rearing areas by chinook parr.

With the exception of one carcass sighted at Pelly Crossing, all chum salmon were restricted to the Yukon River. Arctic grayling were found to utilize almost every stream along the proposed route which exhibited fishery potential. Streams which exhibited large populations of arctic grayling were: Klondike River, McQuesten River, Crooked Creek, Mica Creek, McCabe Creek, Tatchun River, Nordenskiold River and Richthofen Creek.

Lake whitefish were found in spawning condition at the end of October in Fox, Little and Twin Lakes.

Includes maps of the Klondike Highway section of the pipeline showing adjacent and crossed waterbodies, as well as chart form summaries of fish habitat utilization.

#97
Foothills Pipe Lines (Yukon)., 1978. Fishery investigations along the proposed Dempter Lateral pipeline route, 1978. Prepared by Beak Consultants Ltd., Calgary. 2 volumes.

Abstract

An inventory of the fish fauna of the watercourses within the influence of the proposed Dempter Lateral Pipeline was conducted from May 11 to September 22, 1978. The purpose of this investigation was to identify the presence and abundance of fish fauna, to document the routes and timing of fish migrations, as well as to determine the location of spawning sites, nursery, rearing and potential overwintering areas. In addition, this investigation provided site evaluations, photographic documentation and water chemistry data. Habitat utilization by important fish species (chinook salmon, chum salmon, Arctic char, Arctic grayling, lake trout, lake whitefish, broad whitefish and inconnu) are summarized in a table. Volume 2 includes a list of fishes indigenous to the waterbodies under consideration, a series of cross-sectional profiles of the proposed stream crossings, data summary sheets listing physical and chemical information as well as fish species and fish habitat utilization for the pertinent waterbodies, and maps of the proposed pipeline route showing locations of the proposed waterbody crossings.

#98
Foothills Pipe Lines (Yukon) Ltd. 1978. A summary of fishery investigations in waterbodies within the influence of the proposed Alaska Highway gas pipeline in Yukon Territory, 1976-77. Prepared by Beak Consultants Ltd., Calgary. 267 + pages.

Abstract

This report presents a summary of the fisheries information collected in previous inventories (fall 1976, winter 1976-77, spring, summer and fall 1977). Fisheries utilization, including migration routes, spawning areas, nursery and/or rearing areas, summer habitat and potential overwintering areas is presented in chart form, along with an indication of the most sensitive time period for each waterbody and the nature and degree of impact on the fishery resource should the waterbody be crossed during a sensitive time period. General and specific concerns and recommendations relating to pipeline construction and operation are discussed. Appendix I includes data summary sheets listing fish species presence, physical and chemical information for each waterbody studied. Appendix II contains maps showing the locations of Alaska Highway pipeline stream crossings.

#99
Foothills Pipe Lines (Yukon) Ltd. 1977. Fall and early winter aquatic resource inventory: Dempster Lateral pipeline route. Prepared by Aquatic Environments Ltd., Calgary. 2 volumes.

Abstract

This report presents the results of a fall (August 31 to September 9) and early winter (November 15 to 22) survey of watercrossings along that portion of the Dempster Lateral pipeline route from Dawson City, Y.T., north to the Peel River, N.W.T. Physical and chemical data were collected, in addition to data on presence of benthic invertebrates. Fish species were also included, where information was available. There is a discussion of the possible impacts of the proposed pipeline on the aquatic habitat. Volume II contains a lake and stream catalogue.

#100
Foothills Pipe Lines (Yukon) Ltd. 1977. A spring inventory of fishery resources along the proposed Alaska Highway Pipeline in Yukon Territory, 1977. Prepared by Beak Consultants Ltd., Calgary. 54 + pages.

Abstract

An inventory of the fish fauna in watercourses crossed by the proposed pipeline was conducted from April 23 to June 3, 1977. Watercourses were visited and sampled to determine fish fauna and degree of fish utilization. Waterbodies which exhibited fishery protential were visited on an average of seven times.

Watercourses which were found to possess arctic grayling spawning locations at or below the pipeline corridor were: Beaver Creek, Koidern River, Edith Creek, Lake Creek, Donjek River, Jarvis River, Marshall Creek, Aishihik River, Ibex River, Fish Creek, Wolf Creek, Cowley Creek, Yukon River, Squanga Creek, and Morley River. Smart River was judged likely to contain spawning grounds in the vicinity of the pipeline as were Deadman Creek, Lone Tree Creek, and Rancheria River. Populations of migrating arctic grayling were found in Swede Johnson Creek, an unnamed creek (Pipeline Mile 37.8) and Fox Creek. White River, Koidern River, Donjek River, Takhini River, inlet of Squanga Lake (Pipeline Mile 322.5) and the Teslin River were judged likely to support grayling spring migrators. Arctic grayling rearing areas were identified in Beaver Creek, Koidern River, Swede Johnson Creek, Jarvis River, Aishihik River, Cracker Creek, Takhini River, Wolf Creek (Pipeline Mile 277.9), Cowley Creek, M'Clintock River, Squanga Creek, Smart River and Swift River. The Donjek River, Marshall Creek and Ibex River were judged likely to support rearing areas.

Watercourses found to be utilized by chinook salmon parr were: Takhini River, Ibex River, Wolf Creek, Cowley Creek, Brooks Brook, Lone Tree Creek, Ten Mile Creek, Tax Creek, Hays Creek, Strawberry Creek, Morley River, and Logjam Creek.

No chum salmon were encountered during the spring investigation.

Likely fish overwintering areas were documented at White, Koidern, Donjek and M'Clintock Rivers.

Dolly varden were found to populate Beaver Creek, Smart River and Boulder Creek. Round whitefish used Cracker Creek and M'Clintock River as rearing areas. Lake whitefish were found in the inlet of Squanga Lake. Edith Creek, Lake Creek, Wolf Creek and Cowley Creek were used by longnose suckers for spawning.

#101
Foothills Pipe Lines (Yukon) Ltd. 1977. A summer inventory of the fishery resource along the proposed Alaska Highway pipeline in the Yukon Territory, 1977. Prepared by Beak Consultants Ltd. Calgary. 44 + pages.

Abstract

An inventory of the fish fauna in watercourses within the influence of the proposed pipeline was conducted from July 4 to August 8, 1977. The purpose of the investigation was to document fishery utilization of the aquatic environment during the summer season with emphasis on the identification of nursery, rearing and feeding areas, as well as potential fish overwintering areas in watercourses directly affected by the proposed pipeline alignment. Each watercourse was visited at least once, with larger streams and rivers being visited on two to seven occasions. A variety of methods were used to capture fish, depending on the nature of the watercourse. Sub-samples of fish were weighed, measured and sexed. The reproductive condition of the fish was recorded, fish stomach contents were recorded and scales were collected to be used for age determination.

The results of the study revealed that summer fishery utilization of watercourses was highly variable. Arctic grayling were collected in many watercourses and were the most numerous of all species collected. Chinook salmon parr, lake trout, lake whitefish, round whitefish, broad whitefish, least cisco, mountain whitefish, dolly varden, northern pike, longnose sucker, inconnu and burbot were also observed. A summary of summer fish utilization at proposed pipeline crossing locations is presented in a table. Chemical and physical characteristics of waterbodies and fish presence information are presented in data summary sheets.

#102
Foothills Pipe Lines (Yukon) Ltd. 1977. A Survey of Fall Spawning Fish Species in Waterbodies Within the Influence of the Proposed Alaska Highway Pipeline in Yukon Territory, 1977. Prepared by Beak Consultants. 40 pages.

Abstract

An inventory of the fish fauna in selected watercourses within the influence of the proposed pipeline was conducted from August 20 to November 10, 1977. The study was intended to identify waterbodies used by fall spawning fish populations and to determine migration times, routes and spawning areas that might be impacted by the proposed pipeline. Potential overwintering areas were also identified. The emphasis of the study was on chinook and chum salmon. Watercourses examined during this investigation were those which were utilized by fall spawning species as determined from previous studies. A summary of salmon utilization in the investigated area, is presented in chart form. In the southern section of the Yukon Territory chinook spawning activity peaked near the end of August. Chum salmon were first observed near spawning sites in southern Yukon on September 27 and spawning activity peaked about October 22.

In Bear Creek, dolly varden spawning activity began in early September. In Kluane Lake, lake trout spawning appeared to have occurred from mid-September to late October but no spawning sites were located during this investigation.

In Koidern River, lake whitefish and round whitefish spawned in October. Spawning in Kluane began in late October and whitefish spawned along the shore. Broad whitefish were found in Teslin Lake and Nisutlin Bay. No spawning areas were identified in these waterbodies. In early September broad whitefish congregated near spawning areas in Nisutlin River.

#103
Foothills Pipe Lines (Yukon) Ltd. 1976. Application to the National Energy Board. Public interest volume 5B-1 Environmental statement. 537 + pages.

Abstract

The environmental statement addresses the various physical and biological components of the proposed Alaska Highway pipeline route and adjacent areas which may be affected by the project. The report is divided into six chapters: introduction, conclusions, project description, environmental setting, environmental concerns and protection measures and impact prediction. The chapters are subdivided into eight subjects: terrain and hydrology, air and water quality, archaeology, aesthetics and land use, vegetation, aquatic systems, birds and mammals. The section on aquatics systems include a partial inventory of fish in waterbodies crossed by the pipeline, a discussion of possible environmental disruptions caused by the pipeline and a list of proposed measures to prevent these disruptions. The fisheries material included is a summary of information from existing studies and there is very little site specific data.

#104
Foothills Oil Pipe Line Ltd. 1979. The Alaska Highway Oil Pipeline Project. Amended Application Volume 4 and 5 - Part E - Environment. Vol. 4-492 + pages, Vol. 5-767 pages.

Abstract

Volume four presents a description of the project setting, construction, and environmental setting, including a section on fisheries resources which incorporates information from Foothills and Fisheries reports and includes summaries of location, life histories and habitat utilization for chinook salmon, chum salmon, Arctic grayling, lake trout, lake whitefish and Dolly Varden char.

Volume five discusses the environmental impacts of the pipeline and suggests mitigative measures, including a section on pipeline impacts on the fishery resource.

#105
Foothills Oil Pipe Lines Ltd. The Alaska Highway Oil Pipeline Project. Volume 6 - Environmental Atlas.

Abstract

Includes maps showing the Alaska Highway Oil Pipeline route and surrounding area. One set of maps illustrates the various types of terrain found along the pipeline route, another map set shows land use and forest cover and a third map set includes fishery and wildlife information. Fish species and habitat utilization along the pipeline are illustrated.

#106
Garrett, C.L., L.A. MacLeod, H.J. Sneddon, 1980. Mercury in the British Columbia and Yukon Environments - summary of data to January 1, 1979. Environmental Protection Service. Regional Report. 80-4, 456 pages.

Abstract

The report provides an overview of the uses and environmental levels of mercury in British Columbia and the Yukon. The main objectives of ths report were to identify current uses of mercury in the Pacific Region and potential sources of release to the environment; assess the potential for environmental impacts as a result of these releases; document all existing information on mercury levels in the British Columbia and Yukon environments and compare these levels to those reported for other areas of the world; and identify localized areas of particularly high contamination.

Data was obtained from government agencies, universities and research organizations.

Mercury concentrations of fish sampled in the Yukon Territory are summarized in a chart. Within the Yukon River Basin the sampled areas include; Klondike River, Yukon River at Dawson Ferry, South McQuestern River, Mayo Lake, South MacMillan River, Mt. Nanson, Little Salmon Lake at Drury Creek, Pelly River at Anvil Creek, Rose Creek and Pelly River near Faro, Tagish Lake, Little Atlin Lake, Squanga Lake, Quiet Lake, Bonanza Creek, Kluane Lake, Fox Lake, Bennett Lake, Lake Laberge, Shwatka Lake, Francis Lake and Teslin Lake. A map illustrates the mercury concentration levels of fish in the Yukon. A general discussion of the possible effects of mercury on fish is included.

#107
Gordon, R. N., R. A. Crouter and J. S. Nelson, 1960. The fish facilities of the Whitehorse Rapids power development, Yukon Territory, 1960. Can. Fish. Cult. 27: 43-56.

Abstract

This paper includes a description of the Whitehorse Rapids Power Development which came into service in June of 1959. There are also descriptions of the fish facilities present at the powerhouse and the spillway dam. Results of the spring salmon study in 1959 at the facilities are given. The total escapement from July 28 - September 24 was 1,054. Attempts to tag were unsuccessful and no measure of overall delay was obtained. Attraction of the salmon into the fishway, as indicated by a lack of any significant accumulation in the area immediately downstream, was very good. the pattern of migration as well as the general condition of the salmon indicate that migration was not seriously affected. Large numbers of grayling were reported using the fishway between June 26 and July 17 - this may have been a spawning migration. Lake trout, least cisco, round whitefish, northern pike and longnose suckers were also sighted at the facilities. Aerial spawning surveys of the M'Clintock River system and Michie Creek failed to locate any salmon.

#108

Hardy Assocs. Ltd., 1981. Fish and wildlife habitat recovery in the placer mined areas of the Yukon. Hardy Associates (1978) Ltd., Vancouver, B.C. Prepared for Department of Indian Affairs and Northern Development. 131 pages.

Abstract

This project was initiated to determine the rate and degree of natural fish and wildlife habitat recovery in placer mined areas of Yukon. Methods for enhancement of wildlife and fish habitat are suggested for both operating and abandoned placer mining areas. The study areas were the Sixty Mile area, Dawson area, Mayo area, Thistle Creek area, and the area south of the Indian/Klondike River divide. Placer mined sites were characterized by method of mining and age since abandonment. The sites were then assessed for their capability for supporting selected species of fish and wildlife - Arctic grayling was selected as the indicator species for fish habitat capability assessments. Fish habitats were assesed for their physical characteristics, water quality and benthic community parameters. Physical characteristic parameters such as velocity, substrate and channel characteristics were found to be the greatest limitation to grayling habitat recovery in the study sites. From 29 to 72 years were found to be required to restore the physical habitat of placer mined wide valleys to control levels and, in narrow valleys, no trends of physical habitat recovery were observed. Water quality in wide valleys was found to return to control levels in approximately 20 years whereas no predictable trend to recovery was observed in data from narrow valleys. Benthic community densities in wide and narrow valley streams recovered in approximately 5 years, however, the narrow valley streams did not respond to control diversities and became dominated by sediment tolerant organisms.

#109
Hardy Associates Ltd. 1980. Kluane Lake hydrometric and geophysical surveys Kluane Lake, Yukon Territory. Prepared for Foothills Pipe Lines (Yukon) Ltd.

Abstract

A hydrometric survey of Kluane Lake was made to obtain information on bottom contours, sub bottom characteristics, water velocities and temperatures for Foothills, as an aid to designing a Kluane pipeline crossing. Phase I of the study includes a benthymetric map of the southern portion of Kluane Lake. Phase II consists of a sub bottom seismic survey, sidescan survey and water current and temperature measurements. This report contains no fisheries data.

#110
Hawley, C.C. and Associates (D. Weir), 1981. A comparative study of the use of disturbed and undisturbed areas by wild vertebrates in the Klondike, Yukon Territory with recommendations for placer mining reclamation practices. Prepared for Klondike Placer Miners Association. 59 + pages.

Abstract

The aim of the study was to investigate the consequences to wildlife of placer mining in the Klondike and to suggest reclamation and mitigation practices. The method was to compare wildlife populations in disturbed and undisturbed areas.

Aquatic investigations were conducted along three tributaries of Dominion Creek namely: Jensen Creek (an active mining area), Gold Run Creek (an abandoned mining area and Burnham Creek (undisturbed). In addition Indian River, Dominion Creek and several tributaries were briefly visited. Water samples and fish samples were taken, information on riparian vegetation, turbidity fish stomach contents and parasites was recorded.

An aerial survey was conducted to locate potential barriers to fish movement and to document the extent of fish utilization in the study streams.

Grayling, round whitefish and slimy sculpins were found in the Dominion Creek drainage. Previously disturbed reaches of Jensen and Gold Run Creek provided feeding and rearing areas for young of the year, juvenile grayling and slimy sculpin. It is suggested that reclaimed areas may provide better habitat than undisturbed or actively disturbed areas. The effects of turbidity on fish spawning and feeding habits in Indian River and Dominion Creek are discussed. Recommendations relating to reclamation procedures are made and it is suggested that more studies of fish and wildlife of disturbed and undisturbed areas are needed to assess the impact of mining monitor the effectiveness of reclamation and refine reclamation practices.

#111
Hoos, R.A.W. and W.N. Holman. 1973. A preliminary assessment of the effects of Anvil Mine on the environmental quality of Rose Creek, Yukon. Environmental Protection Service Surveillance Report EPS. 5-PR-73-8. 49 pages.

Abstract

A field survey was carried out in September, 1973 to: 1. determine the water quality conditions at key sites in Rose and Anvil Creeks and Pelly River; 2. assess the benthos community characteristics; 3. determine the physical and chemical constituents of the sediments and; 4. determine fish species composition and trace heavy metal levels in fish. Gill nets, seine nets and angling techniques were used at all stations but only one fish, a grayling, was caught. It is suggested that a more intensive fisheries survey be carried out, utilizing electrofishing. There is a discussion of the detrimental effects of the Faro Creek diversion of Ross and Anvil Creeks and the influence of high sediment content on graylings is noted, with reference to previous studies.

#112

Horler, A. 1982. On estimating the productivity of select Yukon lakes using the morphoedaphic index Yukon lake survey 1981. Draft. Department of Fisheries and Oceans, Whitehorse, Yukon. 35 pages.

Abstract

The morphoedaphic index (M.E.I.) was calculated for fifty-two Yukon lakes, and used as described for north-termperate lakes to estimate annual productivity. Lakes were selected either because they were accessible by road, because they were known to be used by sports fishermen, or because there was an application to lease land for commercial purposes (ie. fly-in fishing lodges) at the lake site.

A list of known species in each lake under study is presented in tabular form.

Metric M.E.I. values for the lakes under study ranged from 0.79 to 54.5. Productivity ranged from 1.25 kg/ha to 8.34 kg/ha. Applications and limitations of the productivity estimates are discussed. The results of the study indicate a relatively low level of productivity. Recommendations are made for further studies; to evaluate the accuracy of the M.E.I. as applied to the Yukon, investigate fish species composition and growth rates in Yukon lakes, and to identify lakes which are in danger of overfishing.

Bathymetric maps for all lakes studied are included.

#113
Horler, A., 1980. Arctic Grayling (Thymallus arcticus) Investigations. Whitehorse, Yukon. Unpublished. 75 pages.

Abstract

The objectives of this study were to gain some understanding of the conditions under which grayling could be spawned, the eggs incubated and the young fry handled, when removed from their natural habitat. A feeding system for the fry had to be determined which would ensure a growth rate similar to that of fry in the wild. The grayling for the study came from Richthofen Creek.

This study was not successful because the fish used died off so quickly that there were too few left to proceed with the study. A feeding system for the fry was not determined. In this case commercial starter mash was used and found unsatisfactory as the particle size is too large for the smaller fry. The feeding of live food should be considered as an alternative.

#114

Interdisciplinary Systems Limited, 1978. Initial impact assessment Dempster Corridor, physical and biological environments - Dempster and Klondike segment, background report 1, North Delta segment, background report II. Alaska Highway Pipeline, Winnipeg, Manitoba.

Abstract

This report presents information on the possible impacts of the Dempster Road and Dempster Lateral pipeline on the physical and biological environment. Volume I covers the area from the north end of Campbell Lake, 22 km southwest of Inuvik, south to a point 32 km east of Dawson City and along the Klondike Highway south to a point outside Whitehorse. Volume II covers the segment from Richards Islands in the McKenzie Delta, along the east side of the delta, running south past Inuvik.

The fisheries section of the report lists the fish known to exist along the corridor and outlines possible problems that may be caused by the pipeline. Eight species (chum, chinook, inconnu, arctic char, lake trout, whitefish, grayling, arctic cisco) are selected on the basis of their sensitivity to disturbance, ability to recover from disturbances and economic importance to man. These eight species are used as key species in the report, to indicate potential effects of highway operation and pipeline development. Habitat utilization and potential impacts on fisheries from highway operation and pipeline development are listed. Some mitigative measures are suggested.

The following additional surveys are recommended:

(1) survey to identify chum spawning areas in Klondike and McQuesten River;

(2) field survey to identify inconnu spawning and overwintering areas, particularly in the Klondike section of the corridor;

(3) field surveys to locate spawning and overwintering areas near pipeline crossings;

(4) field surveys to identify whitefish spawning areas and possible spawning migrations;

(5) field surveys to locate spawning and overwintering near borrow sites.

#115
Interdisciplinary Systems Ltd., 1977. Initial environmental evaluation of the proposed Alaska Highway Gas Pipeline - Yukon Territory. Report prepared for Alaska Highway Pipeline Panel, May, 1977. 690 pages.

Abstract

This initial environmental evaluation reviews, for the Yukon section of the proposed pipeline, the existing physical, biological, and human environmental settings, identifies major environmental problem areas, and provides preliminary recommendations for controls and mitigative measures. Project-induced stream sedimentation could affect up to 280 km. but potential sedimentation could be reduced 20-40 percent with effective controls. Adverse effects on water quality would be caused by wastewater and potential accidents and spills. Effects on permafrost soils include thaw settlement and frost heave. Additional terrain problems include erosion and slope instability. An estimated 3-10 percent of available fish habitat in the corridor could be potentially degraded by varying amounts of siltation in streams. Mitigative measures as they apply to chinook salmon, chum salmon, dolly varden, lake trout, whitefish and arctic grayling are presented. Little site-specific information is included.

#116
Jack, M.E. 1981. Baseline study of the watershed near Venus Mine, Yukon and Venus Mill, British Columbia. Environmental Protection Service. Regional Program Report. No 81-18.

Abstract

A study was made of the watershed near Venus mine (site of a former mining and milling operation) in the Yukon, and Venus mill in British Columbia in 1980 before their planned development and start of operation in June 1981. Water quality parameters, were documented at twenty three sampling stations. These included streams and the lake adjacent to an abandoned tailings pond from which Venus plans to recover the tailings, mine water, a creek into which mill tailings ponds will discharge, a control creek and the southern end of Windy Arm (Tagish Lake) into which the creek receiving tailings pond discharge flows.

The environmental quality of the unaltered part of the water-shed was similar to other unaltered lakes and creeks in the Yukon River Basin.

The environmental quality near the abandoned tailings pond continues to be adversely affected by this past mining operation. Significantly higher concentrations of cadmium, lead and zinc were found in sediment samples in the lake near the abandoned tailings pond than in sediment samples from unaltered parts of the watershed. Arsenic concentrations in abandoned tailings pond drainage and mine water were significantly higher than those in water in the unaltered part of the watershed. The report contains no fisheries information.

#117

Johnston, R.A.C. 1981. Distribution and abundance of chum salmon (Oncorhynchus keta) in the Upper Yukon river system in 1978, as determined by a tagging program. Unpublished Canada, Department of Fisheries and Oceans, Whitehorse, Yukon.

Abstract

A total of 427 chum salmon were tagged at the mouth of the Forty-mile river, forty-six miles below Dawson City, in 1978. The salmon were captured by fishwheels, their lengths and sex were recorded, they were tagged with Peterson discs, and released. Rewards were offered for the returned tags. Fisheries and Katimavik personnel operated seven gillnets upstream of Dawson City from August 27 to October 13, to sample the chum runs for relative abundance, size and age composition and tagged /untagged ratios. Sampling sites were situated on the Stewart and White Rivers above the confluence of these rivers with the Yukon River. A third site was located on the Yukon River at Minto Landing.

Chum salmon utilize the Stewart, White, Pelly and mainstem Yukon as far up as Hootalinqua. The tag recovery program revealed that there was a lack of preferential bank orientation. Population estimates were 8,291 chum to reach the mouth of the Forty-Mile River, of which only 3,495 were able to reach spawning grounds upstream from Minto Landing. Information on sizes, weights, and age compositions of chum salmon are given.

#118
Kelso, B.W., W. Robson, K. Weagle. 1977. An assessment of the pre-development water quality and biological conditions in the watershed around the Minto ore body. Environmental Protection Service Manuscript Report 77-9. 29 pages.

Abstract

In June of 1975 and September 1976 a baseline water quality study was conducted on several creeks draining the proposed copper mining area known as the Minto deposit which is situated approximately 150 miles north of Whitehorse. Data was collected on water quality, benthic macroinvertebrates and fish species. Electrofishing with barrier nets downstream was used to collect fish. Slimy sculpin, arctic grayling and chinook salmon were found in Big Creek and Wolverine Creek. The authors of the report recommend that a more detailed study should be made of fish populations in the area, including work on salmon spawning and rearing areas in the Yukon River.

#119
Kendel, R.E. 1973. The Effect of the hydro dam on the Mayo River fish stocks. Unpublished. Department of the Environment, Fisheries Service, Whitehorse. File 5430-82-12. Mayo Dam. 12 pages.

Abstract

In 1954 a hydro power dam was placed on the Mayo River. The structure poses a complete barrier to upstream migration of fish from the Stewart River. Chinook salmon that had spawned above the dam are now reported to spawn in small numbers below, in the short section of river between the Stewart River and the dam. The object of the study was to assess the situation created by the dam and acquire information about other fish species that use the stream. Sampling stations were located in various areas between the dam and the Stewart River and also in the Mayo River above the dam, at Roop River and at Duncan Creek.

Seine nets and mesh nets were used to capture fish and species, length, weight and sex were recorded for most samples collected. These are listed in a chart. The species observed in the Mayo River between the dam and the Stewart River were chinook salmon fry, Arctic grayling, inconnu, lake whitefish, round whitefish, lake trout, longnose sucker, northern pike, lake chum and slimy sculpin. Interviews with local residents revealed that the chinook salmon spawning stock has been reduced to a fragment of the original stock. The effects of the dam on fish populations are discussed.

#120
Kendel, R. E., 1973. Growth and survival of rainbow trout (*Salmo gairdneri*) and coho salmon (*Oncorhynchus kisutch*) in four pothole lakes of the Yukon Territory in 1972. D.O.E., Fish. Mar. Ser. Rep. No. PAC/T-73-7.

Abstract

Four lakes near Whitehorse which were low in oxygen during the winter were selected for planting with young rainbow trout and coho salmon. The mortality of coho salmon was greater than the mortality of trout during transport to Whitehorse. The mortality of coho salmon was also greater in the lakes. There appeared to be an inverse relationship between the initial planting density and growth of rainbow trout. Gill nets were effective in harvesting the fish. There was a good market for the harvested trout in restaurants and supermarkets in Whitehorse.

#121
Klohn Leonoff, 1979. Adanac Mine - Environmental and Socio- economic assessment. Prepared for Placer Development Ltd., Vancouver.

Abstract

Development of the Adanac Mine, near Atlin, B.C., has been proposed by Placer Development Limited. This report describes the existing environmental setting, provides a detailed description of the proposed project development, and assesses the probable environmental and socio-economic impacts, including proposed mitigation procedures. Descriptions of the populations and habitats of the fish resources of the area have been based on a review of existing data and previous sampling surveys and two field investigations carried out during May and July 1979. Additional data collection will continue during late 1979. Sampling programs were carried out in Surprise Lake (near the outlets of Ruby, Otter, Cracker and Cup Creeks), in Ruby Creek 1-3 km. above the mouth, in Pine Creek at the outlet of Surprise Lake and 3 km. downstream, and in Molly Lake. Fish populations were sampled by monofilament gill nets and by electrofishing. Some fish were retained for detailed analysis including fork length, weight, sex, gut contents, and state of maturity. The only fish species identified in the sampling programs were Arctic grayling and slimy sculpin. Grayling are predominant in Surprise Lake but slimy sculpin were found near the outlet to Pine Creek. During the spawning run in May the grayling population appeared to be concentrated in the immediate viscinity of the inflowing creeks and the outlet of Surprise Lake. Comparison of the total number of ripe or spent fish caught, the catch-per-unit effort, and the fish stomach contents suggested that, of the sites sampled, Cup Creek was the most important spawning area, followed by Cracker, Pine, Otter and Ruby Creeks in order of decreasing importance. No fish were observed or collected in Ruby Creek upstream of the mouth, although high water levels in Surprise Lake may encourage grayling to move upstream short distances (a few hundred meters) to feed. It appears that the moderately high gradient of Ruby Creek in its middle reaches acts as a barrier to fish movement from Surprise Lake. There was no evidence of a fish population in Molly Lake. Final determination of impacts upon fishery spawning and habitat areas will depend upon the adopted method of plant operation, and the resulting magnitude and timing of lake level fluctuation.

#122
Knapp, W., 1976. Yukon placer mining study. Department of the Environment. Fisheries Service, Vancouver, B.C. (unpublished report).

Abstract

This report is a presentation of the results of the 1974 and 1975 placer mining studies in the Yukon. No new data is given but there is a discussion of the two studies. A bibliography on the effects of suspended sediments on stream life is included in the report.

#123
Knapp, W. 1975. Summary and analysis of fish capture data from placer mining study, 1975. Environment Canada, Fisheries Service. 15 pages (unpublished memorandum report).

Abstract

A comparative study was made of fish populations and habitat characteristics in Bonanza and Sulphur Creeks, during the summer of 1975. Stream temperature data, water samples, daily precipitation records and benthos samples were collected. Fish counting fences were installed in the lower reaches of Bonanza and Sulphur Creeks to monitor fish population and migrational behaviour data from June to mid-October. The results are discussed, and the effects of heavy sedimentation and strip mining on fish populations of the two streams are documentd. The resident fish population of Sulphur Creek, during the ice-free season was found to be several times larger than that of Bonanza Creek. Those fish resident in Bonanza Creek throughout the summer reside primarily in the relatively clear section of Bonanza Creek above Grand Forks or in the clear areas adjacent to the mouths of unmined tributaries. This difference in fish residency in the two streams is considered to be a result of the fact that the turbidity levels of Bonanza Creek were several orders of magnitude higher than those in Sulphur Creek. Secondly, fish move upstream and downstream throughout the summer in Sulphur Creek whereas those fish, especially grayling and whitefish on Bonanza Creek, exhibit downstream migration with very few fish moving upstream. Third, Sulpher Crek has a much higher proportion of sports fish (grayling and whitefish) as compared to the coarser fish (suckers) than is evident on Bonanza Creek. Finally, there is a general trend on both creeks for fish to begin moving downstream in the fall. However, in the case of Sulphur Creek, this downstream migration consists of mainly grayling and whitefish, whereas on Bonanza Creek it primarily consists of suckers.

#124
Knapp, W. MS., 1974. Summary and analysis of sediment and preliminary biological data from placer mining operations - 1974. Fisheries Service, Vancouver, B.C. (Unpublished Memorandum Report.)

Abstract

During the period of May - October, 1974, four sampling surveys were conducted on streams supporting active placer mines. The surveys were centered on Bonanza and Hunker Creeks, tributary to the Klondike River, and Dominion and Sulphur Creeks, tributary to the Indian River. Several other creeks were also sampled, mostly small tributaries of these creeks, and Flat Creek, a tributary to the Klondike River. Two sites were also located on the Klondike River. In each survey sampling was carried out for bottom fauna and fish and the physical/chemical conditions of the streams were monitored. Presence of fish was determined by visual observations, gill netting, electroshocking and angling. Grayling, whitefish and suckers were the only species noted. Maps of mining sites and sampling sites are included, and all data is tabulated. Captured fish were sampled for scale, length and sex and the stomach contents were analyzed. Sulphur Creek supports more fish of all life stages, has better water quality and has a greater variety of fish food organisms available all because of man-made and natural settling capability. This reinforces the wisdom of requiring the provision of settling ponds for all placer mining operations.

#125
Landucci, J.M., 1978. An environmental assessment of the effects of Cassiar Asbestos Corporation on Clinton Creek, Yukon Territory. E.P.S. Regional Program Report No. 79-13. 36 pages.

Abstract

In the summers of 1974 and 1975 chemical and biological studies of Clinton Creek (located northwest of Dawson City) were undertaken. Fish were sampled, using electrofishers and barrier nets, and bioassays were done, using juvenile coho salmon raised in an aquarium full of water from Hudgeon Lake near the mine pumphouse.

The most numerous fish species found in the area included longnose sucker and arctic grayling. Two chinook salmon were captured on the Forty Mile River, above the confluence of Clinton Creek and Forty Mile River. Other species found in the area include lake whitefish, round whitefish, and slimy sculpins.

The results of the study indicate significant adverse effects on the invertebrate communities and fish populations as a result of substrate alteration and the presence of asbestos fibres in the vicinity of the mine.

#126
Lebida, R., 1972. Yukon Territory king salmon studies, 1971. Alaska Department of Fish and Game, Anchorage, Alaska. 11 pages.

Abstract

From July 26 to September 4, chinook salmon data was gathered at the Whitehorse Dam fishway. Aerial surveys were also conducted to assess chinook salmon escapement magnitude in major spawning systems. Approximately 43% of the run was sampled for age, sex and size information. A total of 856 chinook salmon were counted. Males were most abundant in the 5_2 age group while females dominated the 6_2 age group. A .0:1.1 sex ratio in favour of females was evidenced. Males preceded females on their migration through the fishway. Information is presented in tables. A discussion of the results follows.

#127
Leverton, D. 1982. (in preparation). Ecology and biogeography of the headwater lakes region. Gladstone and Isaac Creek, Ruby Mountains, Yukon Territory. MSC Thesis, Queen's University, Kingston Ontario.

Abstract

Fish inventories were made and physical parameters (e.g. water temperature, water quality) collected. Benthic invertebrates, insects and zooplankton were collected in order to monitor the feeding behaviour of fish species. Growth rates were determined and an attempt was made to relate these to seasonal temperatures. Gillnets were set at various times of the day in order to establish diurnal feeding movements. Fish samples from within the study area were compared with samples taken in other regions in a discussion of post-glacial dispersal patterns.

Note: This information was gathered in a phone conversation with the author. The report is in preparation and has not been seen.

#128
Lindsey, C.C., 1964. Problems in zoogeography of the lake trout, *Salvelinus namaychush*. J.F.R.B.C. 21-5: 977-994.

Abstract

The lake trout poses a distributional problem unique among North American fishes. It is the only freshwater species that ranges into the far north of Canada and Alaska, but that does not extend westward across Bering Strait into Siberia. This paper assembles the available evidence concerning lake trout zoogeography, and attempts to assess the conflicting hypotheses concerning where the lake trout originated and why it is confined to its present range. An appendix of the sources of locality records of *Salvelinus namaychush* is included.

#129
Lindsey, C.C., 1963. Sympatric occurence of two species of humpback whitefish in Squanga Lake, Y.T. J. Fish. Res. Bd. Can. 20(3) 749-767

Abstract

This study was launched in June, 1960 with over 1200 fish specimens being collected over 2 weeks with gillnets and seines. Information was obtained on depth distribution and feeding of humpback whitefish. The lake contains two species of humpback whitefish, characterized by medal first arch gill raker counts of 23 and 28. The species are separated more sharply by raker counts of the second arch, also with modes of 23 and 28. The high count species spawns in inlet and outlet streams - samples of spawning whitefish were obtained from Squanga Creek below Squanga Lake (November 5 and 24, 1960) and from the creek below Little Squanga Lake (November 28, 1960). Gonad development in June suggests that the low count form spawns later, in localities unknown. In addition to these whitefish, burbot and northern pike were found in the lake. Arctic grayling were largely confined to the streams.

#130
Lindsey, C.C., K. Patalas, R.A. Bodaly and C.P. Archibald. 1981. Glaciation and the physical, chemical and biological limnology of Yukon lakes. D.F.O., Canadian Technical Report of Fisheries and Aquatic Sciences, no. 966. 37 pages.

Abstract

A limnological study of ninety one Yukon lakes within the Alsek, Liard, Peel and Yukon drainage basins was conducted from 1960 to 1978. Data was collected on lake morphometry, temperature, chemical composition, zooplankton species and abundance and fish species presence. Twenty three fish species were sampled, using a gillnet. Part of the drainage basins of the Alsek and Liard drained into the Yukon basin during deglaciation and these areas received much of their fauna from the Yukon river basin. Lake whitefish electrophoretic patterns in Alsek, Upper Liard and Peel basin populations are similar to populations found in the Yukon. Inconnu, broad whitefish and least cisco are found in the Yukon River, and are excluded from the Alsek, Liard and Peel system. There is a discussion of the way in which post-glacial alterations in drainage patterns have affected the distribution of fish species in Yukon lakes.

#131
Lister, D.B. 1960. Whitehorse rapids power development 1960 assessment program (Unpublished report). Department of Fisheries and Oceans, Canada. 15 + pages.

Abstract

The 1960 assessment program included enumeration of the total numbers of fish using the fishway, a chinook salmon spawning ground survey, including sampling for fry at various locations and an attempt to determine the level of fish turbine mortality by capturing, marking, and recapturing fish moving downstream through the turbine.

The methods and results for all components of the study are detailed.

The principal species using the fishway were chinook salmon, Arctic grayling and suckers.

Graphs are included indicating the daily and cumulative migrations of chinook salmon, and the migration of grayling through the fishway.

#132

Mathers, J. F., N. O. West and B. E. Burns, 1981. Aquatic and wildlife resources of seven Yukon streams to placer mining. Government of Canada, Departments of Fisheries and Oceans, Indian & Northern Affairs and Environment. 183 pages and 175 pages appendix.

Abstract

Aquatic and wildlife resources of Yukon streams subject to placer mining were surveyed in March, May, July and September, 1980. Seven streams (South Big Salmon River, Big Creek, Thistle Creek, Scroggie Creek, Clear Creek, Johnson Creek, and Duncan Creek) in the south central Yukon Territory ranging from small tributary creeks to major rivers were examined. Placer mining activity observed in the study area included sluicing, dredging and hydraulic mining. For each system, data on water quality, benthic invertebrates, fish distribution, relative abundance, vital statistics, food habits, wildlife distribution and habitat utilization were collected.

Suspended sediment concentrations in areas not disturbed by placer mining ranged from <5 mg/1 to 100 mg/1 depending on stream discharge. Downstream of most placer mines, suspended sediment levels were between 1,000 and 4,000 mg/1 during sluicing. In general, numbers of benthic invertebrates were lower below placer mines compared to other similar areas. Arctic grayling (Thymallus arcticus) were found in all streams studied; however, spawning and rearing were apparently limited in streams with high sediment loads. Dipterans, trichopterans, ephemeropterans, and plecopterans were most common in grayling stomach samples. Age, growth and sex ratios of grayling were similar to other northern populations. Additional fish species found in the streams were slimy sculpin (Cottus cognatus), chinook salmon (Oncorhynchus tshawytscha), round whitefish (Prosopium cylindraceum), burbot (Lota lota), and longnose sucker (Catostomus catostomus). Moose habitat quality in all stream valleys surveyed was very good to excellent; however, the amount of habitat in some areas was limited by valley size and mining activity. The number of bird species found in a valley was related to the amount of riparian habitat available. A stream classification scheme based on the resources present and the extent of mining activity was developed.

#133
McAllister, D.E. and E.J. Crossman. 1973. Guide to freshwater sport fishes of Canada. National Museum of Canada, Nat. History Ser. No. 1. 89 pages.

Abstract

Fifty three species of freshwater fish found in Canada and considered to be sought after by anglers, are included in this report. The book is intended to help the angler identify the fish he has caught. The figures and brief text indicate its maximum size, describes its habits and the usual angling methods used to catch it. Maps provide information on distribution. Supplementary material on making fish prints, knots, cleaning and cutting fish and recipes is included. Information on angling licences and regulations, travel and maps, canoeing and boating are included.

#134
McComas, T., 1980. Department of Environment, Department of Fisheries and Oceans Task Force, Northern Pipelines Program, Vancouver, B.C. 95 pages.

Abstract

The aim of this report is to evaluate existing knowledge and concerns associated with the potential environmental impacts of oil spills on northern freshwater ecosystems, and the construction of 2 pipelines (gas and oil) within the Alaska Highway corridor. For this analysis, it has been assumed that the two pipelines will not be built simultaneously, because of resulting shortages of labour, materials, equipment, and capital. Project design and scheduling is examined, and potential concerns at Beaver Creek, Pickhandle Lake, the West shore of Kluane Lake, Ibex pass, Squanga Lake, Rancheria, and Swift River are examined. In areas where more than one route alternative has been suggested, a gas pipeline route should not be chosen without assurances that the same route will be used for the oil pipeline. Areas where adverse impacts on birds, fish, and mammals are expected are outlined. Gravel is an important resource in the Yukon Territory, particularly in the Beaver Creek section of the right of way. Therefore, sufficient borrow material should be located for both projects prior to construcion. The report contains an annotated bibliography on oil spills in northern environments. The bibliography deals with: spreading and degradation of crude oil; cleanup techniques; effects on flora, aquatic fauna, terrestrial fauna, birds, and the ecosystem.

#135
McPhail, J.D. and C.C. Lindsey. 1970. Freshwater fishes of northwestern Canada and Alaska. Bull. Fish. Res. Bd. Canada, no. 173. 381 pages.

Abstract

This report begins by outlining the geological history, dispersal patterns and zoogeographical patterns of freshwater and anadromous fish in northwestern Canada and Alaska. The bulk of the report contains species accounts, including description, distribution, biology and taxanomic notes. An excellent study.

#136
McPhail, J. D. and C. C. Lindsey, 1958. Preliminary list of fish collections in the Institute of Fisheries from Alaska, Yukon River and the Gulf of Alaska drainages. Mimeo list. Inst. Fish., U.B.C. 16 pages.

Abstract

This publication consists of a list of all species of fish which were found in the Alaska, Yukon River and Gulf of Alaska drainages. For each species the rivers and lakes which it was found in are listed. Following this is a list of map references and Institute of Fisheries Museum collection numbers.

#137
Monenco Consultants Ltd., 1982. Aquatic studies at MacMillan Pass, Yukon Territory. Prepared for Pan Ocean Oil Ltd., Calgary. 100 + pages.

Abstract

This baseline environmental survey of Pan Ocean Oil Ltd.'s Jason property at MacMillan Pass took place during June, August and October, 1981. The study emphasized the description of stream habitat in the area and investigated the fish, benthic macroinvertebrates, and periphyton communities in the South MacMillan River and other smaller streams. Chemical determination of trace elements in the river bed sediment was also carried out. Pan Ocean Oil Ltd. supplied laboratory analyses of the water from the South MacMillan River and its tributaries as well as from the various tributaries of the Hess River draining the Jason Property.

The fish community was sampled by electroshocking. No population estimates were made. The benthic invertebrate community was sampled by Surber and artificial substrate samplers. Attached algae from rock surfaces were also collected. Both the density of individuals and the diversity of species within the benthic macroinvertebrate communities of the South MacMillan River and Hess tributary were found to be low. The same was fourd to be true for the periphyton community. In both cases, a significant decrease in density occurred in the area of the South MacMillan River immediately downstream of Rust and Tom Creeks. The fish community was found to be composed of one species (Arctic grayling) with only 27 specimens collected during the entire sampling program. All fish were collected from 2 sites in the South MacMillan River. Other sites either had no fish present or the fish were at such low densities that electroshocking could not determine their presence.

#138
Montreal Engineering Company Ltd., 1978. Kerr-CNRL Grum Joint Venture, environmental baseline monitoring, results of 1977 program. Prep. for Kerr Addison Mines Ltd., Toronto. 20 + pages.

Abstract

During 1977 the Grum Joint Venture completed its underground exploration work near Faro. Montreal Engineering Company Ltd. have been monitoring water quality and quantity, wind speed and direction and providing waste management services as required. This report outlines the work performed in 1977 and summarizes the results of the year's monitoring.

The fisheries work undertaken in 1977 involved essentially a repetition of the 1976 project. Fish were taken from Vangorda Creek near the confluence of the Pelly River by electrofishing in August. The samples were analyzed for copper, iron, lead, maganese, mercury, nickel and zinc. Only arctic grayling and slimy sculpins were caught. There was insufficient tissue in most of the fish to analyse each fish individually so the fish were grouped with the exception of the larger grayling which could be analysed separately. The results indicate levels which are similar to those found in previous years sampling. Benthic samples were obtained from Rose, Vangorda and Shrimp Creeks by the use of rubble filled mini-pot samplers. These samplers were installed in June and sampled in August with the intervening time left to allow colonization. Diversity indices were calculated using the Shannon-Weaver diversity index.

#139
Montreal Engineering Company Ltd., 1977. Kerr-CNRL Grum Joint Venture, environmental baseline monitoring, results of 1976 program. Prep. for Kerr Addison Mines Ltd., Toronto. 43 pages.

Abstract

During 1975 the Grum Joint Venture continued its underground exploration work near Faro, Yukon Territory. As an integral part of this, Montreal Engineering Company have been monitoring water quality and quantity, wind speed and direction and providing waste management services as required. This report outlines the work performed in 1976 and summarizes the results of the years monitoring.

The fisheries work this year centered on determining heavy metal backgound levels for samples taken from Vangorda Creek at its confluence with the Pelly River. Fish samples were collected on August 10, 1976, using electrofishing gear. The species collected were arctic grayling, slimy sculpin, round whitefish, burbot and chinook salmon. The averages for all metals compare quite closely with the species averages from 1975 except in the case of mercury, where laboratory errors are suspected. Benthic pot samplers were used to sample three streams: Rose Creek, Vangorda Creek and Shrimp Creek. The data suggests that all streams are oligotrophic systems. Results of a survey of Shrimp Lake indicate that the lake is oligotrophic with limited production. The lake was netted for 2 successive days with 3 sizes of monofilament net and baited minnow traps were set in littoral areas at the south end of the lake. No fish were taken in either the nets or the minnow traps.

#140
Montreal Engineering Company Ltd., 1976. Kerr-AEX Joint Venture, 1975 biophysical and socio-economic program. Prep. for Kerr Addison Mines Ltd., Toronto.

Abstract

The 1975 biophysical investigation provides background data on water quality, benthos, fisheries, wildlife, soils and vegetation, in addition to reviewing the socio-economic implications of the proposed development.

Only preliminary results are available for the benthic analysis. These results indicate that the upper stations on Rose and Vangorda Creeks have more diverse communities than the lower stations. This indicates that the lower stations are more stressed than the upper stations. Fish surveys were conducted in May and August, 1975, using electrofishing gear. Sites were located on Pelly River, Vangorda Creek, Rose Creek and Blind Creek. Sampling, however, had to be restricted to quieter pools and somewhat sheltered areas where field personnel could safely enter the water. Another limitation of the sampling procedure was that not all specimens stunned by the electrofishing gear could be collected using the available dip nets. Thus, the data can not be interpreted as reflecting the relative abundance of the various species, but only their presence. Fish caught were arctic grayling, slimy sculpin, longnose sucker, burbot and round whitefish. Heavy metal concentrations are given for the species taken at the various sites. These are intended to provide a standard of existing conditions, against which post-development monitoring could be measured.

#141
Munro, D. and K. Weagle. 1976. An environmental study of the White Pass and Yukon route pipeline from the B.C.-Alaska border to Whitehorse, Y.T. Environmental Protection Service, Whitehorse, Y.T. 35 + pages.

Abstract

White Pass and Yukon Route (WPYR) operates a pipeline from Skagway to Whitehorse to transport furnace oil (diesel) and stove oil (number one).

This report presents the results of a survey conducted from July 19-22, 1976, to evaluate the condition of the pipeline from an environmental point of view. The pipeline was evaluated for the following and problem areas were documented with photography: 1. depth and protection at road and rail crossings 2. type and location of stream crossings 3. areas of unsupported pipe 4. areas of pipe susceptible to rockslides, avalanches, mud slides and glaciation 5. distance of pipe from lakes and soil conditions between the pipe and the lakes 6. distance of pipes from the railway tracks 7. areas of pipe with dents or damage 8. additional situations that may result in spills from the pipe.

Seven areas are identified along the pipeline route that were considered environmentally sensitive.

The report concludes that the pipeline was in an unsatisfactory condition from an environmental point of view at the time of the study. Areas in need of immediate attention are listed. The authors of the report suggest that the pipeline either be upgraded or rebuilt or that fuel be transported by tank car on the WPYR railway or that fuel be transported by road after the Carcross-Skagway road is completed. No fisheries information is included.

Appendices include photographs of unsatisfctory sections of the pipelines, reports of spills, maps indicating pipeline route.

#142
Northern Natural Resource Services Ltd. 1979. Fisheries investigations of Clear Creek, Yukon Territory, in 1977 and 1979. Submitted to Crescent Mines Ltd. and Sundance Gold Ltd. 20 pages.

Abstract

The report summarizes previously unpublished data collected by N.N.R.S. in the Clear Creek area in 1977 and presents new data collected between June 14-18, 1979 on Clear Creek, near its confluence with the Stewart River and also within the property of Sundance Gold and Crescent Mines Ltd.

Fish were sampled using a variety of methods incuding gill nets, beach seines and angling.

In those areas of Clear Creek to which Crescent Mines Ltd. and Sundance Gold Ltd. hold rights to placer mining, high stream gradients exist and the area was found to have a low potential producitivity. Low numbers of arctic grayling, slimy sculpin and longnose suckers were captured. Suitable spawning, rearing and feeding habitat for fish were found to be very limited. Pacific salmon were not observed to spawn in Clear Creek. This part of the creek showed no overwintering potential.

The lower reaches of Clear Creek near its confluence with the Stewart River was found to be more productive. Large numbers of arctic grayling, longnose suckers and round whitefish and low numbers of burbot and slimy sculpin were observed. Chinook juveniles utilized the lower reaches of the river for summer feeding purposes.

Water quality data collected from Clear Creek, McQuesten, Stewart and Pelly Rivers is presented in tabular form.

#143
Northern Natural Resource Services Ltd. 1978. A compilation of fisheries data from waterbodies adjacent to the Alaska Highway from kilometer 1008 to kilometer 1635. Prepared for Department of Fisheries and the Environment. 445 pages.

Abstract

This report presents an evaluation of fisheries resources adjacent to the proposed paving and realignment of the Alaska Highway from the Yukon/British Columbia border to Haines Junction. A review of existing Fisheries and Marine Service data and Foothills Pipeline data was made and summarized in tables. Data on fish species composition, life cycles stage, and timing of fish passage was collected, using gill nets and seine nets at stations in 113 waterbodies on, or adjacent to the Alaska Highway from July 7 to September 7, 1978. Preliminary engineering surveys of some culvert installations along the Alaska Highway right-of-way were also undertaken. The site-specific data is presented in catalogue format. Photographs of watercrossings and culverts are included.

#144

Northern Natural Resources Services Ltd. 1977. Collection of fisheries information from waterbodies along the proposed Alaska Highway Gas Pipeline Route to July 15, 1977. Submitted to DFO Fish. and Mar. Serv., Vancouver, B.C. 333 + pages.

Abstract

This report contains a summary of fisheries information collected by Foothills Pipe Lines and the Department of Fisheries in the waterbodies crossed by or adjacent to the Alaska Highway Gas Pipeline. In addition, the results of fisheries surveys conducted by N.N.R.S. in June and July to investigate the fish species composition and utilization by different life cycles of fish in water bodies on or adjacent to the proposed pipeline are presented. Detailed information on species composition and utilization as presented for each waterbody studied. A discussion of the potential impact of the pipeline on the aquatic habitat is included. The largest single impact on the aquatic habitat could come from the introduction of large volumes of sediment during and after construction. The Appendix I includes a preliminary assessment of proposed alternate pipeline routes. A very thorough study.

#145
Northern Natural Resource Services Ltd., 1977. A collection of fisheries information from water bodies associated with pipeline routes in the Yukon Territory from Dawson to Watson Lake, September 1, 1977. Submitted to DFO, Fish and Mar. Serv., Vancouver, B.C. 404 pages.

Abstract

The purpose of this study is to gather information on the fishery resource on or adjacent to proposed pipeline routes associated with the Klondike and Robert Campbell Highways, and to continue ongoing studies of selected streams on the Whitehorse to Watson Lake section of the Alaska Highway. Information on nursery and rearing areas, spawning ground locations, species composition and relative abundance of freshwater species, timing of migration of freshwater species, enumeration of adult chinook in selected streams, and the timing of migration of anadromous species was collected. This report presents site-specific data collected at or near the proposed pipeline crossing sites. The usefulness of this study should extend beyond the pipeline project, as baseline information concerning the fish stocks of the Yukon Territory is significant in its own right. No attempt has been made to evaluate the relative importance of particular species in terms of human utilization. A comparative assessment of value of the fishery resource and the pipeline development is likely to be futile, because the dollar value of a pipeline development exceeds that of the Yukon fishery harvest. Emphasis is therefore placed on the biological components of the fishery resource in order to minimize impact through study of the existing populations and planning of pipeline development. Certain life stages of fish are more vulnerable than others; the presence of incubating eggs downstream of the construction of a pipeline crossing is an extreme example. Salmon spawning areas are often very localized, magnifying risks to a population caused by disturbance. Life cycle components, amount and quality of habitat, timing of utilization, and general state of the population subject to impact should be considered in the assessment of impact. Sedimentation of areas used by fish, blockage of fish passageways, direct fish mortality during construction, drainage of habitat, depletion of oxygen during winter, and modification of groundwater flow all present potential impacts from pipeline construction. The report contains recommendations for mitigation of pipeline related impacts to the fishery resource. Site specific information on fisheries includes a review of the pertinent literature, a stream description close to the crossing, a summary of biological data, fish utilization and habitat assessments, and a statement of anticipated hazards and suggested safeguards at the site.

#146
Northern Natural Resource Services Ltd., 1976. Additional environmental information of the Sixty Mile River in summer. Submitted to Cogasa Mining Corporation, Vancouver, B.C. 17 pages.

Abstract

The information presented in this report represents a segment of a continuing study being conducted by Cogasa Mining Corporation of the potential environmental impacts of its proposed placer mining operations on the Sixty Mile River. Water samples were taken from sites along the Sixty Mile River and surrounding tributaries. Fish samples were taken from points upstream and downstream of Miller Creek. Fish sampling methods included gillnets and seine nets. The area upstream of Miller Creek was found to be preferred habitat area for Arctic grayling, as evidenced by good water clarity, excellent spawning and rearing gravels, lack of mining and reduced water temperatures. The authors of the report recommend that the area upstream of Miller Creek be protected from further mining operations. The fish mentioned in this report are arctic grayling and sculpins.

#147
Northern Natural Resource Services Ltd. 1976. An Environmental Assessment of the Sixty Mile River in Summer. Submitted to Cosaga Mining Corporation. 115 pages.

Abstract

This report presents results of a survey conducted on the Sixty Mile River in June, 1976. Sampling stations were located at various points along the Sixty Mile River from half a mile upstream of the mouth of Glacier Creek to the confluence of Sixty Mile River with the Yukon River. The primary objectives of the June survey were to collect data pertinent to the fishery resource of the entire Sixty Mile River with emphasis on those sections of the river on which Cogasa retains the rights to placer mining. Data was collected on the distribution and abundance of fish, age and growth relationships, length-weight relationships, sex ratios, size of fish at maturity, food habits, spawning habitat and migrational pattern. Four species of fish (arctic grayling, round whitefish, longnose sucker and slimy sculpin) were captured. Fish population was found to be greatest downstream from the mine. Growth rates for grayling and whitefish were not significantly different from other comparable populations but the size of arctic grayling in the population were smaller than in other comparable areas. The diets of the fish were restricted to a few organisms thus making it important to maintain high waster quality levels. Pacific salmon are not thought to utilize the river for spawning. Heavy metal concentrations are low in the upper reaches of the river but specific conductance, suspended solids and turbidity are substantially higher near the confluence of creeks where placer mining is in progress. Cogasas anticipated mining program does not constitute a threat to fish populations from the point of view of heavy direct introduction to silt.

#148

Northern Natural Resource Services, Ltd. 1976. An environmental assessment of the Sixty Mile River in winter, April 1976. Submitted to Cosaga Mining Corporation, Vancouver, B.C. 26 pages.

Abstracts

The purpose of this study, conducted in April, 1976, was to assess the suitability of the Sixty Mile River and tributaries as winter habitat for fish. Sampling was conducted at 10 locations on the Sixty Mile River and selected tributaries from the Alaska/ Yukon border to the mouth of Fifty Mile Creek. At each site, one or more holes were drilled through the ice across the flood plain of the river to measure ice thickness and consistency and to determine the presence or absence of water (as overwintering habitat) beneath the ice. Water quality measurements were taken at those sites where water was found beneath the ice. Water was located at 6 locations under the ice but in only one location was the dissolved oxygen concentration over 5.0 ppm which was taken to be the level necessary to sustain fish life throughout the winter. Gill nets were not set under the ice here because of insufficient water depth and a constricted river channel at this point.

#149
Paish, H. and Assocs. Ltd. 1981. The Yukon sport fishery. A policy oriented assessment of sport fishing in the Yukon. Prepared for Dept. of Fish. and Oceans, Gov't. of Yukon and Dept. of Indian and Northern Affairs. 233 pages.

Abstract

Essentially, this is a status report on:

1) The nature of the aquatic ecosystems that support fish in the Yukon and a first look at their biological productivity.

2) The historic and geographic perspective on the various uses that we make of fish in the Yukon, with an obvious emphasis on sports fishing, leading up to a more detailed examination of the current status of the fishery.

3) An examination of the various legislative and institutional aspects of the sports fish management.

On the basis of this information, an assessment is made of the likely future for sports fishing in the Yukon, and the kind of management systems that will be needed to ensure that Yukon waters continue to produce fish to meet the demands of Yukon anglers as well as the commercial, domestic and subsistence fisheries. The emphasis is on lakes because that is where most of the existing problems lie and where there is the greatest need for a management program.

#150
Palmer, R.N. and E. Hollet. 1957. Fisheries survey of the Upper Yukon River drainage in 1955-1957. Memorandum. Environment Canada. 23 pages.

Abstract

During the years 1955 to 1957 surveys were conducted on waters of the Yukon River drainage with respect to proposed power developments. Limnological data was obtained from Marsh, Tagish and Bennett Lakes and the Yukon River, from the outlet of Marsh Lake to the junction of the Takhini and Yukon Rivers. These surveys were of a superficial nature and were designed to obtain preliminary information as to the morphology and productivity of the system and to determine the species of fish indigenous to the system. Data was collected on lake areas, transparency, water temperatures, plankton, bottom fauna, and fish species.

Gillnets, beach seines and angling were used to capture fish.

Grayling were the most common fish in the section of the Yukon River that was sampled, making up more than half the net catch. From the results of the gillnet sampling, it appears that there is a greater density of fish in Marsh Lake than in Tagish or Bennett Lakes. All three lakes are oligotrophic. The three lakes supported domestic fisheries, had supported commercial fisheries in the past, and were considered important recreational areas. The chinook salmon population of McClintock River is considered valuable, not only as a food source for native settlements, but as it contributes to the fisheries of Alaska.

Fish species sampled in these surveys include: Arctic grayling, round whitefish, common whitefish, lake herring, northern pike, longnose sucker, and lake trout.

#151

Patterson, D. and R. Kendel (compilers). 1978. Winter river surveys 1975 Yukon - Takhini confluence, Yukon - Klondike confluence. Unpublished Memorandum Report. Dept. of Fisheries and Oceans. 8 + pages.

Abstract

River surveys were conducted at the Yukon - Takhini confluence between December 30 and January 4 and at the Yukon - Klondike confluence between February 18 and 22, and also between March 23 and 27th. The objectives of the surveys were to test and refine equipment and techniques for use in rivers during the period of ice cover and expand knowledge of winter working conditions and the limitations encountered during river sampling procedure. The secondary objectives were to determine fish species presence, habitat utilization, life history and feeding habits. Equipment is described and fisheries results are presented. In total the fish species observed during the study included burbot, longnosed sucker, lake whitefish, round whitefish and slimy sculpin.

#152
Pharo, C. 1982. (in preparation) Morphometry and morphology of Southern Yukon Lakes. Department of the Environment. Inland Water Directorate, Vancouver.

Abstract

This report will document the results of bathymetric field work on five Yukon lakes. The lakes to be studied are: 1. Laberge 2. Marsh 3. Tagish 4. Bennett and 5. Atlin. More extensive field work on lake ecology will follow the bathymetric studies.

NOTE: This information was gathered in a phone conversation with the author. The report has not been seen.

#153
Renewable Resources Consulting Services. Ltd. 1975. An environmental overview study of the proposed Marsh Lake Dam. Prepared for N.C.P.C. Edmonton. 132 pages.

Abstracts

This environmental overview is presented in two sections: Wildlife and Fisheries. Most of the field investigations conducted for the study were concerned with the wildlife segment. The purpose of the fisheries study was to review and summarize existing data concerning the fisheries resource of the Marsh/Tagish/Bennett lakes system, to identify deficiencies in the information, and to design and recommend appropriate field studies to fulfill information gaps necessary for an assessment of the impacts of the project. Site inspections of areas by road were conducted between 15 and 18 June and between 6 and 16 August, 1974. Site inspections and habitat assessment were conducted by boat between 6 and 16 August, 1974. Interviews were conducted with fishery biologists working with Environment Canada and with residents of the study area to obtain information concerning the fishery resource in the area. Potential effects of increased water levels on fish are outlined, critical or sensitive areas and conditions are described and a map showing potential spawning areas for northern pike, lake trout, whitefish and cisco is included. Due to the very limited amount of relatively up-to-date information dealing with the fishery resource of the Marsh/Tagish/Bennett lakes system, an extensive field program is required before an assessment of the environmental impact is possible.

#154
Rennie, P.J. 1977. A broad environmental comparison of northern Canadian natural gas pipeline routes, third edition with specialist appendixes. 150 + pages.

Abstract

This report aims to compare the relative merits of 12 different routes for moving Arctic natural gas to points in southern Canada. The report constitutes a broad synoptic analysis based on the specialist contributions of an appreciable number of environmental scientists. Eleven main areas of environmental concern are dealt with: permafrost and engineering problems, vegetation and vegetation relationships, hydrology, terrestrial mammals, avifauna, fish and marine mammals, conservation areas and recreation lands, native use of resources, and environmental design. More specifically the analysis does three things: 1. identifies the main environmental sensitivities associated with each route, 2. ranks routes in order of preference based on which route would create the least environmental impact, and 3. stratifies route comparisons so that routes of similar gas transmission capabilities can be compared. Major routes considered are the Mackenzie Valley and northern Yukon routes, the Alcan Highway route, the Alcan highway with a spur to reach Canadian Arctic gas, and an Arctic Islands route. Of all the routes there was almost universal agreement that the Alcan route poses the fewest environmental problems. It traverses very little permafrost and rarely crosses major rivers. In addition several decades of experience with the highway has provided considerable understanding of the terrain and hydrological characteristics along the route. Presence of the highway means few new haulage roads would have to be built. Eleven appendices discuss each of the major environmental topics seperately.

#155
R. L. and L. Environmental Services. 1982. (in preparation) Preliminary environmental and socio-economic evaluation of potential hydro-electric projects in the Yukon. Aquatic resources component. Prepared for Northern Canada Power Commission.

Abstract

This is one component of a feasibility study that is being prepared by Reid Crowthers and Partners for N.C.P.C. as an aid to hydro-electric site selection. The five potential hyroelectric sites under consideration are Hoole Canyon, Granite Canyon, Mid-Yukon (Five-fingers rapids) Ross Canyon and False Canyon.

The fisheries component of the study consisted of a survey of the five potential sites. Fish were sampled in late July and August upstream of the proposed dams and within reservoir areas, using electrofishers and beach seines. Some aerial surveys were also conducted upstream and downstream of the proposed sites in order to classify habitat areas. Information was collected on fish species composition and data on lengths, weights and maturity were recorded.

An additional survey was made in October to determine preferred chum spawning habitats between Carmacks and the confluence of the Pelly River, and to locate additional chum spawning areas in the Pelly and Yukon mainstem between Carmacks and Whitehorse. An attempt was also made to determine preferred chinook rearing and spawning areas on the Pelly River.

NOTE: This information was gathered by a telephone conversation with the author. The report is in preparation and has not been seen.

#156
Robson, W. and Weagle K. 1978. The effect of the abandoned Venus Mines tailings pond on the aquatic environment of Windy Arm, Tagish Lake, Yukon Territory. Environmental Protection Service. Regional Program Report 78-13. 22 pages.

Abstract

Information on benthic macroinvertebrates, fish and water quality was collected from Windy Arm, Tagish Lake, and water chemistry data were collectd from the abandoned Venus Mine tailings pond in 1975 and 1976. The object of the study was to ascertain whether the abandoned mill, and, in particular, the tailings pond, were having any adverse effects on the water, fish, macroinvertebrates and sediments of Windy Arm.

Gill nets were used to collect fish and fish samples were sexed, aged, and measured for length and weight. Tissue samples were analyzed for heavy metals. The species of fish previously recorded in Tagish Lake system are: lake trout, grayling, least cisco, humpback whitefish, burbot, longnose sucker and northern pike (R. Kendall, 1976, Fisheries Management Biologist, personal communication). Lake trout, humpback whitefish and round whitefish were the only fish captured in the gill net sets.

The levels of lead, zinc, copper, mercury, and arsenic in fish tissue were well below the levels set in the Food and Drug Regulations. The data is summarized in tabular form for lake trout, humpback whitefish and round whitefish.

The report concludes that the abandoned mill site had an effect on the lake in the immediate area of the overflow. The effects in the area of the outfall were seen in the reduced diversity of the bottom fauna community and the higher concentrations of lead and zinc in the sediments, as compared to similar areas in the lake.

#157
Roxborough, V.A. and G. Zealand, 1981. Russell Creek placer mining operation. Unpublished note. Department of Fisheries and Oceans, Whitehorse. 3 pages.

Abstract

A brief fisheries survey of Russell Creek (a tributary of the MacMillan River) was conducted from July 16-18, 1981.

The purpose of the study was to identify species presence, and to determine the extent to which the tributary was used as a chinook spawning and rearing area.

In total seven grayling, one sucker and numerous chinook fry were captured. Information on grayling sex, length, condition, stomach contents, egg size and gonad size are presented.

Results indicated that Russell Creek is a good chinook spawning and rearing habitat with a healthy grayling population. Chinook probably spawn in the upper reaches of the stream but salmon fry only rear in lower reaches of the creek. Environmental encroachments, such as the presence of suspended solids and the deposition of silt loads could destroy the salmon population.

Subsequently the D.F.O. has requested that Dawson Mining Equipment Ltd. implement total re-circulation of their water and that there is to be no direct discharge into Russell Creek from the operation.

#158
Schultz International Ltd. 1972. Environmental impact study of the Dempster Highway. Prepared for D.P.W. Canada, Vancouver. 68 pages.

Abstract

The objective of this study was to broadly assess the ecological and environmental impact of the proposed Dempster Highway. Road construction, operating techniques and roadway design are examined with regard to the possible impairment of the environment and adverse effects on the ecology. No fish sampling was done during the study. Impacts which can result from highway construction are given and the report lists some of the impacts which have resulted from portions of the highway already constructed.

#159
Scott, W. B. and E. J. Crossman, 1973. Freshwater Fishes of Canada. Fisheries Research Board of Canada. Bulletin 184, 966 pages.

Abstract

A thorough study of the species of freshwater fish found in Canada. Includes a survey of each species; its biology, description, relation to man, and distribution. Drawings of fish and distribution maps accompany the text. An invaluable guide.

#160
Sergy, G., K., Weagle, W. Robson, B. Ruggles. 1976. Water quality and biological investigations in the vicnity of the MacMillan tungsten property. Amax Northwest Mining Company Ltd. Environmental Protection Service Manuscript Report NW-76-1. 21 + pages.

Abstract

Sampling stations were set up within the Mactung property, along tributaries to the Hess River, at Cirque Lake, Dale Creek, and on the Tsichu River, to collect data on water chemistry, benthic invertebrates and fish species presence. Methods used to capture fish included electrofishing on the Tsichu River about one mile above Camp 222, electrofishing with a barrier net downstream on various tributaries of the Hess River and gill netting in Cirque Lake. Fish stomach contents were examined. Slimy sculpin and arctic grayling were collected in the Tsichu River and three immature dolly varden were collected in Hess Creek. Although no fish were captured in Cirque Lake, tricopterans, oligochaetes and chironomids were observed. The water quality at all stations were excellent.

#161
Shepard, C.D. 1981. Parasites of fish from four Yukon Lakes. Helmco Research. 11 + pages.

Abstract

A total of 18 fish (lake trout, lake whitefish, round whitefish, arctic grayling, northern pike, longnose sucker) were collected from Annie, Tarfu, Lewes and Summit lakes in the summer of 1980. This report presents the results of the examination of these fish for parasites and provides life history and etiological information on the parasite genera encountered.

#162
Shepard, C.D. and D.J. Davies. 1982. A suvey of lakes in the MacMillan Pass area with estimates of potential fish production. Government of Yukon, Department of Renewable Resources, Resource Planning and Management Branch. 10 + pages.

Abstract

Seventeen lakes in the MacMillan Pass-Ross River area were surveyed in the summer of 1981. The lakes under consideration were: Blue Birch, Bulgar, Dragon, Echo, Fuller, Gulf, Itsi, John, Lagoon, Lapie, Lewis, Niddery, Oly, Orchay, Otter and Sheldon Lakes. The major emphasis of the study was to determine fish species composition and potential fish production. Data was collected to calculate the morphoedaphic index: a formula that can be used to predict potential fish yields. All lakes were sampled for fish by gill net to determine species composition and relative abundance. In addition, dissolved oxygen concentration, temperature and water clarity were measured. The results of the study and the applicability of the MEI for the lakes under study are discussed. The authors conclude that with the exception of Dragon and Orchay lakes, the productivity estimate for the other lakes are probably not accurate enough to be used for management purposes.

Of the seventeen lakes examined, only Dragon and Orchay are likely to be subjected to any degree of sportfishing at present. Other lakes which have good potential for sportsfishing are Fuller, Itsi, Lagoon, Lapie, Oly and Otter.

Accompanying fish sampling results are presented on graphs. Two maps illustrate fish species presence.

#163
Sigma Resource Consultants Ltd. 1979. Environmental studies, Clear Creek gold dredging, preliminary findings. Prepared for Queenstake Resources Ltd. 29 + pages.

Abstract

The left and right forks and the main channel of Clear Creek were observed by helicopter, from the headwaters to the mouth at the Stewart River. The areas of Clear Creek that were inspected from the ground were as follows:

1. near the mouth at the Stewart River

2. near Barrett Pond in the lower reaches of the Tintina Valley

3. from two miles below the forks to the upper reaches of both the left and right forks.

Water, sediment, aquatic insects, vegetation and fish were sampled. Further information on the fisheries and wildlife aspects of the area were obtained from a literature review and from discussions with staff of the Yukon Game Branch and the Department of Fisheries and Oceans.

The physical setting and the biological resources of the project area are presented in the initial part of the report. This is followed by a discussion of possible concerns and mitigation procedures, and includes a section on fishery concerns.

The sampling method(s) used to capture fish are not described. The report concludes that the Clear Creek system is a fairly simple aquatic environment, supporting grayling, sculpins and possibly migrating chinook and chum. The benthic invertebrate population of Right Clear Creek above the mine reflects a clean, unpolluted atmosphere, while those in lower Clear Creek (near the mine) are indicative of an area unnaturally high in suspended solids.

Potential spawning areas for grayling (or salmon) are limited by suspended and deposited sediment loads in left Clear Creek and the main stem downstream. Right Clear Creek above the forks contains a few areas below the present mining activities where spawning is possible, while the areas upstream of present developments are relatively good (gradient, gravel types and water quality in natural state).

The grayling population in the right fork of Clear Creek is estimated to be about one hundred and the sculpin population is believed to number in the few thousands. Based on the available habitat the economic value of salmon in the area is believed to be low.

#164
Sigma Resource Consultants Ltd. 1975. The development of power in the Yukon Appendix 3 Environmental and social aspects. Prepared for N.C.P.C. 190 pages.

Abstract

The present study is the first phase of a planning program to establish the power resources of the Yukon territory, and the alternative sequences of development of such resources to meet a range of possible power demands to the year 1990. This report presents a preliminary assessment of the important renewable resources of the territory covering fish, wildlife, forests, recreational values and tourism, and historic and scientific values. The present distribution of Yukon communities is described along with the effects which future power developments might have on them. Salmon are utilized primarily in commercial fisheries on the Yukon and domestic fisheries on both the Yukon and porcupine rivers. Whitefish runs in the Peel are the basis for a major domestic fishery there. The developments proposed for the Peel, Porcupine, Tatshenshini and Yukon Rivers will have significant effect on their fish resources. The impact of hydro development on communities varies from complete inundation as at Carmacks, Mayo, Pelly, and Stewart crossings, to loss of viability such as at Old Crow and Fort McPherson if the fish and caribou are seriously affected. The impact on wildlife and fish resources has implications in communities where a significant part of the population depends on these resources for food or income. Hydro power planning should consider the impact it will have on tourism. Development on the Yukon, Porcupine and Pelly Rivers could severely impair the long term future of tourism in the Yukon territory. The development of large reservoirs on the Stewart, Pelly or Frances Rivers will require that the communities forego the renewable value of the forests in these areas.

#165
Sinclair, W.F. and O. Sweitzer. 1973. The economic value of the Yukon sport fishery. Economics Unit, Northern Operations Branch, Fish. Mar. Ser. Pacific Region D.O.E. Tech. Rep. NOB/ECON 2-73.

Abstract

This paper reports on a study conducted during the winter of 1971 and the summer of 1972 to determine the economic value of the Yukon sport fishery. Information is given on the net economic benefits for both the Yukon and Canadian economies. The value of the Yukon sport fishery to residents of the Yukon is enhanced considerably because of lack of alternative recreational pursuits - sport fishing is an important social pastime for Yukon residents. The main influx of fishermen to the Yukon takes place during the months of June, July and August. Those lakes, rivers and streams located near Whitehorse or close to roads leading to Whitehorse are the focus of most of the sport fishing activity.

#166

Slaney, F.F. and Company Ltd. 1978. Catalogue of fishery information and concerns at proposed pipeline stream crossings. Prepared for Foothills Pipe Lines (South Yukon) Ltd., Calgary. 18 pages.

Abstract

This report is a summary of information found in eight reports dealing with fish populations in waterbodies crossed by or adjacent to the proposed Alaska Highway Gas Pipeline. Data on fish occurrence, habitat utilization as given for each watercrossing in addition to the reference(s) from which information is taken, the dates of the studies and the existence of any conflicting information. Included are maps which identify all stream crossing sites along the pipeline route (and along alternate routes). An appendix summarizes available fisheries information for waterbodies adjacent to the proposed pipelines.

#167
Slaney, F.F. and Company Ltd. 1976. Supplemental environmental considerations and partial field investigations. Volume I Fish and aquatic resources. Prepared for Gulf Interstate Engineering Company, Houston. 34 pages.

Abstract

A field survey was carried out between July 8 - 25, 1976 of ninety-four watercourses potentially affected by the proposed Alcan Pipeline from Beaver Creek, Yukon to Zama Lakes, Alberta and from Empress to Monchy, Saskatchewan. Data sheets present chemical and physical data for all watercourses studied, inventories of fish species and fishery habitat utilization where this information is available. In general the fisheries information is sketchy: many waterbodies were not sampled for fish or were only sampled briefly.

#168
Slaney, F.F. & Company Limited. 1974. Environmental considerations Carcross to Skagway Highway, Mile 65.6 to 85.2. Submitted to the Department of Public Works, Vancouver. 400 + pages.

Abstract

An environmental study was made of that section of the Carcross to Skagway Highway which crosses the Tutshi Valley (mile 65.6 to 85.2). Environmental concerns for vegetation, wildlife, archaeology and aquatic systems are discussed. A section on fisheries describes studies conducted between July 25 and 27, 1974 in watercrosses including Tutshi Lake and Tutshi River, Lower Tutshi River, Summit Lake and Bernard Lake. Gillnetting, beach seining and angling were used to conduct a cursory sampling project. Species captured in Lower Tutshi River included lake trout, longnose sucker, slimy sculpin and round whitefish. In Tutshi Lake round whitefish, lake trout, longnose sucker, slimy sculpin and arctic grayling were observed. Spawning was not observed in either waterbody but both appeared to provide good spawning environments. In Bernard Lake, arctic grayling and lake trout were found. Two lake trout were captured in Summit Lake and round whitefish, arctic grayling and longnose sucker were caught in Tutshi River. Assessment maps outline fish species presence, fishery potential, habitat utilization and potential disturbance to the fishery resource, where known, at watercrossings along the road route. The major impact from the highway is expected to be the excessive exploitation of fish stocks by anglers.

#169
Slaney, F.F. & Company Limited. 1974. Environmental considerations proposed northern railway extension Whitehorse to Pelly River. White Pass and Yukon Corporation Limited, Vancouver, Canada.

Abstract

A 189 mile extension of the White Pass and Yukon Railway into the upper Pelly drainage area is envisaged. The most feasible mainline route is the "Mandana-Little Salmon" location which parallels the Robert Campbell Highway for most of its route.

This study was designed to describe the environmental setting, identify elements of particular concern, and recommend alternative routes or procedures to protect or mitigate potential adverse environmental effects of the proposed railway. Environmental data was assembled from published literature, resource and topographic maps, airphoto interpretation, interviews with government personnel and concerned individuals, field reconnaissances along the proposed route.

The chapter on environmental concerns includes subdivisions into: regional considerations, topography, climate, hydrology, vegetation, wildlife, aquatic concerns, archaeology concerns for native people, aesthetics and recreation, the Yukon tourist industry.

A section on fisheries reviews the fish species found in the area (Arctic grayling, lake trout, northern pike, Dolly Varden, chinook salmon, chum salmon, humpback whitefish, broad whitefish, round whitefish, inconnu, least cisco, lake chub, longnose sucker, burbot, slimy sculpin). Reproductive phases of known and possible fish species along the proposed rail route are shown in tabular form. There is some discussion of the waterbodies which will be affected by railway construction. Charts outline the species, role, abundance, habitat, life history, sensitive elements, resilience, and mitigative recommendations for fish species living in the study area. Possible construction disturbances and their effects on fish resources are outlined.

The last section of the report outlines all environmental concerns along the route of the pipeline. Includes photographs of sensitive areas along the proposed pipeline route.

Contains very little site-specific fisheries information, and present no new fisheries information

#170
Sweitzer, O. 1974. Distribution and abundance of chinook (Oncorhynchus tshawytscha) and chum (Oncorhynchus keta) salmon in the upper Yukon River system in 1972, as determined by a tagging program. D.O.E., Fish and Mar. Ser., PAC/T-74-20, 24 pages.

Abstract

Totals of 425 chinook and 1,067 chum salmon were tagged and released in the Yukon River 39 miles downstream from Dawson City.

Chinook salmon were found to utilize the Stewart, Pelly and upper Yukon River systems for spawning but generally did not use the White River system to any extent. Mainstem spawning grounds were located in the Yukon River amongst the Ingersol Islands.

Chum salmon favoured the White River system and main Yukon River for spawning. Spawners in the Yukon River were scattered in the mainstem from Fort Selkirk to Teslin River and through the Teslin River to Teslin Lake. Chum salmon in the White River system spawned in Klaune Lake and River.

Age V was dominant in chinook males representing 61.9% of the sample and Age VI in female fish at 62.7%. Chum salmon ages for both sex were IV,V and VI.

An estimated population of 29,054 chinook and 39,669 fall chum passed through the Yukon River system above Dawson City. Population limits (95%) are 48,521 - 19,173 and 49,464 - 32,548 respectively.

#171
Sweitzer, O.D. 1971. Report on rainbow trout planting program, 1971. Prepared for J.A. Summers. Unpublished. Department of Fisheries and Oceans. 15 pages.

Abstract

In June 1971 juvenile rainbow trout were transplanted from a commercial hatchery in British Columbia to selected lakes in the Yukon. The lakes planted with trout included Wye Lake, Long Lake, an unnamed lake locally referred to as Scout Lake, and an unnamed lake located at the junction of the Faro cutoff and Campbell Highway.

There was some fish mortality in the transport tank. Trout introduced into the four lakes seemed to adjust rapidly to temperature and environment.

Mortality on individual plants was as high as 25% but overall mortality was 12% for the four lakes. Recommendations are made to increase the size of plants for the four lakes in 1972. It is recommended that only new lakes which can support a spawning population be selected for trout planting. The incubation of eyed eggs in the Yukon is suggested.

#172
Tanner, A. 1966. Trappers, hunters and fishermen: Wildlife utilization in the Yukon Territory Northern Co-ordination and Research Centre. Department of Northern Affairs and National Resources (Ottawa), No. 5 in Yukon Research Project Senes.

Abstract

This is a survey of the hunting, trapping and fishing industries of the Yukon Territory. Data was collected in the summer of 1964 through interviews with wildlife and fisheries personnel as well as individuals engaged in the industry. A chapter on fisheries describes the Indian domestic fisheries, types, techniques and quantities caught and makes a brief survey of the status of commercial and sports fisheries in the territory.

#173

United Keno Hill Mines Ltd., 1980. Stage I report for Venus Mines. Submitted to the Metal Mines Steering Committee, Victoria, B.C. May 1, 1980.

Abstract

This report includes sections dealing with project description, environmental and social conditions, environmental and social impacts, and geotechnical studies for the proposed tailings area and plant. The environmental and social sections were prepared by F.F. Slaney and Co. Ltd. while the geotechnical section was prepared by Geocon (1975) Ltd.

The report deals only with the watershed south of the B.C./ Yukon border. No new fisheries information is presented but existing knowledge is reviewed - most of which comes from E.P.S. reports. The aquatic habitat which would be affected by the proposed Venus mill includes a small oligotrophic stream flowing into the south end of Windy Arm, the small bog lake between the site and Windy Arm, Windy Arm itself, and Tagish Lake. Species of fish recorded from the Tagish Lake system include lake trout, arctic grayling, least cisco, humpback whitefish, round whitefish, burbot, longnose sucker and northern pike. Variations in species composition in Windy Arm have not been documented but none are expected. The composition of populations resident in streams flowing into Tagish Lake have likewise not been reported. It is expected that small streams of the type draining the proposed mill site do not support fish resources of particular economic or social importance. It is likely that the lower reaches of these streams provide spawning and/or rearing habitat for grayling, round whitefish, and northern pike, and may also serve as a migratory route to a small bog lake.

#174
United Keno Hill Mines Ltd., 1980. Report in application for a reclamation permit, Venus Mines. September, 1980. 144 pages.

Abstract

United Keno Hill Mines Limited desires to reopen the Venus Mine for production of gold, silver, lead and zinc. An extensive feasibility study was carried out by the Falconbridge and United Keno Mining/Engineering and Metallurgy divisions with two consulting firms commissioned to report on the impact of environmetal and social conditions relating to a proposed new mineral processing plant located at a site in British Columbia. This report presents the relevant information obtained from the various feasibility studies.

The fisheries section is almost identical to that in the report from May, 1980 except that an investigation was conducted in August, 1980 to determine the fish population in the oligotrophic stream and the bog lake. It was found that the total population was extremely low, bordering on nil. There is no detail given as to how this investigation was carried out, which species were found, or how many individuals were found.

#175
U.S. Department of Interior. 1958. 1956 field investigations Fishery resources of the upper Yukon River basin between Eagle, Alaska and Carmacks, Yukon Territory. Prog. Rep. #3 on fish and wildlife resources, Yukon River, Alaska. U.S. Fish and Wildlife Service.

Abstract

The field work for this study was initiated in 1955 and continued in 1956. The objectives of the studies were:

1. to ascertain portions of the basin which are utilized by the various species of salmon for spawning and rearing purposes;

2. to gather information concerning the timing of spawning runs;

3. to evaluate the personal-use fishery of the area.

The number of fishermen operating gillnets and fishwheels in the upper Yukon River basin are given. Incomplete data indicated that more than 4,000 king salmon and more than 2,000 chum salmon were taken in the area studied during 1955. Data in 1956, known to be more complete, showed a catch of 6,200 kings and 5,300 chums. Other species, notably suckers and whitefish, were taken in smaller quantities incidentally to the salmon fishery. A sample of 363 king salmon taken from the catch of the fishery of the upper Yukon River contained 73% males and 27% females. Most of the females were 6 year olds while the majority of the males were 4 and 5 year olds. The peak of the king salmon migration at Eagle in 1956 occurred approximately July 17. At Dawson the peak occurred between July 22-28. For chum salmon, the peak at Eagle occurred between September 1-15, and at Dawson, between September 7-12. Maps showing known locations of chum and king salmon spawning areas are presented.

#176
U.S. Department of the Interior. 1956. A special report on the salmon resources of the upper Yukon Basin above Carmacks. Prog. Rep. #1 on fish and wildlife resources, Yukon River, Alaska. U.S. Fish and Wildlife Service.

Abstract

There were 2 phases to this investigation which were carried out in the summer of 1955. The first phase was a familiarization trip by vehicle to the upper Yukon drainage during the last two weeks in May to obtain general information on the magnitude, distribution and time of arrival of salmon runs. Local residents possessing some knowledge of the existing and historical salmon spawning areas were sought out and interviewed. The second phase was an aerial survey of reported salmon spawning streams during the latter part of August to determine spawning areas and numbers of salmon utilizing them. Weather conditions were adverse and on several occasions the surveys were discontinued because of poor visibility or extreme turbulence. The Mc'Clintock River is the only stream in the Yukon River system above Whitehorse that was found to support a run of king salmon. A total of 43 kings were observed from the air in Michie Creek and 40 kings were taken for personal use at the mouth of the Mc'Clintock during the last 2 weeks of August. Takhini River was reported to support a king salmon run of unknown magnitude but aerial surveys failed to reveal their presence. Kings were reported to use the Teslin River basin extensively for spawning; however, they were observed only in the Teslin River below Johnson's Crossing, at the upper end of Teslin Lake and in the Nisutlin River. Over 300 kings were taken by 3 families three miles below Johnson's Crossing during latter part of August.

#177
Usher, P.J. 1979. Status report on fur, fish and game harvest data in the Yukon Territory. A report to the Northern Pipelines Branch, Department of Indian and Northern Affairs. Ottawa. 36 pages.

Abstract

The report describes the data collection systems used by the Yukon Wildlife Branch and Department of Fisheries and Oceans to assess annual fur, fish and game harvests. The difficulties associated with data collection and the accuracy of data are assessed.

It is concluded that there is some basis for estimating the total volume of fur, fish and game harvested by non-native Yukoners, although there is little basis for allocating these estimates regionally within the Yukon. There is some basis for estimating the total harvest of furs but none for estimating the total harvest of fish and game by Yukon Indians. The author of the report notes that research in the area of land use and occupancy, harvesting and diet including cultural and sociological aspects are needed to understand the dynamics of Yukoners; and especially native peoples' reliance on traditional lands and resources. Without this knowledge the assessment of development impact on existing resources use will be reduced to guesswork and self-fulling prophecy or trivialized by the necessity of dealing with extreme time and site specific data.

#178
Walker, C.E. 1976. Studies on the freshwater and anadromous fishes of the Yukon River within Canada. D.O.E.. Fish. Mar. Ser. Rep. PAC/T-76-7.

Abstract

Characteristics of the adult chinook and chum salmon populations of the Yukon River within Canada are described. Significant differences were found between the composition of the catch and the composition of the escapement in terms of sex and length. It is hypothesized that any change in the fishery from wheels to gillnet will result in greater catch of larger size fish and also of females. Such a change may have severe consequences on the reproductive capacity of the populations.

Ten fish populations, other than adult salmon, are described for an area from Tatchun Creek to Fort Selkirk. Salmon is the most abundant fish, with longnose sucker, Arctic grayling and slimy sculpin occurring in this order. Growth data are presented from some species. In general, the growth of fish in the Yukon River is slower than that in tributary streams and other waterways of the territory.

The predictive value of pre-determining the location of salmon spawning habitat based on pinpointing areas that are ice-free and/or have ground water intrusion was nil. Areas of salmon spawning were found by observing spawners. The areas are classified into five groups. Mainstem water temperatures and transparencies are given. It is postulated that any change in the water temperature and, particularly water level in the mainstem of the Yukon River, may have serious detrimental effects on salmon spawning and reproduction.

Historically, the fish resources of the Yukon River system were of tremendous value to the native, explorer, farmer and developer. The salmon is the most important fish today with an estimated 10,000 - 20,000 taken annually in the commercial, subsistence and domestic fisheries.

#179
Walker, C.E. M.S. 1975. History of the chinook salmon run at Whitehorse Dam. Fisheries Sevice, D.O.E., Vancouver, B.C. File No. 31-5-17-4.

Abstract

In 1959 a fishway was installed at Whitehorse Dam. In the first four years the run varied from 660 to 1,500 with an average of 1,070 fish. From that time to the present the abundance of chinook at Whitehorse has decreased and now is in the range of 200-400 fish. At this rate of decline the run would cease to exist in 1981. The combined loss of adult reproduction (not proven) and the existence of turbine mortality for juveniles (proven) may explain this population trend. An additional introduced loss at the juvenile stage may be predation which may occur in Schwatka Lake before fish move through the dam or on "dazed" fish following passage through the facility.

#180
Walker, C.E. M.S. 1974. Assessment of fish stocks at Whitehorse Dam in 1973. Fisheries Service, D.O.E. Vancouver, B.C. Unpublished memorandum report. 21 pages.

Abstract

The objectives of the 1973 assessment program were to 1. determine the species and size composition of the fish stocks immediately above and below Whitehorse dam 2. to determine downstream migrating habits of the stocks and 3. to assess injures and/or mortality due to passage through the hydroelectric facility.

Gillnets were set to catch fish above the dam and 3 traps were used to capture fish migrating down the fishway.

The study concludes that increased loss of migratory fishes will accompany further development of the dam. Mortality will occur principally to the fingerling - size fish during downstream passage through the powerhouse. Further loss may be suffered during migration in Schwatka Lake due to increased predation. To minimize losses at the powerhouse it will be necessary to screen the fish from the power canal and release them below the facility, operate the turbines within defined limits during the time of downstream migration; and attract or guide fish to the regulatory gate, spillway gate and/or fishway entrance.

#181
Walker, C.E., R.F. Brown, and D.A. Kato. 1974. Catalogue of fish and stream resources of Carmacks area. D.O.E., Fish. Mar. Ser. Rep. PAC/T-74-8.

Abstract

This catalogue reports on 194 river miles of the Yukon and four major tributaries, namely, Big Salmon River, Little Salmon River, Nordenskiold River and Tachun Creek. Fifteen species of fish inhabit the waters of the study area. The chinook and chum salmon are most important for subsistence fisheries at this time. Information on biological critical areas and abundances are unknown except for salmon in which case some spawning areas have been identified and spawner populations enumerated. The total population of spawning chinook salmon in the study area is estimated at 3,000 - 7,000 fish distributed as follows: Big Salmon River 2,000 - 5,000; Little Salmon River 500 - 1,000; Tachun River 100 - 200 and Nordenskiold River 0 - 50. Chinook and chum spawning has been observed in the Yukon River but numbers are unknown.

#182
Walker, C.E., J.E. Bryan, and R.F. Brown. 1973. Rainbow trout planting and and lake survey program in the Yukon Territory 1956-1971. D.O.E., Fish. Mar. Ser. Tech. Rept. PAC/T-73-12.

Abstract

Thirteen lakes or ponds were surveyed to determine their suitability for trout, and rainbow trout were introduced into 11 of them to diversify sport fishing opportunities. This report presents the survey information and describes the history of the trout plantings and populations. Although rainbow trout survived in 9 of the 11 lakes or ponds stocked, only 5 of the lakes (Long, McLean, Ruth, Scout and Wye Lakes) developed populations adequate to provide good angling. Case histories of the trout plants suggest that, given a good seeding of fry, the absence of competing fish species and adequate lake depth and volume to prevent winterkill were instrumental in the establishment of trout populations adequate for angling. Some alternative management practices for an aquaculture program are described.

#183
Weagle, K. Environmental Consultant Ltd. 1982. An evaluation of aquatic resources of Big Creek, Yukon Territory. Prepared for Queenstake Resources Ltd., Vancouver, B.C. 25 + pages.

Abstract

This report was intended to provide background aquatic data on Big Creek (Little Klondike River) which would be used to support the dredging application by Queenstake Resources.

The aquatic habitat of Big Creek was examined in the summer of 1981 and classified using the B.C., R.A.B. aquatic inventory system. Four reaches were found on Big Creek, above Losthorses Creek; several reaches on tributary systems were also included. Data on water quality, and benthos was collected and fish samples were collected using a Coffelt BP-3 electrofisher.

Fish species captured included slimy sculpin, and arctic grayling. The numbers and site locations of observed fish are indicated. Some reaches of the Big Creek drainage are utilized as adult feeding areas for arctic grayling. It is likely that little or no grayling spawning or rearing occurs in Big Creek, nor were there any signs of salmon spawning in the creek.

An appendix summarizes the results of a winter reconnaissance of Big Creek conducted on March 9, 1981. Data on water chemistry was collected. No fish sampling was done. Note: Big Creek is a local name for Little Klondike River.

#184
Ken Weagle Environmental Consultant Ltd. 1981. Benthic Community monitoring program at Cyprus Anvil Mine, Faro. Yukon, 1981. Prepared for Cyprus Anvil Mining Corporation. 19 pages.

Abstract

The benthic communities of the water sheds in the area of Cyprus Anvil Mine, Faro were sampled using artificial substrate samplers. There were six sampling sites on Rose Creek, three sites on Vangorda Creek, two sites on Blind Creek and one site on Swim Creek. In comparison with 1980 and 1978 the benthic communities of Rose Creek downstream of the mine were in a healthier state in 1981. The benthic communities of the baseline sites sampled on Vangorda, Blind, and Swim Creeks clustered in a similar manner in 1981 as compared to 1980. This was considered to be sufficient data on which to base future monitoring programs. Water quality parameters downstream of the mine on Rose Creek as measured during the survey in 1981 were similar to 1980. An appendix consisting of the invertebrate data is included.

#185
Weagle, K. Environmental Consultant Ltd. 1981. Fish spawning survey on Teenah and Dalayee Lakes October 1981: Squanga hydro investigations. Prepared for Northern Canada Power Commission Purchase Order: 19868 NCPC, 23 pages.

Abstract

The purpose of the study was to gather data on lake trout and humpback whitefish spawning habits in Teenah and Dalayee Lakes, Yukon Territory in relation to the Squanga Hydro Investigations.

Gill net surveys were conducted on Teenah Lake from October 6 to October 9, 1981 and Dalayee Lake from October 9 to October 12, 1981.

The fish species encountered in Teenah Lake were northern pike, humpback whitefish, arctic grayling and burbot. The prime spawning area for humpback whitefish was believed to be in a bay at the north end of the lake and spawning was believed to have begun on October 8. The fish species encountered in Dalayee Lake were lake trout, arctic grayling, burbot, and slimy sculpins. Lake trout spawning was completed on arrival at Dalayee Lake by October 9, but spawning areas were believed to be in 2-5 meters of water over varied substrate of sand and cobble.

#186
Ken Weagle Environmental Consultant Ltd. 1981. Draft Report. The impact of the Scheme 2 abandonment plan on the grayling population of Rose Creek. Prepared for Cyprus Anvil Mining Corporation, Faro.

Abstract

The arctic grayling populations from two isolated sections of Rose Creek were sampled from May to September 1981. Section 1 was the pumphouse pond, the North Fork and upstream to the freshwater pond; Section 2 was the freshwater pond and Rose Creek upstream to the mine access road crossing. Section 2 was considered to closely approximate the Scheme 2 abandonment lake from a physical point of view. A total of 405 arctic grayling were captured during the survey.

The data indicated that fish in Section 1 were longer lived than those in Section 2 while the growth rates in the two sections showed no significant differences at the 1% level of significance. In section 2 males began to mature earlier than Section 1 with females maturing at the same age and all fish being mature at age 5+. The utilization patterns of adults, juveniles and fingerlings in both sections were noted with the conclusion that after completion of the scheme 2 lake sufficient habitat for all life history stages would be available to support the grayling populations upstream of the lake.

The water quality parameters predicted for the scheme 2 lake were found to be no higher than the background levels in Rose Creek.

The impacts of the scheme 2 lake were summarized in a matrix as follows:

Positive Impacts - a) increased habitat for fry and fingerlings rearing.

b) increased adult and juvenile feeding areas.

Negative Impacts - a) if the D.O. in the scheme 2 lake falls below 7.25 mg/l in winter there will be an impact on overwintering of grayling in the pond.

The conclusion was that the scheme 2 lake with no fish passage could support a grayling if winter D.O.'s remain above 7.25 mg/l.

#187
Weagle, K. 1981. A preliminary evaluation of the environmental impacts of a bed load mining method. Unpublished feasibility study. Prepared for Allen Falle. 21 pages.

Abstract

A brief description of the bed load mining method under development is provided. Production testing and environmental monitoring of the mining method will be carried out in the Stewart River near Steamboat Bar (McQuesten Airfield) in 1981.

According to the available literature eleven fish species have been identified in the area. The existing literature suggests that those that may be impacted by the operation are chum and chinook salmon, Arctic grayling and long-nosed suckers. The known habitat requirements for these three species are discussed. The projected impacts of the mining method are summarized as: degradation of the water quality through increased sediment loads, degradation of the bottom habitat through hydraulic scour, and aggrading of the river bottom downstream of the site. These three impacts are discussed and it is concluded that the water quality of the Stewart River should not be degraded unless extensive scour occurs. The extent of bottom scour cannot be predicted because of the lack of data on bottom velocities and substrate composition. On a theoretical basis, aggrading of the channel downstream of the operation was not considered a problem.

An environmental monitoring program for 1981 is proposed which covers water quality, scour and fish habitat utilization.

NOTE: The 1981 environmental monitoring program was never carried out, and the mine has not been developed.

#188
Ken Weagle Environmental Consultant Ltd. 1981. The surveillance and transportation of fish during the completion of the diversion canal at Cyprus Anvil Mine, Faro, Y.T. May 4 to May 15, 1981. Prepared for Cyprus Anvil Mining Corporation, Faro.

Abstract

Cyprus Anvil Mining Corporation is in the process of building the Down Valley Tailings Disposal System. When completed the system will involve a tailings pond which will be approximately two km long and a four km diversion canal to carry the flow of Rose Creek around the tailings pond. Ken Weagle Environmetal Consultant Ltd. was contracted to provide fish surveillance and transporation services for Cyprus Anvil Mine. The fish surveillance indicated that there were no upstream fish movements from May 4, 1981 until the diversion of Rose Creek was made on May 11, 1981 and that grayling spawning began upstream of the mine on May 8 and lasted until May 15. After the diversion was completed a total of 23 grayling were captured in the former Rose Creek channel and released in the pump house pond. On May 12, 1981, the day after the diversion, the canal was capable of passing fish.

#189
Ken Weagle Environmental Consultant Ltd. 1980. Benthic community monitoring program at Cyprus Anvil Mine, Faro, Yukon. Prepared for Cyprus Anvil Mining Corporation. 24 + pages.

Abstract

Benthic invertebrate communities were sampled on Rose, Vangorda, Blind and Swim Creeks using artificial substrate samplers. There were six sampling sites on Rose Creek, three sites on Vangorda Creek, two sites on Blind Creek and one site on Swim Creek. The examination of the benthic invertebrate communities in Rose Creek during 1980 indicted that the impact of the mine effluent on the creek had lessened when compared to the 1977 and 1978 data. The water quality in 1980 was approaching that of 1977. The improved water quality had probably led to the recovery of the benthic communities. At least one additional year of benthic invertebrate monitoring needs to be conducted on Vangorda, Blind and Swim Creeks. An appendix consisting of all the invertebrate data is included.

#190
Weagle, K. 1980. Report on the technical workshop on suspended solids and the aquatic environment. Department of Indian Affairs and Northern Development. 24 + pages.

Abstract

A workshop was held to address the effects of suspended solids on the aquatic environment in the Yukon. Seven technical experts, all with field experience, participated in the workshops. The workshop began with the presentation of the following series of papers:

1) H.F. McAlpine. 1980. "Water quality data at several representative Yukon placer mining operations".

2) O. E. Langer. 1980. "Effects of sedimentation on salmonid stream life.

3) S. Atkins-Baker. 1980. "In situ bioassay study at a gold placer mining area in Yukon using Arctic grayling".

4) D.M. Rosenberg. 1980. "Responses of benthic macroinvertebrates to sedimentation in freshwaters".

5) P.H. Whitfield. 1980. "Some observations on suspended sediments in natural systems".

All of these reports are presented in the appendix of this document.

Discussion groups were held on a variety of related topics and the conclusions are presented.

With regard to fisheries studies it was concluded that the biological reactions to increased sediment loading are not fully understood, because of a lack of detailed reserch. Furthermore effective water quality criteria for suspended solids and the aquatic environment could not be set at this time. In areas with critical fish populations, detailed research on all aspects of the aquatic ecosystem would be necessary before acceptable concentrations could be set. In other areas, levels should be set only after completion of a two phase study. The first phase would examine in detail the life histories and distribution of a variety of fish and macroinvertebrates in Yukon. The second phase would be a long-term environmental evaluation of an industrial operation having a suspended solids effluent. These studies should be supplemented by laboratory work dealing with the effects of suspended solids on pertinent aquatic species.

#191
Weagle, K. and A. Pearson. 1980. Initial environmental assessment of the Baroid of Canada MacMillan Pass Project, June to August Data. 1980. Prepared for Baroid of Canada, Ltd. Calgary. 41 pages.

Abstract

This report presents the results of a preliminary environmental assessment made of the area surrounding the Baroid of Canada, MacMillan Pass project. The Baroid of Canada project will involve the construction of an access road (which crosses the South MacMillan River and various waterbodies associated with a tributary of Hess River), the development of an open pit barite mine, and the operation of a primary crushing plant. Data on water quality, climate, terrain, vegetation, wildlife and fisheries was summarized from existing reports and three field trips were made to the proposed area in the summer of 1980.

The results of fisheries surveys in the area are reported. Angling in an unnamed lake below the mine site resulted in the capture of four dolly varden. A helicopter survey of a creek, tributary of the Hess confirmed the presence of spawning chinook salmon. Surveys along the waterbodies crossed by the proposed access road indicted that some creeks would be suitable as fish habitat while others were too steep or too turbulent to provide suitable habitat. Means of offsetting potential impacts on fish are suggested.

#192
Weagle, K. and A. Pearson. 1980. Initial environmental assessment of the Baroid of Canada MacMillan Pass Project, September to October Data 1980. Prepared for Baroid of Canada Ltd., Calgary. 13 pages.

Abstract

Wildlife and fisheries surveys were conducted in the vicinity of the CATHY claims area in the fall of 1980. Fish were collected on October 9, using a beach seine. It appeared tht some streams and lakes in the vicinity of the mine site contained Dolly Varden and that some areas on these streams were probably used for spawning and rearing. One stream contained only arctic grayling.

The schedule for instream construction is suggested on the basis of the fisheries data collected.

Identification and enumeration of benthic invertebrate samples are included in an appendix.

#193
Weagle, K., W. Robson and K. Gullen. 1976. Water quality and biological survey at Arctic Gold and Silver Mines Ltd., Yukon Territory, summer 1975. Environmental Protection Service Report No. EPS 5-PR-76-10.

Abstract

Data on water chemistry, benthic invertebrates and fish species were collected from the Tank Creek watershed near the abandoned Arctic Gold and Silver Mines Ltd. mill site during the summer of 1975. Electrofishing was used to capture fish but no fish were captured. The abundance and type of bottom fauna present in the waterbody would provide ample food for fish. The apparent lack of fish was thought to be related to the size, elevation and resultant gradient of the stream. The deformed appearance of fish taken from Bennett lake by the local people was probably due to the unproductive nature of the lake and not a result of pollution emanating from the abandoned mill site. The report concludes that the abandoned mill site had no adverse effect on Bennett Lake or the watershed of Tank Creek.

#194
Weir, D.N. 1979. An ecological reconnaissance of Klondike gold placer mining and discussion of interim guidelines for the issuing of authorizations to use water for the purposes of placer mining in the Yukon Territory. Prepared for the Klondike Miner's Association. 45 pages.

Abstract

The aims of the study were to investigate the impact of placer mining on riparian life, the impact of mining on the natural ecosystem as a whole and some problems of mine process water treatment. The report presents the results of fieldwork in summary form and discusses the Yukon Territory Water Board 'Interim Guidelines' with special emphasis on effluent discharge and on landscape restoration. Fieldwork was conducted on Ten Mile Creek (at Oak Bay Manor - active mine), on Black Hills Creek (Territorial Gold Placer Mine - active mine), on Dominion Creek (Ace Placer Mines - active), on Hunker Creek (Miben mines - active) on Jensen Creek (old mining area) and on Sulphur Creek (old and active mining area). In addition other former mining areas were briefly visited by truck from the Klonkike River, on Hunker and Discovery Creeks, along Dominion Creek, out to Black Hills Creek and back to Sulphur Creek. A number of areas were inspected from the air. Settleable solids in water samples around the mine site were determined. Streamside and stream biota were noted. The soils and vegetation of disturbed or altered ground were examined. Fish presence was established through seine netting, fly fishing or interviews with local people. Fish sampling was cursory, some areas were not sampled at all. The only fish species obvserved was grayling. All four actively mined creeks showed reduced suitability for supporting fish, although there were indications that some restoration would be possible on both Jensen and Sulphur Creeks. By contrast, all six creeks showed heavy utilization by wildlife and it is suggested that the economic value of terrestrial wildlife had increased.

A discussion follows on the regulations concerning effluent limits and landscape restoration.

#195
Wickstrom, R.D. (C.W.S.). 1977. Fish distribution in Kluane National Park and peripheral area. Prepared for Parks Canada, 31 pages.

Abstract

This report presents information on fish distribution within Kluane National Park and peripheral areas. The geography of the area, previous drainage history, pertinent literature on fish species presence, and general theory of fish dispersal into the area are reviewed.

Fish were caught by gillnetting, seining, angling, dip netting and electrofishing during the summers of 1973 to 1976. Previous records and information from local sources provided additional information. Discussions follow on the distributions and locations of kokanee, Dolly Varden rainbow trout, steelhead trout, lake trout, round whitefish, Arctic grayling, pygmy whitefish, northern pike, lake chub, chum salmon, coho and chinook salmon, least cisco, broad whitefish, inconnu, lake whitefish, longnose sucker, burbot, slimy sculpin, in the Kluane Lake region.

The report concludes that the species distribution of the Alsek and Yukon drainages in the Kluane area resulted from the post glacial drainage connections (outlined in the report). The difference between the large numbers of species found around the park periphery and the small number within the park probably reflects the geologically later time of glacial ice melting within the park. Temporary and seasonal flows and higher silt loads of some flowing waters are a deterrent to more extensive distribution in flowing waters, and the general lack of lake habitat contribute to the reduction in species diversity within the Park.

#196
Withler, I.L. 1956. A limnological survey of Atlin and southern Tagish Lakes. B.C. Game Commission. Management Publication No. 5. 36 pages.

Abstract

The purpose of this limnological survey was to evaluate the present fishery and the potential fishery on Atlin and Tagish Lakes and to predict the effects of their use as water storage reservoirs in a proposed Yukon River-Atlin Lake-Taku River power project. Information was collected on drainage basin characteristics, morphometry, physical and chemical conditions, bottom fauna, fish species and spawning facilities. Fish were sampled using gillnets. At Atlin Lake, or in its tributaries, Arctic grayling, lake trout, lake whitefish, round whitefish, northern pike, lake chub, finescale sucker, burbot, lake herring were observed. Gillnetting at a sample station in Tagish Lake and near to Taku in Graham Inlet during August produced samples of lake trout, grayling and round whitefish.

The report concludes that use of Atlin Lake as a storage reservoir would probably have little effect on the biological productivity of the lake although production of lake trout might be adversely affected in the years immediately following flooding, due to a loss of previous spawning grounds. A lake level increase of over sixty feet in Tagish Lake might create adverse conditions for the production of most bottom organisms. The spawning areas of lake trout and whitefish would be disrupted.

#197
Wynne-Edwards, V.C. 1947. Northwest Canadian fisheries surveys in 1944-45, Chapt. 2 (The Yukon Territory) Bull. Fish. Res. Bd. Can., No. 72.

Abstract

This preliminary reconnaissance of Northwestern Canada was required for the proper formation of fishery regulations for the Yukon Territory as well as for more purely scientific reasons. The survey took place over 8 weeks beginning July 7, 1945 and was done by airplane, boat and truck. The report describes the commercial fishery in operation for chinook salmon on the Yukon River and Alsek tributaries. There is a brief section on fishing in lakes for whitefish, lake trout and least cisco. Included are notes on how the fish distribution in the Yukon reached its present state. The conclusion is that the Yukon fishery resources will probably never be exploited for export to outside markets because they are insignificant compared with those of neighbouring B.C., however, there is a great opportunity for developing sport fishing for the tourists.

#198
Yates, D. 1979. Wellesley Lake Program - 1978. Memorandum letter. Department of Fisheries and Oceans, Whitehorse. File 5830-5. Wellesley Lake Program. 3 pages.

Abstract

The Wellesley Lake program was designed to evaluate the impact of sport fishing on Wellesley Lake as well as providing a preliminary biological data base. During the period between August 25 and September 16, 144 creel census interviews were conducted with fishermen. In addition, some preliminary limnological data was collected and some netting conducted to supplement the creel census information. The fish observed on the lake were lake trout, pike and lake whitefish. This 3 page memorandum contains very preliminary data.

ANNOTATED AISHIHIK TITLES

#199
Arthur, J. R., L. Margolis and H. P. Arai, 1976. Parasites of fishes of Aishihik and Stevens lakes, Yukon Territory, and potential consequences of their interlake transfer through a proposed water diversion for hydroelectrical purposes. J. Fish. Res. Board Can. 33:2489-2499.

Abstract

Forty-six species of parasites were recovered from 383 specimens of 8 species of fishes collected from Aishihik and Stevens lakes during the summer of 1973. Substantial differences in the parasite fauna of Aishihik and Stevens lakes are noted. The possible consequences of the introduction of species of potential economic and pathogenic importance into new areas as a result of the proposed diversion of Stevens Lake (Yukon River system) into Aishihik Lake (Alsek River system) are discussed. Factors that must be given attention include preventing the transfer of infected fishes, infected intermediate hosts, free-swimming larval stages, and contaminated host waste products.

#200
Brown, R. F., 1975. Food content in the stomachs of Aishihik fishes. Fish. Mar. Ser. Tech. Rep. PAC/T-75-4, Chapter 6, 14 pages.

Abstract

Fish were sampled in the Aishihik system and their stomach contents were analyzed to determine the kinds and quantities of food consumed. Stomachs analyzed from arctic grayling, round whitefish and humpback whitefish are summarized in tables. Stomach contents of Arctic grayling from the Sekulman River are presented but not analyzed.

#201
Bryan, J. E. and D. A. Kato, 1975. Spawning of lake whitefish, Coregonus clupeaformis, and round whitefish, Prosopium cylindraceum, in Aishihik Lake and East Aishihik River, Yukon territory. J. Fish. Res. Bd. Canada, 32(2):283-88. (Also: Fish. Mar. Ser. Tech. Rep. PAC/T-75-4, Chapt. 3, 23 pages.)

Abstract

Spawning grounds in Aishihik Lake used by lake (Coregonus clupeaformis) and round whitefish (Prosopium cylindraceum) were discovered in the course of an investigation on effects of hydro-electric development. The spawning period for lake whitefish extended from early November to at least mid-December 1973. Lake whitefish spawned over silt and Potomageton in water which had little current and was 2.0 - 2.5 m deep. Lake whitefish spawned at night. Spawning of round whitefish was probably completed in November. Round whitefish spawned during the day. Eggs were apparently broadcast over a variety of substrata ranging from silt and Potamogeton to gravel and boulder. Round whitefish eggs were deposited in both fast and slow current at depths ranging from 0.7 to 2.5 m. Although deposited in a range of habitats, round whitefish eggs seemed to be most abundant on gravel in fast currents at depths less than 1 metre. Drifting eggs were captured both downstream of Aishihik Lake and downstream of Canyon Pond.

#202
Canada, Department of the Environment, 1975. Presentation to the Yukon Territorial Water Board inquiry into the Aishihik Hydroelectric Project. Fish. Mar. Ser., August 1975. 55 pages.

Abstract

This presentation to the Yukon Territorial Water Board inquiry is divided into two sections:

1. Comments on the Northern Canada Power Commissions assessment of the Fisheries Service biological and engineering data; and

2. Comments on the economic assessment of the environmental implications of the Aishihik River power development.

The first section is a critical review of two engineering consultant reports prepared for N.C.P.C. which attempt to discredit the Fisheries Services for incomplete research on the Aishihik River system. The review discusses fish population data, minimum flow requirements from Aishihik Lake to Canyon Lake, the value of the native and sport fisheries, Aishihik Lake water levels, erosion problems and siltation. Evidence is presented showing that N.C.P.C. exceeded the limits of its water licence and flooded Aishihik Lake. The author concludes that N.C.P.C. is unwilling to provide sufficient flow releases from Aishihik Lake to maintain fish production and is unwilling to expend resources to maintain the environmental and aesthetic qualities of the lake.

The second section of the presentation discusses the suitability of economic data used by Fisheries Service and N.C.P.C. consultants. Included is an economic assessment of environmental damages inflicted by N.C.P.C. in the Aishihik area, a discussion of the costs to N.C.P.C. of protecting fisheries, environmental and recreational values and an analysis of N.C.P.C.'s projected future power demands in the Yukon. This section concludes that N.C.P.C. can meet its electrical power requirements in the Yukon while still protecting the environment.

#203
Canada, Department of the Environment, 1974. Report on investigations carried out by Pacific Biological Station Fish Health Programme personnel in the Yukon Territory in 1973. Fish. Mar. Ser., Tech. Rep. PAC/T-74-19, Appendix G, 10 pages.

Abstract

The diversion of water flow from Lake Stevens into Lake Aishihik, proposed within the framework of hydroelectric development on the latter lake, introduces possibilities of faunal exchanges between these two lakes, and potentially, between the entire Yukon River system and Alsek river system. To look into possible danger of such transfaunation to fish health, several members of the Fish Health Programme from the Pacific Biological Station in Nanaimo, B.C., at the request of Fisheries Operations Branch, carried out investigations in both lakes during the summer of 1973.

The investigations were directed at detection of possible differences between the parasite fauna, bacterial flora and viruses of fish in Aishihik and Stevens Lakes.

Results show that several dangerous parasites exist in each system which are absent in the other river system. The consequences of water diversion are thus potentially detrimental to fish populations in the two lakes and could extend to the larger river systems.

#204
Canada, Department of the Environment, 1973. Fisheries problems associated with the proposed Aishihik River power development. Interim report, January 1973. Environment Canada Fish. Ser. 45 + pages. (Also D.O.E. Fish. Mar. Ser. Tech. Rep. PAC/T-74-19, Appendix A.)

Abstract

This report represents the first interpretation of raw data collected during the 1972 field season. Biological field investigations were conducted by two separate teams. One team studied nine streams and rivers through netting, trapping, observing and sport fishing, while another crew sampled stations around Aishihik Lake using gillnetting, plankton and dredge sampling and water chemistry testing.

In addition to biological studies, some engineering field work was done including an Aishihik Lake shoreline study, stream surveys and lake sounding. A questionnaire and shoreline enumeration constituted a small socio-economic survey of the Aishihik area.

The results of these studies are used to detail the most significant effects of the proposed power project on the streams, rivers and lakes in the Aishihik system. A list of recommendations regarding fisheries concerns is noted for use by the Yukon Territorial Water Board.

#205
Canada, Department of the Environment, 1972. Brief presented to Yukon Territory Water Board Public hearing into proposed Aishihik hydro-power development. Fish. Mar. Ser. May, 1972. 19 + pages.

Abstract

This report consists of a list of fish species found in Aishihik Lake, and discussions of possible Aishihik system river diversions and their affect on fisheries populations. Potential fisheries problems associated with reservior design, stream tributaries, power canal construction and gravel removal are indicated. Recommendations are listed for minimizing damage to fisheries.

The fish species present in Aishihik Lake included humpback whitefish, round whitefish, arctic grayling, lake trout, burbot, rainbow trout and northern pike. Water diversions under consideration included Stevens Creek (Yukon R.), Long Lake (Yukon R.), Nisling River (Yukon R.), and West Aishihik River (Alsek). The possible impacts of water diversions on fisheries populations were: 1. the mixing of discrete populations, 2. the introduction of disease and parasites, and 3. the alteration of predator prey relationships.

#206
Cleugh, T. R. 1977. Aishihik lake fishway investigation. Unpublished memorandum. In file 5430-82-12. Aishihik dam, D.F.O., Whitehorse, 5 pages.

Abstract

On June 27-28 fish were sampled at the outlet of Aishihik lake. Net catch above the impoundment was seven humpback whitefish (feeding exclusively on zoo plankton) and one grayling (feeding on insects). Net catch below the impoundment amounted to thirty-seven round whitefish (feeding on drift insects), one grayling and one humpback whitefish.

#207
Crippen Engineering Ltd., 1975. Aishihik River power development. Aishihik-Whitehorse systems study. Phase II report, May, 1975. 66 + pages.

Abstract

Computer programs were developed representing mathematical simulation models of the hydrological and operational characteristics of the integrated Aishihik-Whitehorse system. Different values for maximum fisheries flows from Aishihik Lake, Canyon Lake levels, and Otter Falls flows were inserted into the program to evaluate their effects on power losses to the system.

#208
Gallup, D. N. and A. Clements, 1975. Base-line studies on the invertebrate drift populations of a section of the Aishihik River. Fish. Mar. Ser., Tech. Rep. PAC/T-75-4 Chapt. 5, 46 pages.

Abstract

A survey of the invertebrte drift populations in the Aishihik River between Aishihik Lake and Canyon Pond was conducted between May and July, 1974. Drift populations were collected, species composition was noted and drift daily and seasonal peaks were determined for various macro invertebrate and zoo plankton species.

The hydro dam will result in a reduction of the water flow rate which will affect the drift behaviour of invertebrate populations. The relationship between fish feeding habits and the drift pattern of invertebrate organisms is discussed. It is suggested that altered flow rates will upset the equilibrium of the aquatic habitat system, possibly resulting in the reduction of invertebrate and fisheries populations in the Aishihik system.

#209
Kato, D., R. Kendel, P. Savoie and O. Sweitzer, 1975. Aishihik survey of January, 1973. Fish. Mar. Ser. Tech. Rep. PAC/T-75-4, Chapt. 1. 22 pages.

Abstract

A survey was done in the Aishihik system during January 8 - 19, to obtain biological data on stocks during winter conditions and physical measurements of ice, snow and water conditions. The survey was an experiment in carrying out field work under cold conditions. The maximum temperature varied from +2.1 to -55°F. Gill nets with seven mesh sizes ranging from 2.5 to 6.5 inches fished in nine lakes and stream locations for a total of 855 hours. The catch was 72 fish. Longlines caught 2 fish. Seventy percent of the catch were lake trout. Lake whitefish, burbot and pike were also present. Lake trout were found only at lake sampling sites and round whitefish only at stream sites. Data on fish weights and lengths is presented. Lake ice was generally two feet thick however the outlet from Aishihik Lake was unfrozen. A discussion is included on general considerations for winter survey.

#210
Kendel, R. E., 1975. Fisheries resource inventory of Giltana Creek, Yukon Territory, with particular reference to spawning populations of the longnose sucker (Catastomus catastomus). Fish. Mar. Ser. Tech. Rep. PAC/T-75-4, Chapt. 4, 24 pages.

Abstract

Due to the complex relationships between Giltana Creek and the upper Aishihik system, any detrimental effects resulting from Aishihik Hydro Development would be reflected in the fish stocks of the creek. The report presents basic biological and physical information which will be useful in the assessment of such affects.

Seven species of fish are known to inhabit Giltana Creek. Some species use it as a migration route, it provides spawning habitat for longnose suckers and arctic grayling. It is used as a rearing and feeding area for grayling fry, longnose suckers, northern pike and slimy sculpins. It's shores provide fry with shelter from predation, and it provides feeding areas for adult grayling, humpback whitefish, pike and sculpins.

#211
Kussat, R., 1973. Report on the 1972 Aishihik Lake, Yukon Territory, limnological survey. Fish. Mar. Ser., Rep. 1973-1. (Also: D.O.E. Fish. Mar. Ser. Tech. Rep. PAC/T-74-19, Appendix E, 35 + pages).

Abstract

In January, 1973, the Fisheries Service presented an interim report to the Yukon Territorial Water Board reporting cursory results and interpretations of the fisheries studies conducted in the summer of 1972. This report summarizes and reviews the results obtained from the limnological survey of Aishihik Lake and attempts to assess the impact of the power proposal on the fish resource of the lake.

The 1972 limnological study consisted of four parts:

(1) A fish inventory program to determine the abundance and geographic location of the major fish species in the lake;

(2) A sampling program of the fish populations in different lake areas in terms of age, weight, length, condition factor, stomach content, and sex ratio;

(3) The productivity of various lake areas by the collection and enumeration of benthic organisms and zoo-plankton; and

(4) limnological determinations in the main lake basin consisting of physical and chemical parameters.

The report concludes that the change in biological productivity due to minimal changes in lake level will probably not be measurable.

#212
Lawley, B.C., 1974. Memorandum Report - Aishihik Lake Studies, 1973. Fish. Mar. Ser. Tech. Rep. PAC/T-74-19, Appendix D. 2 + pages.

Abstract

This short report notes some field study results from the summer of 1973. Several different studies were carried out. Extensive beach seining was done around the Aishihik Lake shoreline to determine areas of fry rearing. Tow netting was also tried but no fish were obtained using that method.

All adult fish collected at Aishihik and Stevens Lakes were analyzed for parasites, bacterial and viral diseases, and sexual maturity. A diving survey was done in several locations to locate spawning locations for lake trout, Arctic grayling and whitefish. Two dredging surveys were carried out in further studies on benthos composition.

#213
Pearse Bowden Economic Consultants, 1972. Aishihik River Power Development Yukon Territory. A preliminary environmental impact statement. Prepared for Northern Canada Power Commission.

Abstract

The purpose of the preliminary environmental impact statement was to determine the impact of the hydroelectric development on the environmental and recommended design and operating parameters that would minimize adverse environmental effects.

The study was divided into chapters which dealt with impacts on human, aesthetic and scientific resources, fisheries, wildlife, waterfowl and flora. The three page section on fisheries impacts was admittedly based on guesswork as very little field data existed for the Aishihik system. This section concluded that a definite loss in fish production in the Aishihik River below Otter Pool was likely, but that elsewhere on the system the impact could be either detrimental or beneficial.

#214
Pearse Bowden Economic Consultants Ltd., 1975. Environmental Implications of Aishihik River power developments in Yukon Territory. A compendium report. Prepared for Northern Canada Power Commission.

Abstract

This report reviews previous studies and summarizes the environmental effects of the Aishihik hydro-electric development proposal on various components of the environment; Aishihik village, archaeological sites, vegetation, wildlife and fisheries. The environmental costs associated with power development are assessed.

In the section dealing with the fisheries resource, the authors of the report conclude that it will not be possible to maintain fish stocks in the Aishihik River between Canyon Lake and the powerhouse tailrace at their present level due to the diversion of flows to the powerhouse. Potential parasite transfer problems associated with diversion of water from Stevens Lake could be dealt with by preventing pike from entering Stevens Lake and poisoning pike resident in the lake before a diversion took place. Fisheries Service investigations are criticized for providing little quantitative data relating to either the size of fish populations or the value of the fisheries they support.

#215
Robertson, R.A. and B.R. Eliasen, 1974. Environmental engineering studies associated with the Aishihik River hydroelectric development. Fish. Mar. Ser. Tech. Rep. PAC/T-74-18. (Also: D.O.E. Fish. Mar. Ser., Tech. Rep. PAC/T-74-19, Appendix B). 31 + pages.

Abstract

A number of engineering studies were conducted in the Aishihik area from 1972 to 1974. This report details the results of some of these studies and outlines the implications of various physical aspects of the proposed power development.

The engineering studies include water surface profiles, depth soundings, shoreline classifications and controlled stadia surveys. The report presents results of work on the Sekulman and upper Aishihik Rivers and the Aishihik Lake shoreline study. Maps, photographs and water profile charts accompany the text. The engineering study results are used to detail the effects that raising and lowering the lake levels will have on Aishihik Lake and its tributaries.

#216
Schowenburg, W. J. (ed.), 1974. The Aishihik hydroelectric development. Implications for fisheries resource maintenance. D.O.E., Northern Operations Branch. Pacific Region, Fisheries and Mar. Serv., Tech. Rep. PAC/T-74-19. 19 pages.

Abstract

This publication brings together a number of technical and manuscript reports prepared from field studies carried out on the proposed Aishihik hydroelectric development. The field investigations were done between the spring of 1972 and the fall of 1974. The implications of the project for fisheries management is the focus of an assessment of problems posed by the hydro project.

The discussion places emphasis on the following concerns:

(1) Effects of water storage on Aishihik Lake and its tributaries;

(2) Implications of regulation of flow from the Aishihik Lake reservior;

(3) Implications of water diversion into the powerhouse intake canal;

(4) Implication of diverting Stevens Lake into Aishihik Lake.

A number of recommendations concerning the operation of the hydroelectric facility are noted for use by the Yukon Territorial Water Board in discussions of the water licence for the power project.

#217
Synergy West Ltd., 1975. Aishihik Village impact analysis. Prepared for the Yukon Native Brotherhood, Whitehorse, Yukon.

Abstract

The report summarizes the impacts of the Aishihik hydroelectric power plant on the Aishihik village.

The "new" Aishihik village is inhabited on a seasonal basis and is used as a base for moose hunting and food fishing activities. The "old" Aishihik village is a historically important trading post. The Aishihik hydro development has resulted in the reduction of fish in the Aishihik Lake. High water levels have caused the erosion of shoreline areas and parts of the village have been flooded. The power development has not produced any benefits for the village.

#218
Walker, C. E., 1975. Field activities in the Aishihik area during four winter surveys from November 1973 to March 1974. Fish. Mar. Ser. Tech. Rep. PAC/T-75-4, Chapt. 2, 8 pages.

Abstract

Four winter surveys (Nov. 6-8, Nov. 11-15, Dec. 10-13, March 11-15) were carried out in the area from the outlet of Aishihik Lake to Canyon Lake, to determine overwintering fish presence.

Round whitefish, humpback whitefish, grayling, sculpin and pike are present year round in the watercourse between Aishihik and Canyon Lakes. Populations of whitefish and pike don't vary much throughout the year, while grayling are scarce in the winter but abundant in summer. Longnose suckers and trout don't utilize the stream waters during the winter but are present in summer.

Whitefish were caught in varying stages of spawning. Conditions ranged from fully mature nonspawned fish to fully spawned fish.

A note is included on vertebrate drift populations (in March). Species diversity was low, Simulidae and Tendipedidae made up the majority of the macro-invertebrate populaton. The invertebrate and plankton biomass was unusually high.

#219
Walker, C. E., 1975. Minimum flow requirements from Aishihik Lake. Fish. Mar. Ser. Tech. Rep. PAC/T-75-4, Chapter 7. 39 pages.

Abstract

This report presents the results of a study made in the Aishihik Lake system to determine the minimum flows arising from Aishihik Lake that are required to maintain fish stocks in this area. Structures were prepared to provide experimental flows. The lesser than normal discharge resulted in significant reductions in depth, velocity and shoreline habitat. Habitat loss forced fry and fingerling to nurse and feed, respectively, in areas not normally utilized and in conditions not previously encountered. Increased stream temperature is a potential adverse effect of low flow as is predation. Minimum flows are presented on a monthly basis.

#220
Walker, C. E., 1974. Memorandum report on minimum flow requirements in upper Aishihik River, 1974. Fish. Mar. Ser., Tech. Rep. PAC/T-74-19, Appendix F. 3 + pages.

Abstract

This short report examines minimum flow requirements for fish populations between Aishihik and Canyon Lakes that will be affected by the hydro-electric power project. Four locations were studied under controlled conditions using constructed facilities to simulate low flow situations. Temperature changes and fish movements were closely monitored. The results of these observations were used to propose several remedial measures to mitigate losses to fish populations during periods of low flow.

#221
Walker, C. E. MS. 1972. Aishihik Lake program. Memorandum to A. Gibson, Fisheries Service, Pacific Region. File 31-5-17-5, Vol. 2, 26 pages.

Abstract

This memorandum outlines studies made and the results obtained within the Aishihik watershed during 1972. Observing/netting/trapping activities were carried out in eight streams considered to be important in the Aishihik system. Nine species including grayling, rainbow trout, lake trout, round whitefish, humpback whitefish, lake whitefish, pike, burbot and suckers were found. Spawning areas were not identified but were suggested by the presence of sexually ripe fish, spawned eggs and/or newly emerged juveniles.

#222
Walker, C. E. and R. F. Brown, MS. 1975. Apparent growth, age and maturity of Aishihik stream fishes. D.O.E., Fish. Mar. Ser., Vancouver, B.C. 26 pages.

Abstract

Biological data on lengths, ages and sexual maturity of nine fish species of the Aishihik River system were collected in the course of field studies from 1972 to 1974. This report synthesizes this data and presents some preliminary conclusions on the biology of these species.

The majority of fish sampled were of four species: humpback whitefish, round whitefish, Arctic grayling and longnose suckers. Also sampled were lake trout, northern pike, burbot, slimy sculpin and rainbow trout. Raw data on lengths, ages and maturity is included.

#223
Walker, C. E. and R. F. Brown, 1974. Stream studies in the Aishihik system with reference to hydroelectric development. Fish. Mar. Ser. Tech. Rep. PAC/T-74-9. (Also: Tech. Rep. PAC/T-74-19, Appendix C, 46 + pages).

Abstract

The Aishihik River system located approximately 100 miles west of Whitehorse is under development for the production of hydro power. The Fisheries and Marine Service, Department of the Environment, carried out studies in 1972 and 1973 to assess the effects of the proposals on the productivity of the aquatic ecosystem. Responsibility was divided between agencies. This report discusses the program to evaluate streams and stream populations which was conducted by the Northern British Columbia and Yukon division.

Gillnetting, seining, dipnetting, fyke netting and sport angling were used to capture fish between Aishihik Lake and Canyon Lake. Analysis of stomach contents, scales invertebrate drift samples and bottom samples were completed. A number of tributary streams flowing in the Aishihik system were also sampled.

The results of these studies provide information from which to assess the effects to fish populations of the changing water levels in the system due to hydro dam construction. The effect of flow reduction on an all-purpose, all-season waterway and the fish therein is discussed.

YUKON RIVER MAINSTEM SUMMARY

Introduction

The Yukon River is the fifth largest river in North America and the third largest salmon producing river in North America. The salmon resource supports a native subsistence fishery, a commercial fishery, as well as a domestic and sport fishery. The river arises on the British Columbia side of the north coastal mountains within 30 km. of the Pacific Ocean; but it flows northward and westward for over 3000 km. draining the southern portion of Yukon Territory and crossing the international boundary to continue through Alaska to the Bering Sea. Major headwater lakes include the Kluane, Bennett, Tagish, Atlin and Teslin.

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The following references include information on the Yukon mainstem.

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#2 Acres Consulting Services Ltd. in association with IEC International Environmental Consultants Ltd. 1981. McIntyre Creek Power Plant No. 3 Prefeasibility Study Report prepared for the Yukon Electrical Company Limited. 42 pages.

#3 Alaska Highway Pipeline Panel 1979. Initial impact assessment: Dempster Corridor. 300 pages.

#6 Baker, S. 1980. Environmental concerns and recommendations: Alaska Highway reconstruction and maintenance, Kilometre 1008-1635. Environmental Impact and Assessment Report. EPS 8-PR-80-1. 94 pages.

#7 Baker, S. A. 1980. In situ bioassay study at a gold placer mining area in Yukon using Arctic grayling. In: Weagle, K. 1980, Report on the technical workshop on suspended solids and the aquatic environment. Department of Indian Affairs and Northern Development. 10 + pages.

#8 Baker, S. A. 1979. An environmental survey of selected Yukon river headwater lakes impacted by communities. Canada Department of Environment. EPS Regional Program Report No 79-26. 75 pages.

#9 Baker, S. A. 1979. A survey of environmental mercury concentrations in Yukon Territory. Canada Department of Environment, E.P.S. Regional Program Report No. 79-21. 74 pages.

#13 Beak Consultants Ltd. 1979. Fisheries investigations along the Dawson Trail segment of the Dempster Lateral gas pipeline route, Yukon Territory, 1979. Report prepared for Foothills Pipe Lines (Yukon) Ltd., Calgary, Alberta. 32 + pages.

#14 Beak Consultants Limited 1979. Fishery resource investigations of waterbodies within the influence of the Alaska Highway gas pipeline. Alternative alignments, 1978. 2 Volumes. Report prepared for Foothills Pipe Lines (South Yukon) Ltd. Calgary, Alberta. 267 + pages.

#15 Beak Consultants Ltd. 1979. Surveillance of selected watercourse crossings along the Alaska Highway gas pipeline route in Yukon Territory, Year 1 (1978). Prepared for Foothills Pipe Lines (South Yukon) Ltd., Calgary. 57 + pages.

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#19 Bodaly, R. A. 1979. Morphological and ecological divergence within the lake whitefish (Coregonus clupeaformis) species complex in Yukon Territory. J. Fish. Res. Board Can. 36:1214-1222.

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Species Distribution

Chinook salmon are found throughout the Yukon River all the way to spawning grounds in the M'Clintock River.

This spawning area is the farthest upstream from Whitehorse which is known. Some of the known spawning areas in the Yukon system for chinook include the Takhini River, Big Salmon River, Nordenskiold River, Tachun Creek, Big Creek, Klondike River and North Klondike River. Chinook also spawn in the mainstem Yukon River in many locations. Chum salmon spawn in the Nordenskiold River, Big Creek and in the mainstem Yukon. Coho salmon are known to spawn upstream of Dawson but it is not known exactly where.

Freshwater fish species found in the Yukon River include arctic grayling, lake trout, burbot, northern pike, whitefish, and rainbow trout. Many of the lakes in the Yukon River drainage support overwintering fish populations as do some of the larger tributaries.

Report Summary and Evaluation

Many reports are available for the more southerly sections of the Yukon River. Many of these reports relate to the Alaska Highway Gas Pipeline which will pass through the area. Stream inventories for the pipeline project were often done superficially. Overwintering studies have been done in some of the areas.

Spawning areas, critical habitat areas and timing of migration have not been documented sufficiently. Fisheries and Oceans have produced stream catalogues covering the Yukon River from Carmacks and above. These reports are quite detailed giving physical, chemical and biological information for each stream but basically only fish presence is recorded. Mining studies in this area include those done for the Adanac Mine near Atlin, B.C., Venus Mines on the B.C./Yukon border and Arctic Gold and Silver Mines near Bennett Lake. The Adanac report is quite detailed and includes locations of spawning areas for grayling in the Surprise Lake area.

The construction of Whitehorse Rapids Dam has initiated several studies, many of which were done in the late 1950's and 1960's. Monitoring of chinook migration through the fishway has taken place every year since 1959. Population estimates were done in 1973 and 1974 by the Fisheries and Marine Service for chinook and chum in the Yukon River by means of a tagging program.

Fisheries studies are much more limited in the more northerly areas of the Yukon River Basin. As there is more mining activity in the Klondike region there have been more placer mining environmental studies done in the north but they are not usually very detailed reports. Overwintering studies are lacking except for in the Sixty Mile River. Pipeline reports have been produced involving the proposed Dempster Lateral to the Alaska Highway Gas Pipeline.

In 1975 and 1976 Northern Natural Resource Services conducted chinook salmon studies near Dawson involving migration and possible spawning locations.

Data Gaps and Recommended Studies

Geographically, there are several data gaps in the mainstem Yukon River. The Yukon River between Lake Laberge and Big Salmon River has very little information except that a few of the small lakes are used for overwintering and spawning. Between Fort Selkirk and the Sixty Mile River the Yukon River is virtually unstudied. Hardly any studies are available for the Klondike and North Klondike Rivers or for the Yukon River downstream of Dawson to the Yukon/Alaska border.

(a) Energy

With respect to hydroelectric development, much more information is needed about the Mid-Yukon Project at Five Finger Rapids. With this development comes potential disruption in up/down stream migration of adult and juvenile salmon and indigenous freshwater species. Spawning and rearing habitat within the area could be lost. A complete inventory of fish resources in the Yukon River and tributaries between Tachun Creek and Hootalingua in the proposed reservoir area is required as is identification of spawning grounds rearing and overwintering areas for all fish species. A survey made in the summer of 1981 included fish sampling upstream of the proposed Mid-Yukon site (R.L. and L: 1982 in progress). A study of minimum flow requirements of rivers and streams below the dams is also needed. Another proposed development is the Primrose-Kusawa proposal which is located in the headwaters of the Takhini river system. The development involves an upstream storage dam, 70 feet high, at the outlet of Primrose Lake and a storage diversion dam on Rose Lake, 40 feet high. The same information is needed for this area as for the Mid-Yukon Project. Studies are still needed at Whitehorse Rapids. There is a lack of general inventory in the storage area and tributaries. Follow-up to studies on juvenile mortalities at Whitehorse Rapids should be done.

(b) Transportation

The proposed Alaska Highway Gas Pipeline will pass through the Yukon River basin. As mentioned above, more information is needed for spawning areas and timing, migration timing, and critical habitat areas with respect to the pipeline crossings. The same information is needed for the proposed Dempster Lateral section of the pipeline.

(c) Mineral Development

Placer mining is carried out along the length of the Yukon River and has been increasing over the past few years. Mining activity is especially evident in the Klondike region. Studies on the streams affected by placer activity have been brief. Biophysical inventories of streams are needed and methods of stream rehabilitation should be investigated. Experiments should be conducted to determine specific effects of placer effluents on all life stages of northern fish species. Overwintering studies are needed on affected streams.

(d) Tourism, Parks and Recreation

Sport fishing pressures have increased with the Yukon Territory population. Fishing is especially heavy in the Whitehorse area as approximately one half of the Yukon Territory population live in or near Whitehorse, and because of the ease of accessability of the rivers and lakes nearby. Lake trout and arctic grayling are the native species most actively sought. Lake stocking programs have been investigated in the past using rainbow trout and coho salmon but this has not been carried on. More of these programs should be implemented if they are found to improve the sport fishing. Information on number of angler days and catches is needed. Portions of the Yukon river are open to commercial salmon fishing by means of gillnet or fishwheel and regulations restrict the waters, the time, the species and the method of fishing, as well as determining the eligibility for a commercial fishing license. The lakes which are open to commercial fishing have poundage quotas established for each of them. More site-specific information is needed on the commercial fisheries. Future native food fish requirements also should be assessed.

WHITE RIVER SUMMARY

Introduction

The White River is one of the principal western tributaries of the Yukon River. With headwaters in the St. Elias Mountains the river drains northeastward, joined by the Donjek River, Koidern River and the Ladue River. The Donjek is supplemented by major river tributaries draining northwestward - the Nisling and Klotassin Rivers. The White River empties into the Yukon River about 130 kilometres south of Dawson City.

The White River is extremely turbid due to the input of glacial flows from mountain glaciers. It is a highly braided, broad flat river with many changeable channels.

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#195 Wickstrom, R.D. (C.W.S.). 1977. Fish distribution in Kluane National Park and peripheral area. Prepared for Parks Canada, 31 pages.

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Species Distribution

Chinook and chum salmon both ascend the White River to headwater tributaries such as the Koidern and Donjek Rivers to spawn. Chum salmon spawning has been documented in significant numbers in the Kluane River. Overwintering potential exists in the White River, several major tributaries and in Kluane Lake. In addition to the anadromous species, lake trout, Arctic grayling, lake and round whitefish, burbot and northern pike are found in the system.

Report Summary and Evaluation

Of all the reports concerning watercourses and waterbodies within this sub-basin, over 60% of the documentation centers on work done in connection with the proposed Alaska Highway Gas Pipeline routes. Field work is concentrated along the Alaska Highway. The assorted pipeline reports document fisheries information collected at all minor and major watercourses crossed by and/or adjacent to the proposed pipeline route. The body of information collected along the Alaska Highway Gas Pipeline route is probably the most intensive in the Yukon Territory.

The bulk of other information for the White River sub-basin has been collected along the Shakwak Highway route from Haines, Alaska to Fairbanks Alaska. This route overlaps watercourses crossed by the pipeline route.

A few reports have resulted from studies on lakes in the sub-basin. A hydrometric and geophysical survey was conducted on Kluane Lake and a report on the Gladstone Lakes is currently being prepared.

The pipeline reports have all been completed in the last six years, therefore information is up-to-date. Similarily the Shakwak Highway reports have been done very recently. Fisheries information has been gathered in all seasons of the year for both of these development projects.

The majority of pipeline reports containing fisheries information are very superficial. Fish presence is documented at various times of the year, but little or no work has been done on fish biology or life history. Much of the spawning and over-wintering information may be speculative and require further verification.

The cursory nature of most of the fisheries information creates difficulties when discussing mitigative effects of development projects for the fish resources. Without precise and detailed information on fish biology the damage to fish stocks cannot be minimized by timing project construction in less critical seasons of the year. Similarly little detailed information is noted on the mitigation necessary to reduce sedimentation and erosion in watercourses.

The site-specific nature of most of the fisheries information makes it's usefulness limited when studying the river system as a whole. Simple extrapolation of data from one point to another along a river course is not recommended.

Data Gaps and Recommended Studies

Intensive fisheries information has been collected along several linear corridors in the White river sub-basin. Most of the major tributaries of the White River have some documented fisheries data obtained during field work along these corridors.

There has been little research or field work carried out away from the development corridors. The lower tributaries are virtually unstudied. The area of the White River drainage east of the mainstem (Nisling and Klotassin Rivers) has been given no attention either. The White River mainstem itself has been a low-priority system for fisheries research. Specific spawning areas for salmon species, especially chinook salmon are not well known.

(a) Energy

No proposals for energy development projects have been put forward for the White River sub-basin. The physical nature of the river does not favour a hydro dam site. there has been no field work done related to hydro potential.

(b) Transportation

The existence of the Alaska Highway has created the possibility for an Alaska Highway gas or oil pipeline. the majority of the field work in the White River sub-basin has been done in connection with these proposed pipelines. More detailed information on fisheries biology, such as timing of spawning and migration, productivity and critical habitat areas, is needed to supplement a large amount of inventory information on fish presence at specific pipeline crossing sites. Further studies on fish processes within the sub-basin, and particularly along the development corridors, would provide the knowledge necessary to minimize deleterious effects to fish stocks when corridor development begins.

One often overlooked aspect of development is the increase of workers into the area who will put increasing pressure on fish stocks through angling. An initial survey of lake productivity in the White River sub-basin shows low estimates of fish production compared to lakes in other parts of the Territory. Sport fishing pressure should be carefully monitored to prevent any over-harvesting of stocks.

(c) Mineral Development

Mining exploration in the sub-basin is increasing, and placer mining activity is still important on a number of water courses such as Burwash and Badlands Creeks. However, the fisheries information related to mining activities in this sub-basin is minimal.

Newly staked but unworked placer claims could be the focus of investigations of how stream quality and fish populations are affected by placer mining. Field inventory before, during and after mining activities might finally answer questions of how placer mining affects fish stocks. A broad-brush inventory of fish stocks on some of the major tributaries would also fill in some data gaps. The Ladue, Nisling and Klotassin rivers have not been studied in any respect.

(d) Tourism, Parks and Recreation

The presence of Kluane Park and Kluane and Wellesley Lakes are attractions for large numbers of tourists and fishermen. These two lakes have had a long history of commercial fishing exploration. However, little information exists on the utilization of fish stocks in the White River system by native or sport fishermen. The importance of the salmon resource to the people in the area is not well known.

Studies could be undertaken to gather data on native and sport catches in the sub-basin. This work could be done in conjunction with monitoring of the fishing pressure from construction workers. With the majority of sport fishing pressure coming from fishermen near the Alaska Highway, data collection using a creel census should not be too difficult.

The increase in patrons of fly-in lodges provides another source of pressure on fish stocks. There is a need to monitor the sport fishing pressure on the lakes which support this business, to prevent over use of select fish species.

Regulatory Factors

The remoteness of much of the White River sub-basin discourages extensive research. Most of the tributaries can only be reached by water or air. The large sediment load of the White River negates salmon spawning ground surveys from the air. The changeable nature of the river course makes the environment of the mainstem particularily harsh for fish spawning and rearing. The White River sub-basin is a low-priority area from a fisheries point-of-view, due to distance, lack of financial manpower resources and lack of development pressures compared to other areas.

STEWART RIVER SUMMARY

Introduction

The Stewart River originates near the NWT/Yukon border in the Hess Mountains and flows west for about 400 metric miles, eventually emptying into the Yukon River at Stewart. On its course west it is joined by the Hess, Lansing, Beaver, Nadaleen, Mayo, and McQuesten.

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#132 Mathers, J. F., N. O. West and B. E. Burns, 1981. Aquatic and wildlife resources of seven Yukon streams to placer mining. Government of Canada, Departments of Fisheries and Oceans, Indian & Northern Affairs and Environment. 183 pages and 175 pages appendix.

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Distribution of Species

Chinook salmon migrate through the Stewart River system to its headwaters. Chinook salmon have been noted in the major tributary systems and some spawning locations have been identified. Potential overwintering sites have been located on the Hess and Stewart rivers. Chum salmon have been observed in the Stewart River but populations are believed to be small and the extent of chum migration within the system is not known. Grayling occur throughout the Stewart system and spawning sites have been located on the Stewart and McQuesten rivers and in a number of streams and lakes (Ethel, Mayo, Reid, to name only the larger lakes). Other species found within the system include lake trout, northern pike, burbot, round whitefish, lake whitefish, inconnu, broad whitefish and dolly varden.

Report Summary and Evaluation

There are very few reports concerning waterbodies in the Stewart River system. The largest category of information includes studies of selected streams impacted by mining (eg. Mathers, West and Burns; 1981). A few reports (eg. Foothills Pipe Lines, 1978) document fisheries studies of waterbodies crossed by the Klondike section of the Dempster Lateral Pipeline. A catalogue of fish and stream resources (Elson, 1974) provides data on the east and central sections of the river. Three lake studies (Horler, 1982; Shepard and Davies, 1982, Lindsey, C.C., K. Patalas, R.A. Bodaly and C.P. Archibald; 1981) provide information on selected lakes in the system.

Most of the studies associated with mining sites are preliminary environmental assessments in which fisheries investigations are only one component. There have not been any mining studies which have examined the fishery resource throughout the year. All the reports have been written since 1973.

Data Gaps and Recommended Studies

Of the five Yukon river sub-basins considered in this report, the Stewart River has the least amount of fisheries documentation. Most data have been collected along the mainstem, particularly in the western section of the river, with scattered references at points on the major tributaries and some documentation of streams affected by mining.

(a) Energy

Potential hydro sites are located on the Stewart River at Fraser Falls and Independence (situated 20 miles downsteam of McQuesten). Fisheries data should be collected in these areas. Data required to assess the proposed hydro projects include a biophysical inventory of species, particularly within the proposed impoundment areas, minimum flow requirements, and migrational patterns for all species inhabiting the area.

(b) Transportation

No major transportation projects are planned in the Stewart system.

(c) Mineral Development

There is substantial mining activity within the Stewart River system (eg. Scroggie Creek, Black Hills Creek, Clear Creek, Moose Creek). As noted earlier, fisheries information is limited. More data is needed on fish distribution and identification of critical habitat (spawning, rearing and overwintering) particularly in areas affected by mining activity. Water quality data should also be collected on a continuing basis and in situ experiments are needed to determine the effects of placer effluents on life stages of fish and faunal invertebrate species.

(d) Tourism, Parks and Recreation

A number of Indian food fisheries are located on the Stewart River between McQuesten River and Mayo. Domestic fisheries exist above and below Fraser Falls, at Maisey May Creek and near the confluence of the Yukon and Stewart Rivers. More site specific information is needed on the commercial, sport and native food fish requirements, evaluations of fishery habitat and the extent of sports fishing are needed.

The potential of some of the fly-in-lakes for sports fishing should be determined. The possibility of establishing campgrounds at lakes situated along the Klondike Highway (eg. Ethel Lake has a campground) should be considered.

PELLY RIVER SUMMARY

Introduction

The Pelly River originates in the Selwyn Mountains near the NWT/Yukon border and flows west through the Boreal Forest region to meet the Yukon River at Fort Selkirk. Its major tributaries include the Hoole, Lapie, Ross, Tay, MacMillan and Kalzas Rivers.

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The following list includes references which contain information pertaining to the Pelly River sub basin.

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#38 Canada, Department of Fisheries and Oceans. 1982. Lake and Stream Files, Department of Fisheries and Oceans, Whitehorse, Yukon.

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Species Distribution

Chinook salmon migrate up the Pelly River as far as its headwaters. Migrating chinook have been observed in most tributaries within the system and spawning areas have been located in the mainstem and on many major and minor tributaries near their confluence with the Pelly River. It is probable that spawning also occurs in the upper reaches of tributaries but in most cases studies have not been made to substantiate this conjecture. The most significant chinook spawning areas documented are in Ross River, Kalzas River and Pelly River. The MacMillan River is also suspected of being a major producer but does not lend itself to aerial surveys because of water colour.

Chum salmon have been observed in the Pelly River and are known to ascend the river at least as far as Ross River in most years. However, little else is known about the extent of chum migration or location of spawning grounds.

Arctic grayling are found throughout the system and spawning locations have been identified in the mainstem and throughout a number of tributaries and lakes.

Other freshwater species found throughout the system include northern pike, lake whitefish, inconnu, round whitefish, burbot, lake trout and broad whitefish. Overwintering areas for a variety of freshwater species have been identified in the mainstem and in various lakes throughout the system.

Report Summary and Evaluation

The largest single category of information includes environmental studies of waterbodies in the vicinity of mining activities. In addition a catalogue (Elson, 1974) documents the tributaries in the central and eastern sections of the Pelly river system. Two lake studies (Horler, 1982; Shepard and Davies, 1982) provide information on productivity of selected lakes. Shepard and Davies report includes results of some fish sampling.

Pipeline studies provide some fisheries information for waterbodies associated with the Klondike and Robert Campbell routes, and a salmon tagging study includes some data on salmon migration through the Pelly system. Many of the reports compiled for the mining companies are superficial as sampling is often brief, sample sizes are small, there is little documentation of critical periods (migration, spawning) and studies have not been made throughout all seasons. In a few instances environmental information for waterbodies associated with mining sites has been collected over a number of years but in other cases studies are restricted to one year or one season.

Data Gaps and Recommended Studies

Incomplete fisheries inventories exist for the mainstem and major tributaries. Little information is available on the upper reaches of the tributaries, although there is some data on the

Ross and South MacMillan. Preliminary investigations have been made in conjunction with development projects (i.e. North Canol Road, mining operations) but more detailed information is needed.

While rearing habitat in the Pelly River is excellent and spawning of freshwater fish and chinook salmon is known to occur throughout the system (N.N.R.S., 1977) there is still little information on the extent of spawning in the mainstem. Overwintering is thought to be good but no winter studies have been conducted. However, the Department of Indian and Northern Affairs has hired Envirocon to make a study of overwintering habitats of juvenile chinook salmon in tributary steams along the North Canol Road right-of-way.

(a) Energy

A number of hydro power sites have been proposed within the Pelly River system. These sites include Hoole Canyon (about 20 miles upstream of Pelly Crossing), Granite Canyon (15 miles upstream of Pelly Crossing) and Ross Canyon. Data gaps related to these proposed developments include a lack of biophysical inventories, a lack of minimum flow requirements for the fish species as well as a lack of information on detailed fish distributions. A study is presently being prepared by R. L. and L. Consultants for N.C.P.C. on the five proposed hydro sites including the Mid-Yukon, Granite Canyon, Hoole Canyon, Ross Canyon, and False Canyon sites. This study will include the results of a fish sampling program conducted upstream of the proposed hydro sites.

(b) Transportation

As mining continues to expand in the Ross River and MacMillan Pass areas, improvement and realignment of the North Canol Road and construction of access roads will follow. Studies are needed to determine critical habitats, migration and spawning timing in order to mitigate the impacts of road construction and ensure easy fish passage through culverts and/or bridges.

(c) Mineral Development

There are a number of active and proposed mining operations within the Pelly system, particularly near Faro (lead, zinc, silver) in the MacMillan Pass area (tungsten, lead, zinc) and along Ross River (coal and barite). A placer mining operation is situated on Russel Creek.

A number of studies have been made on Rose Creek/Pelly River near Cyprus Anvil. Preliminary environmental investigations have been made in the vicinity of some mining properties in the MacMillan Pass area including the Jason and Cathy properties. Summer fisheries investigations have been made on the Ross River along the North Canol road. However, more studies are needed in these areas to identify critical fish habitat and migration timing, fish species and distribution, and water quality in order to assess and monitor the impacts of mining.

(d) Tourism, Parks and Recreation

There are substantial commercial, domestic and Indian fisheries located throughout the Pelly system. Commercial, domestic and Indian food fisheries at Minto involve residents from Pelly Crossing, an Indian and domestic fishery operate at the town of Ross River and Indian food fisheries exist at Pelly Crossing and Lapie River. Sports fishing also occurs on the Lapie River and possibly along other tributaries. Site specific information is needed on distribution and species composition. An assessment of native food fish requirements and an evaluation of the fisheries habitat is also needed.

Continued mining developments in the MacMillan Pass, Ross River area will result in more exploitation of the fishery resource by mine personnel. Preliminary studies to investigate lake productivity and exploitation by sports fishing have been made in selected areas within the Pelly system (Shepard and Davies, 1982; Horler, 1982). More lake studies are needed to assess productivity and to assess the amount of sports fishing activity that these lakes can sustain.

TESLIN RIVER SUMMARY

Introduction

The Teslin system drains an area of approximately 12,000 sq. miles located in south central Yukon and northern British Columbia. Teslin Lake, 78 miles long and averaging about 2 miles wide at an elevation of 2239 feet, is the sub-basin's dominant waterbody and is one of the principal headwaters of the Yukon river. The southern part of the area lies in the order of 2000 - 3000 feet above sea level. Mountains prevail in the northeast part of the area with elevation at approximately 700 feet. The climate is severe with long cold winters. The Alaska Highway lies east-west through the center of the area and utilizes in part the north-east shore of Teslin Lake.

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Species Distribution

The rivers and streams flowing into Teslin Lake support spawning populations of chinook salmon, arctic grayling, burbot, dolly varden, lake trout, whitefish and northern pike. Fish are present in many of these waters but spawning areas are unknown. Overwintering takes place in some of the rivers and lakes in this area. In Teslin Lake itself lake trout, arctic grayling and whitefish species spawn. The lake is an overwintering area for these and other freshwater species and for juvenile chinook salmon. The Teslin River between Teslin Lake and the confluence with the Yukon River is lacking information. Arctic grayling and chinook are present in this area. Chinook salmon spawn in the Teslin River near the mouth of the Baswell River and probably spawn in other areas of the mainstem Teslin River but specific areas have not been documented. Chum salmon spawn throughout the Teslin River downstream of Teslin Lake.

Report Summary and Evaluation

Most of the studies done in the Teslin sub-basin have been done as part of the environmental requirements of the proposed Alaska Highway Gas Pipeline. These reports have been completed within the last 6 years but they deal only with the southern portion of the Teslin system. A catalogue of fish and stream resources was produced by the Department of Fisheries and Oceans in 1973 which also covers the southern portion of the system. Studies in 1977 by Northern Natural Resources Services also provides stream inventories for this area. All of these stream inventories use basically the same sampling techniques, electrofishing, beach seining, gillnetting and angling. Overwintering studies are included. A limnological survey of Teslin Lake was done in 1944 by scientists from the University of British Columbia.

Data Gaps and Recommended Studies

Fisheries studies have been concentrated in the southern portion of the Teslin system. Teslin Lake and the rivers leading into it have been the subject of quite a few investigations but there is still much information needed on spawning areas and critical habitat areas for all species. The length of the Teslin River from Teslin Lake to the confluence with the Yukon river is almost totally lacking in data.

(a) Energy

The site of the Low NWPI hydroelectric proposal is about 37 miles downstream from Johnston's Crossing. This proposed development could involve an 80 foot high rockfill dam. Studies are needed to assess the minimum flow requirements for fish species and to assess the importance of the area for fish rearing and spawning.

(b) Transportation

The proposed Alaska Highway Gas Pipeline would run east-west through the Teslin River basin. Much of the information collected in the river basin was collected in order to assess alternate routes for this pipeline. Timing of spawning and migration, productivity and critical habitat areas are still unknown for many of the crossing locations. Some overwintering studies have been done.

(c) Mineral Development

Placer mining is not prevalent in the Teslin watershed at the present time but studies are needed on streams to be affected by placer development. These studies should include biophysical inventories of the streams, gathering of hydrological information, and exploration of methods for stream rehabilitation. Lab experiments could be done to determine specific effects of placer effluents on the various life stages of northern fish species.

(d) Tourism, Parks and Recreation

Sport fishing is important in the Teslin river and more site specific information is required on number of angler days and catch. More information is also required on commercial and native fisheries. Future native food requirements should be assessed. It is important that overfishing does not occur during the construction of the Alaska Highway Gas Pipeline.

AISHIHIK RIVER SUMMARY

Introduction

The Aishihik River system is located approximately 160 kilometres west of Whitehorse. It consists of three major lakes, namely Sekulmun, Aishihik and Canyon, which are interconnected by short streams. A number of lateral tributaries flow into the system. Drainage is by way of the Aishihik, Dezadeash and Alsek Rivers to the Pacific Ocean.

The Aishihik River system is being included in this bibliography because of the possibility of diversion of several Yukon River Basin waterbodies into the Aishihik system. Northern Canada Power Commission is currently doing feasibility studies on the diversion of the following waterbodies:

1) Stevens Lake - diverted into Aishihik Lake;

2) Gladstone Lake - diverted into Issac Creek;

3) Long Lake - diverted into Giltana Creek.

Any of these diversions may introduce foreign parasites into new river systems. Also, introduction of Yukon River Basin species foreign to Pacific drainage (eg. Coregonus sardinella, least cisco) may cause ecological disruption. Least cisco could enter into interspecific competition with Pacific herring in the Pacific drainage; there is also a possibility of least cisco salmon interaction at the fry stages of the life cycle.

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Species Distribution

The Aishihik River system is at the headwaters of the Alsek

River. Though the Alsek has several anadromous salmon species in its lower reaches, no anadromous fish occupy the Aishihik system at the present time. The major lakes of the system contain lake trout, Arctic grayling, lake and round whitefish, burbot and northern pike. Rainbow trout were found in the Aishihik River prior to hydro dam construction.

Report Summary and Evaluation

Because of time constraints, the complete range of documentation on the Aishihik River system was not reviewed. A number of consultant reports done for the Northern Canada Power Commission were not included. However, all known Fisheries Service reports were examined.

The Aishihik river system is probably the most intensively studied river system in the Yukon Territory from a fisheries point-of-view. In 1972 N.C.P.C. brought forward a proposal to construct a hydro-electric dam on Aishihik Lake. This proposal elicited an intensive field studies program by the Fisheries Service to determine the fish species and aquatic invertebrate presence and distribution and biological processes operating in the system prior to development. Limnological surveys, hydro surveys, physical and chemical testing of lakes and tributaries and minimum flow investigations were also undertaken.

Intensive fish inventories have been done using gillnetting, seining and angling in all the lakes and major tributaries. Invertebrate populations have also been studied. Sampling was carried out in all seasons of the year. Laboratory analysis of fish stomach content, parasites, scale samples, invertebrate and benthos samples and water samples was very thorough. All this work was done from 1972 until 1975 when the hydro dam became operational.

The Fisheries Service used this information to prepare comments and recommendations for several Yukon Territorial Water Board hearings discussing conditions for N.C.P.C.'s water licence. In addition to Fisheries Service documentation there are a number of reports completed by N.C.P.C. consultants. The N.C.P.C. consultants did minimal field work in preparing their impact statements. Little work has been done in the area since 1975.

Data Gaps and Recommended Studies

With N.C.P.C. interested in diversion of Yukon River Basin streams into the Aishihik River system, there is a need to learn more about the biophysical aspects of the waterbodies to be diverted. One parasite study was done to compare Stevens and

Aishihik Lake species; this process should be extended to the other waterbodies to reveal potential parasite problems between the different river systems.

Several of the waterbodies to be diverted (eg. Long and Stevens Lakes), provide excellent fly-in and/or land access fishing opportunities. Current use of these lakes should be measured to determine the potential disruption to sport fishermen. All of the potential diversion waters should be further studied for fish species presence and distribution,physio-chemical water qualities and invertebrate populations.

Fisheries maintaince flow stuides should also be undertaken. The role of each existing outflow should be carefully assessed with studies on streamflow patterns, biological regime and influence on downstream habitat. For example, chinook salmon spawning at the outlet of Hutshi Lake may be impacted by reduced outflow from Hutshi Lake due to curtailment of inflow from a diverted Long Lake. These types of linkage relationships should be carefully examined.

A volume of information exists which provides a fairly complete knowledge of the fisheries characteristics of the Aishihik River system prior to the hydro development. Follow-up studies could be done, duplicating the methods of earlier field work, to provide a data base for comparison with pre-constructed information. This would provide an opportunity to assess the actual impacts of the development.

Regulating Factors

The Aishihik River system is very accessible for further studies, as it is close to Whitehorse and approachable by road. The diversion areas are a little more difficult to reach but air access is still relatively inexpensive. There are no major logistical obstacles to further field work.

The river system itself is regulated by very low rainfall in the area, and by the structure of the power project. The dam creates artificial lake conditions and interrupts flows in river courses below the impoundment. Fish populations have probably changed since 1975 due to lake siltation, changing water levels and reduced river flows.

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